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CSE 450 CAPSTONE PROJECT FINAL REPORT

Antiphony

Dual-Perspective Modeling of Persuasion and
Commitment in Asymmetric Dialogue

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CANDIDATES' DECLARATION

We declare that the work presented in this report, titled “Antiphony: Dual-Perspective Modeling of Persuasion and Commitment in Asymmetric Dialogue”, is the outcome of the engineering and research work carried out by us under the supervision of Dr. Md. Shariful Islam Bhuyan, Associate Professor, Department of CSE, BUET.

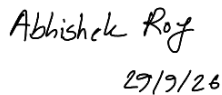
All prior work, third-party code, libraries, data, copyrighted material, and tools used in the project have been properly acknowledged. Any use of generative AI tools has been disclosed in accordance with academic integrity guidelines. We accept accountability and responsibility for the accuracy and integrity of this report and for our individual contributions to the project.



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ABSTRACT

Many everyday conversations are attempts at persuasion: a fundraiser asking for a donation, or an online seller negotiating with a buyer over chat. These conversations are asymmetric. One person argues and makes offers, while the other weighs them and finally decides. For the people running such conversations at scale, two questions matter most: is the other person actually going to say yes, and which persuasion tactics are being used along the way? Existing text classifiers usually treat a dialogue as one undivided block of text, which ignores who is speaking, cuts long conversations short, and often leans on surface cues in the last few lines.

This project, *Antiphony*, builds a framework that answers both questions. For commitment prediction, we manually re-labeled the *Persuasion for Good* charity dialogues, because the dataset’s recorded donation amounts often disagree with what participants actually said. We then designed the *AntiphonyClassifier*, a model that reads the two speakers as separate streams and lets the decision-maker’s side attend to the arguer’s side, following how the decision builds up over the conversation. It was the strongest of the fourteen model versions we built, at about 0.86 Macro-F1. For strategy recognition, we labeled every persuader turn with the tactics it uses under an 11-category, 41-strategy scheme, and built an ensemble of classifiers that gets close to the agreement limit of the labels themselves.

To test whether the approach carries over to commerce, we created a new 700-dialogue buyer–seller negotiation benchmark, including dialogues in which the buyer changes their mind before the end. On these, *Antiphony* holds its accuracy better than a model that only reads the final turns, which shows that it tracks the whole course of the decision rather than its last words.

The resulting models are small enough to run on a single commodity GPU. They can help fundraising and sales teams spot likely commitments early, prioritize promising leads, review which tactics work, and flag manipulative pressure in automated conversations.

Keywords: Persuasion Strategy, Conversational Intention, Asymmetric Dialogue, Antiphony Architecture, Multi-Label Classification, Adversarial Reversal.

Contents

Candidates' Declaration	i
Acknowledgement	ii
Abstract	iii
1 Problem Identification and Formulation	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.2.1 Sub-Problem 1: Grounded Conversational Intention Classification	2
1.2.2 Sub-Problem 2: Multi-Label Persuasion Strategy and Category Attribution	2
1.2.3 Sub-Problem 3: Cross-Domain Transfer and Adversarial Trajectory Tracking	2
1.2.4 Required Engineering Knowledge Profiles	3
1.3 Complexity of the Problem	3
1.4 Project Objectives and Scope	3
1.4.1 Project Objectives	3
1.4.2 Project Scope and Explicit Exclusions	5
2 Preliminary Research	6
2.1 Literature Review	6
2.1.1 Persuasion Theory and Rhetorical Frameworks	6
2.1.2 Persuasion Corpora and Conversational Strategy Annotation	6
2.1.3 Transformer Encoders and Dialogue Language Models	6
2.1.4 Hierarchical and Role-Aware Dialogue Networks	6
2.1.5 Imbalanced Learning and Loss Formulations	7
2.1.6 Ensembling and Stacking Generalization	7
2.1.7 Negotiation Benchmarks and Shortcut Learning	7
2.1.8 Synthesis of Research Gaps	7
2.2 Existing Solutions and Technologies	7
2.3 Market and User Research	8
2.3.1 The Rise of Conversational Social Commerce	8
2.3.2 User Personas and Behavioral Dynamics	8
2.3.3 Industry Consultations and Field Observations	9
2.4 Feasibility Study	9
3 Requirements Analysis	10
3.1 Project Objectives and Constraints	10

3.1.1	Stakeholder Analysis and Conflicting Expectations	10
3.1.2	Functional Requirements (FR)	10
3.1.3	Non-Functional Requirements (NFR)	11
3.1.4	Physical and Operational Constraints	11
3.2	Regulatory Requirements and Standards	11
3.3	Formulation of Solutions	12
3.3.1	Multi-Criteria Evaluation and Decision Matrix	12
3.3.2	Conclusion and Selection Justification	13
4	System Design and Implementation	14
4.1	Functional Design and Sub-problem Partitioning	14
4.2	Persuasion Strategy Annotation and Taxonomy	16
4.2.1	Hierarchical Taxonomy Formulation	16
4.2.2	Contextual Annotation Pipeline and Decision Protocol	16
4.2.3	Empirical Corpus Statistics and Rater Effects	18
4.2.4	Derivation of the Cross-Annotator Agreement Ceiling	18
4.3	Conversational Intention Ground Truth Generation	19
4.3.1	The Ground-Truth Divergence Problem	19
4.3.2	The Three-Pass Annotation Lineage	19
4.3.3	Inter-Annotator Agreement and Boundary Cases	20
4.4	Synthetic E-Commerce Negotiation Benchmark	20
4.4.1	Motivation and Privacy Bottlenecks	20
4.4.2	The Three-Phase Generation Framework	20
4.4.3	Frozen Benchmark Dataset Counts	22
4.4.4	Generic Architecture Diagnostic and Ablation Harness	22
4.5	Strategy and Category Modeling Architecture	23
4.5.1	Multi-Task Category-First Formulation	23
4.5.2	The 19-Member Heterogeneous Model Pool	23
4.5.3	Decision Layer Search: Calibrated Blend vs. XGBoost Stacker	25
4.6	The AntiphonyClassifier Architecture	26
4.6.1	Detailed Mathematical Formulation	27
4.7	Architectural Evolution: The Path to Dual-Stream Antiphony	30
4.7.1	Stage 1: Monolithic Flat Baselines and Forensic Debugging	30
4.7.2	Stage 2: Hierarchical Discourse Modeling	31
4.7.3	Stage 3: Recency Gating and Decoupled Baselines	31
4.7.4	Stage 4: Role-Pure Dual-Stream Antiphony Architecture	31
4.8	Design Refinement and Simulation	31
4.8.1	Capacity Calculations and Sequence Length Budgeting	31
4.8.2	Memory Profiling and Hardware Guards	32
4.9	Application of Tools	32
4.10	Implementation	32

4.10.1	Version Control and Development History	32
4.10.2	Deterministic Seeding and Audit Artifacts	32
4.11	Deployment	33
4.11.1	Prototype Execution Pipeline	33
4.11.2	Resource Consumption	33
4.11.3	Operational Boundaries and Omissions	33
4.12	User Interface	33
4.12.1	Donor and Buyer Side	33
4.12.2	Fundraiser Workspace	33
4.12.3	Responsive Layouts	33
4.12.4	Merchant Workspace	37
5	Investigation and Evaluation	39
5.1	Experimental Investigation Setup	39
5.1.1	Research Questions and Hypotheses	39
5.1.2	Data Splits and Evaluation Protocols	39
5.1.3	Evaluation Metrics	39
5.2	Corpus Dynamics and Strategy Exploratory Data Analysis	40
5.2.1	Key Analytical EDA Takeaways	40
5.3	Intention Classification Results Across Three Passes	41
5.3.1	Part 1: Results on the First-Pass Dataset (Binary + Modifier)	41
5.3.2	Part 2: Results on the Second-Pass Dataset (Modifier Dropped)	43
5.3.3	Part 3: Results on the Third-Pass Dataset (Strict Commitment)	43
5.3.4	Direct Comparison: Decoupled Single-Stream vs. Dual-Stream Antiphony	43
5.4	Strategy and Category Multi-Task Benchmarking	45
5.4.1	The Evolutionary Arc Across Five Modeling Stages	45
5.4.2	Operating Points and Micro-Optimal Decoding Analyses	45
5.4.3	Evaluated Decision Combiners: Stacker vs. Blend	45
5.5	Cross-Domain Transfer and Adversarial Ablation	46
5.5.1	Monolithic Input Ablation Study	46
5.5.2	Treatment Evaluation: Antiphony on Augmented Dialogues	46
5.5.3	Widening of the Sufficiency Gap	46
5.6	Skew Stress Analysis	47
5.7	Forensic Post-Mortems	47
5.7.1	Tokenizer Truncation Directionality Forensic	47
5.7.2	Merchant Accommodation Leakage	49
5.8	Performance Evaluation Against Requirements	49
5.9	Deviations and Design Revisions	49
6	Project Management, Teamwork, and Impact	51
6.1	Project Management and Finance	51

6.1.1	Planning and Risk Management	51
6.1.2	Budgeting and Resource Identification	53
6.1.3	Economic Analysis and Financial Prospecting	53
6.1.4	Sustainability in Business and Commercialisation	53
6.2	Individual and Teamwork	53
6.2.1	Participation and Contribution	53
6.2.2	Collaboration and Conflict Resolution	54
6.2.3	Leadership and Direction	54
6.2.4	Multidisciplinary Engagement	54
6.3	Impact on Society	55
6.3.1	Positive Societal and Commercial Benefits	55
6.3.2	Dual-Use Risks: Coercive Manipulation and Dark Patterns	55
6.4	Environmental Impact and Life Cycle Sustainability	55
6.4.1	Transparent Computational Energy and Carbon Accounting	55
6.5	Ethics	56
6.5.1	Equity and Inclusivity	56
6.5.2	Accountability and Personal Responsibility	56
6.5.3	Proper Use of Intellectual Property	56
6.5.4	Professionalism and Ethical Codes	56
7	Conclusion	57
7.1	Conclusion and Future Work	57
7.1.1	Summary of Achievements Against Objectives	57
7.1.2	Documented Limitations	58
7.1.3	Focused Future Work	58
7.2	Continuous Engagement and Self-directed Learning	58
A	Hyperparameters and Optimization Specifications	63
B	Repository Map and Notebook Lineage	64
C	Data Contracts and Schemas	65
D	Conflict Register and Empirical Audit Log	66
E	Complete 41-Strategy Empirical Prevalence Table	68

List of Figures

4.1	Overall functional system architecture, illustrating data ingestion across three corpora, shared utterance encoding, the dual-stream Antiphony intention network, the 19-member multi-task strategy classification framework with calibrated decision blending, and the diagnostic ablation evaluation framework.	15
4.2	Persuasion strategy annotation pipeline, depicting contextual batching (200 turns per batch, 5-turn history), the 4-tier decision protocol, LLM agent annotation under human checkpoints, automated schema validation, and multi-label compilation into <code>multilabel_merged.jsonl</code>	17
4.3	Three-pass intention annotation lineage, showing the initial 738/279 binary split with modifier, the second pass deprecating the modifier, and the third pass enforcing strict unconditional commitment (694 yes / 323 no via two-step re-checking).	19
4.4	Three-phase synthetic negotiation benchmark generation pipeline, comprising persona-conditioned LLM drafting ($N = 500$ canonical, $N = 200$ adversarial reversals), organic paraphrasing with linguistic jitter, and multi-tier semantic audit.	21
4.5	Strategy and category classification framework, illustrating the 19-member model pool, multi-task representation heads, and decision layer search comparing calibrated power-mean blending (selected, val macro-F1 0.5716) against XGBoost stacking (tested, val macro-F1 0.5508).	24
4.6	High-level conceptual schematic of the AntiphonyClassifier architecture, showing the information flow from alternating conversational turns to speaker stream disentanglement, directional asymmetric cross-attention, dual trajectory and attention pooling context aggregation, and final commitment classification.	26
4.7	Detailed neural architecture and tensor specifications of the AntiphonyClassifier, depicting exact layer dimensions, 2-layer independent stream Transformers, directional cross-attention ($\mathbf{Q}_D \mathbf{K}_P^\top / \sqrt{d_k}$), parallel commitment-trajectory GRU and learned attention pooling, and composite Class-Balanced Focal and Supervised Contrastive loss heads.	28
4.8	Four-stage architectural evolution of intention classification models, showing the main innovations, bug diagnoses, and peak test Macro-F1 across stages.	30
4.9	Donor side of the persuasion app on desktop.	34
4.10	Donor side of the persuasion app on a phone.	35
4.11	Fundraiser overview: summary cards, AI-ranked “most likely to donate” list, donation-intent distribution, and recent conversations with intent badges.	35
4.12	Fundraiser inbox. Left: conversation list with intent badges, unread counts, search, and filters. Centre: chat with strategy labels under each fundraiser message. Right: AI analysis panel with the intent gauge, predicted outcome and confidence, strategies used, intent timeline, and latest donor replies.	36
4.13	Compact (1100px) layout: the AI analysis panel opens as a slide-over drawer instead of a fixed third pane.	36
4.14	Fundraiser workspace on a phone.	37
4.15	Merchant inbox: purchase-intent gauge in the medium band, detected buyer-signal checklist, intent timeline, and quoted evidence.	38
4.16	Buyer-seller app on a phone.	38
5.1	Strategy corpus EDA: label density, category prevalence, and category co-occurrence.	40
5.2	Strategy corpus EDA: strategy deployment by turn position.	41

5.3	Strategy corpus EDA: label density and turn length distributions.	41
5.4	Cross-domain transfer: ablation curves and confusion matrices on canonical ($N = 500$) and augmented ($N = 700$) benchmarks.	47
5.5	Training convergence dynamics and performance summaries across both e-commerce benchmarks.	48
5.6	Verified label distribution of the canonical e-commerce benchmark ($N = 500$), with 262 negative / 238 affirmative instances across the four primary outcome categories.	48
5.7	Skew stress analysis of Antiphony under severe class imbalance, showing that Macro-F1 holds up across affirmative training proportions from 47.7% down to 9.8%.	48
5.8	Tokenizer truncation forensic: left-truncation retains proximal conversational context, recovering Macro-F1 from 0.8133 to 0.9331 and shrinking the apparent necessity gap from 0.1600 to 0.0402.	49
6.1	Project timeline from 20 June to 30 September 2026. Technical work ended on 16 September; the project was presented at the departmental showcase on 26 September.	52

List of Tables

1.1	Complex engineering problem characteristics (P1–P7) and project evidence	4
1.2	Knowledge profiles referenced by P1	4
2.1	Synthesis of research literature and identified gaps	8
3.1	Multi-criteria architectural decision matrix (scores: 1 = Poor, 5 = Excellent)	13
4.1	Modular subsystem decomposition and responsibilities	14
4.2	Empirical prevalence of communicative categories across 10,600 Persuader turns	18
5.1	Test set performance on First-Pass dataset (738 yes / 279 no, test $n = 153$)	42
5.2	Test set performance on Second-Pass dataset (738 yes / 279 no, test $n = 153$)	43
5.3	Detailed class-level test breakdown: Dual-Stream Antiphony on Second-Pass dataset	43
5.4	Test set performance on Third-Pass dataset (694 yes / 323 no, test $n = 153$)	44
5.5	Detailed class-level test breakdown: Dual-Stream Antiphony on Third-Pass dataset	44
5.6	Head-to-head comparison: best encoder of Decoupled Baseline vs. Dual-Stream Antiphony	44
5.7	Evolutionary progression of strategy and category modeling	45
5.8	Comparison of operating points across multi-task ensemble configurations	45
5.9	Cross-validated evaluation of ensemble decision combiners	46
5.10	Monolithic transformer input ablation results	46
5.11	Benchmark performance on augmented negotiation corpus ($N = 700$, test $n = 105$)	46
5.12	Sufficiency gap expansion across canonical and adversarial corpora	47
5.13	Verification matrix evaluating system compliance against requirements	50
6.1	Project risk management matrix, impact levels, and mitigations	52
6.2	Estimated computational energy consumption and greenhouse gas emissions	55
7.1	Objective-by-objective verification of project outcomes	57
A.1	Training and optimization hyperparameters across model families	63
B.1	Primary project artifacts, notebooks, and source code mapping	64
D.1	Project conflict register, discrepancies, and resolutions	67
E.1	Data-verified prevalence of all 41 persuasion strategies (10,600 turns)	68

Chapter 1

Problem Identification and Formulation

1.1 Background and Motivation

Conversational interaction is the primary medium through which human beings negotiate terms, coordinate actions, resolve conflicts, and solicit prosocial commitments. In contemporary digital ecosystems, conversational processes are increasingly mediated by computerized platforms, ranging from social commerce messaging channels (such as WhatsApp, Facebook Messenger, and Instagram Direct) to automated customer support, peer-to-peer micro-lending, and digital philanthropy. In these domains, interaction does not follow the symmetric patterns of open-ended chit-chat or standard task-oriented slot filling. Instead, it has a built-in *communicative asymmetry*: one participant assumes the role of an *Arguer* or *Persuader* (e.g., a merchant pitching goods or an advocate soliciting charitable contributions), while the counterparty assumes the role of a *Decider* or *Persuadee* (e.g., a prospective buyer assessing product-price fit or a donor deciding whether to surrender a portion of their earnings).

Modeling these interactions computationally has real social and economic value. In developing economies such as Bangladesh, conversational commerce represents a multi-million-dollar informal market. Thousands of micro-enterprises and independent merchants conduct daily trade exclusively through unscripted multi-turn chats. Merchants must address customer skepticism, negotiate price concessions, and provide logistical assurances, while buyers must guard against counterfeit goods, non-delivery risks, and aggressive sales tactics. Similarly, in non-profit fundraising and public health advocacy, persuasive dialogues determine whether individuals commit to prosocial actions.

However, building reliable machine learning systems for asymmetric conversational analysis runs into several serious obstacles:

- (a) **Ground-Truth Unreliability in Existing Corpora:** In benchmark datasets such as *Persuasion for Good* (P4G) [1], the recorded numeric transaction metadata frequently contradicts the grounded semantic content of the conversation. For instance, participants who clearly confirmed a monetary donation in the dialogue transcript have \$0 recorded in the dataset’s tabular summary, while others who deferred or hedged are marked with positive values. Automated classifiers trained on raw tabular metadata inadvertently learn to fit logging artifacts rather than linguistic commitments.
- (b) **Structural Failure of Monolithic Sequence Encoders:** Conventional natural language processing pipelines flatten multi-turn dialogues into a single contiguous token sequence passed to a pre-trained transformer (e.g., BERT [2] or RoBERTa [3]). Standard pre-trained models have hard context ceilings (typically 256 or 512 tokens). Natural dialogues often span 300 to 800 words across 15 to 30 turns, so right-truncation silently discards the final turns where the decision is made, capping classification performance at trivial baselines (Macro-F1 $\approx$ 0.65).
- (c) **Neglect of Conversational Role Dynamics:** Standard hierarchical models mix utterances from both speakers into a single self-attention sequence, relying on weak positional or additive role embeddings to distinguish who is speaking. Such architectures do not capture the directional effect by which the decider’s stance shifts in reaction to the persuader’s rhetorical moves.
- (d) **Severe Class Imbalance and Extreme Long Tails:** When annotating the specific rhetorical strategies deployed by the persuader (e.g., Cialdini’s influence techniques [4], emotional appeals, credibility citations), the resulting multi-label distribution exhibits extreme skew. Rare but psychologically potent strategies (e.g., *door-in-the-face*) appear only a handful of times across thousands of turns, and standard multi-label classifiers collapse on them.
- (e) **Data Scarcity and Privacy Barriers in Commercial Negotiation:** Commercial negotiation logs from real social commerce platforms are closely guarded proprietary data subject to strict privacy regulations. Researchers lack access to standardized, high-density negotiation benchmarks that test whether conversational models can track subtle, non-monotonic stance shifts (such as a buyer agreeing to purchase after severe

resistance, or aborting a transaction at the final turn due to delivery friction).

This capstone project addresses these challenges with a mathematically grounded, empirically verified computational framework for modeling conversational intention and persuasion strategies in asymmetric multi-turn dialogues.

1.2 Problem Statement

Mathematical Invariant: Two-Party Asymmetric Dialogue Representation

Let a multi-turn two-party dialogue $\mathcal{D}$ be represented as an ordered sequence of N utterances:

$$\mathcal{D} = ((u_1, r_1), (u_2, r_2), \dots, (u_N, r_N)) \quad (1.1)$$

where each utterance $u_i \in \mathcal{V}^*$ is a sequence of tokens from vocabulary $\mathcal{V}$, and $r_i \in \{\mathcal{P}, \mathcal{D}\}$ designates the speaker role, partitioned between the Persuader/Merchant ($\mathcal{P}$) and the Persuadee/Buyer ($\mathcal{D}$).

We formulate three interconnected computational sub-problems:

1.2.1 Sub-Problem 1: Grounded Conversational Intention Classification

Given the entire multi-turn transcript $\mathcal{D}$, the objective is to learn a mapping $f_\theta : \mathcal{D} \rightarrow \mathcal{Y}$ predicting the final binary commitment state $y \in \mathcal{Y} = \{\text{yes}, \text{no}\}$ of the decider $\mathcal{D}$. The ground-truth label y is defined by the decider's final, unconditional position: $y = \text{yes}$ if and only if the decider explicitly and unconditionally commits to the target action (e.g., donating task earnings or finalizing a product purchase); $y = \text{no}$ if the decider refuses, deflects, defers to an indeterminate future time, or imposes unresolved conditional hedges. The optimization objective maximizes binary Macro-F1:

$$\text{Macro-F1} = \frac{1}{2} \left(F_1^{(\text{no})} + F_1^{(\text{yes})} \right) \quad (1.2)$$

under severe class imbalance (31.8% negative prevalence in the ground-truth corpus).

1.2.2 Sub-Problem 2: Multi-Label Persuasion Strategy and Category Attribution

Given an arbitrary Persuader utterance u_t (where $r_t = \mathcal{P}$) situated within a causal, non-anticipatory historical conversational context of prior turns $\mathcal{H}_t = (u_{\max(1, t-k)}, \dots, u_{t-1})$, the objective is to predict the active subset of rhetorical strategies $\mathbf{y}_t \subseteq \mathcal{S}_{41}$, where $|\mathbf{y}_t| \in \{0, 1, \dots, 6\}$ and $\mathcal{S}_{41}$ represents a closed taxonomy of 41 fine-grained persuasion strategies. The multi-label strategy vector deterministically maps to a higher-level communicative category subset $\mathbf{c}_t \subseteq \mathcal{C}_{11}$ via a surjective mapping $\mathcal{M} : \mathcal{S}_{41} \rightarrow \mathcal{C}_{11}$. The learning system must optimize predictive fidelity across the 11 communicative categories as the primary objective, with the 41 strategies acting as an auxiliary multi-task regularizer:

$$\hat{\mathbf{c}}_t = g_\phi(u_t, \mathcal{H}_t) \in [0, 1]^{11}, \quad \hat{\mathbf{y}}_t = h_\psi(u_t, \mathcal{H}_t) \in [0, 1]^{41} \quad (1.3)$$

subject to the operational constraint that neither future dialogue turns $(u_{t+1}, \dots, u_N)$ nor human annotator identity tags are accessible at inference time.

1.2.3 Sub-Problem 3: Cross-Domain Transfer and Adversarial Trajectory Tracking

Given a target commercial domain with buyer-merchant negotiations $\mathcal{D}_{\text{comm}}$, the objective is to evaluate the cross-domain generalizability of the learned intention architectures under both canonical transactional dynamics and adversarial non-monotonic trajectory reversals. Let τ_{rev} denote an internal turn at which a buyer abruptly reverses their stance (e.g., transitioning from skeptical resistance to sudden purchase commitment, or from tentative agreement to explicit cancellation). We define the *sufficiency gap* Δ_{suff} of a conversational model relative to a localized closing-turn decision baseline:

$$\Delta_{\text{suff}} = \text{Macro-F1}(f_{\text{full}}(\mathcal{D})) - \text{Macro-F1}(f_{\text{last}}(u_{\text{final}})) \quad (1.4)$$

The engineering objective is to demonstrate that trajectory-aware, role-disentangled architectures maintain a positive, widening sufficiency gap ($\Delta_{\text{suff}} \gg 0$) when closing turns are destabilized, whereas monolithic sequence baselines degrade severely.

1.2.4 Required Engineering Knowledge Profiles

Solving this problem draws on several engineering and scientific knowledge profiles (as defined in Table 1.2):

- **K3 (Theory-Based Engineering Fundamentals):** Mathematical probability theory, discrete multivariate distributions, gradient-based optimization dynamics, focal and contrastive loss formulations, and information-theoretic evaluation metrics (Macro-F1, micro-F1, Jaccard similarity, bootstrap confidence intervals).
- **K4 (Specialist NLP and Deep Learning Frameworks):** Theoretical foundations of deep contextual language representation, self-attention and cross-attention mechanics, transformer computational complexity, recurrent gated state tracking (GRU), and out-of-fold stacked generalization.
- **K5 (Architecture and System Design):** Design of decoupled dual-stream neural networks, masked tensor batching, memory footprint profiling for hardware accelerators, and cross-attention routing under asymmetric key-value constraints.
- **K8 (Engagement with Frontier Research Literature):** Critical synthesis of recent peer-reviewed literature in dialogue discourse modeling (HiTrans, DialogueRNN), persuasion taxonomy design, asymmetric multi-label loss functions, and shortcut learning artifacts in natural language inference.

1.3 Complexity of the Problem

The technical challenges addressed in this capstone project constitute a complex engineering problem as defined by the International Engineering Alliance (IEA) and the Board of Accreditation for Engineering and Technical Education (BAETE). The project satisfies characteristic P1 and all six secondary characteristics P2 through P7. Table 1.1 maps each complexity characteristic to the technical evidence from this project. Table 1.2 enumerates the referenced knowledge profiles.

1.4 Project Objectives and Scope

1.4.1 Project Objectives

To address this problem formulation, the project pursued seven verifiable engineering objectives:

- O1. Curation of Ground-Truth Intention Corpus:** Re-annotate the 1,017-dialogue P4G corpus in three passes to resolve discrepancies between recorded metadata and transcript text, producing a cleaned binary benchmark (694 yes / 323 no) with verified inter-annotator consistency (92.21% agreement).
- O2. Systematic Intention Architecture Evolution:** Implement and evaluate 14 successive intention classification architectures across four evolutionary stages, overcoming the 256-token context ceiling of flat transformers and arriving at the role-disentangled *AntiphonyClassifier* (Dual-Stream Cross-Attention Architecture).
- O3. Establish State-of-the-Art Intention Prediction:** On held-out test splits, exceed 0.86 Macro-F1 on second-pass labels and 0.84 Macro-F1 on strict third-pass labels across multiple pre-trained transformer backbones (RoBERTa, DeBERTa-v3, TOD-BERT).
- O4. Fine-Grained Persuasion Strategy Annotation:** Construct a verified multi-label dataset of 10,600 Persuader utterances annotated under an 11-category, 41-strategy hierarchical taxonomy under strict decision protocols, and derive an empirical cross-annotator agreement ceiling (0.79 micro-F1).
- O5. Multi-Task Strategy and Category Modeling:** Build an ensemble modeling framework over an 11-category primary objective with a 41-strategy auxiliary head, evaluate a 19-member model pool, and show that a calibrated power-mean blend reaches competitive validation scores (0.6949 micro-F1 / 0.5597 macro-F1, with a micro-optimal operating point of 0.7062 / 0.5642) and outperforms an XGBoost stacker.
- O6. Synthetic E-Commerce Benchmark Generation:** Construct a high-fidelity 700-dialogue negotiation benchmark ($N = 500$ canonical, $N = 200$ adversarial reversals) using persona-conditioned prompting, organic paraphrasing with linguistic jitter, and 100% human-in-the-loop verification.
- O7. Cross-Domain Transfer and Adversarial Sufficiency Diagnosis:** Run input-ablation experiments on the canonical and reversal negotiation corpora, showing that Antiphony maintains trajectory robustness (0.9619 Macro-F1) and widens the sufficiency gap ($\Delta_{\text{suff}} = +0.0384$) when closing utterances are destabilized.

Table 1.1: Complex engineering problem characteristics (P1–P7) and project evidence

Code	IEA Characteristic	Evidence from this Project
P1	Depth of Knowledge Required Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, or K8.	Required combined command of mathematical loss design (Class-Balanced Focal Loss, Asymmetric Loss), hierarchical transformer attention mechanics, recurrent state tracking, and multi-model calibrated ensemble blending across 19 candidate models.
P2	Range of Conflicting Requirements Involves wide-ranging or conflicting technical, engineering, and other issues.	Severe tension between binary commitment accuracy and minority modifier/hedge recall; optimizing macro-F1 (rare classes) directly degraded global micro-F1 by chasing recall on 2-example labels; processing 2,048-token hierarchical dialogues conflicted with the 16 GB VRAM ceiling of NVIDIA T4 GPUs.
P3	Depth of Analysis Required Has no obvious solution and requires abstract thinking and originality in analysis.	Standard monolithic transformers failed ($\text{Macro-F1} \leq 0.65$); solving the task required originating a speaker-disentangled dual-stream network (<i>Antiphony</i>) wherein the decider queries the persuader via directional cross-attention, coupled with an empirical sufficiency gap analysis.
P4	Familiarity of Issues Involves infrequently encountered issues.	Extremely skewed tail classes with only 2 to 5 instances in 10,600 turns (<i>door_in_the_face</i>); subtle tokenizer truncation side-effects where default right-truncation silently excised closing decisions in 30.4% of dialogues; whitespace-gluing corrupting speaker tags in 15.9% of augmented data.
P5	Extent of Applicable Codes Outside problems encompassed by standard codes of practice.	No standardized engineering codes exist for multi-label persuasion taxonomy annotation or synthetic negotiation generation; required formulating custom 4-tier decision rules (Removal Test, Function-over-Keyword) and three-phase human-in-the-loop verification protocols.
P6	Stakeholder Involvement and Conflicts Involves diverse stakeholder groups with widely varying needs.	Direct conflict between commercial merchants aiming to maximize conversion velocity and consumers requiring protection from predatory manipulation; academic requirements for reproducible open-source benchmarks vs. strict privacy constraints prohibiting the release of private customer chats.
P7	Interdependence High-level problem with many interconnected sub-problems.	Tight cascading dependencies across three datasets ($N = 1,017$ P4G, 10,600 strategy turns, $N = 700$ negotiation benchmark); tightening ground-truth intention labels in Pass 3 necessitated retraining and recalibrating the entire downstream 14-model architectural lineage.

Table 1.2: Knowledge profiles referenced by P1

Code	Description
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
K5	Knowledge that supports engineering design in a practice area.
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
K8	Engagement with selected knowledge in the research literature of the discipline.

1.4.2 Project Scope and Explicit Exclusions

To keep the work focused and within the available compute, the project scope is bounded as follows:

- **In Scope:**

1. *Commitment Prediction:* Offline discriminative modeling of whether the deciding party commits by the end of a dialogue.
2. *Strategy Classification:* Multi-label recognition of the persuasion categories and strategies used in each persuader turn.
3. *Model Design and Training:* Hierarchical and role-aware neural architectures, with multi-label and multi-task loss formulations.
4. *Dataset Construction:* Manual re-annotation of the P4G corpus and a synthetic, manually audited e-commerce negotiation benchmark.
5. *Evaluation:* Input ablations, adversarial stress tests, and post-mortem analysis of failure cases.
6. *Resource Accounting:* Reporting of compute usage and estimated energy consumption.

- **Explicit Exclusions:**

1. *No Live Conversational Chatbot Deployment:* The project develops research-grade discriminative classifiers; it does not deploy an autonomous agent interacting with live human users in production environments.
2. *No Natural Language Generation (NLG):* The primary focus is dialogue understanding and strategy classification; autonomous tactical response generation is excluded.
3. *No Non-English or Code-Mixed Modeling:* All corpora and benchmarks studied are in English. Modeling Bengali, Banglish, or regional code-mixed dialects is left to future work.
4. *No Human Subject Interventional Field Trials:* The research relies on public academic datasets and synthetic simulations; no randomized human behavioral intervention trials were conducted on commercial platforms.

Chapter 2

Preliminary Research

2.1 Literature Review

Computational analysis of persuasive, multi-turn conversational interaction sits at the convergence of behavioral psychology, natural language processing, and statistical machine learning. This section reviews foundational and recent work in these areas, explains how it informs this project, and identifies the theoretical and empirical gaps this capstone addresses.

2.1.1 Persuasion Theory and Rhetorical Frameworks

The systematic study of interpersonal persuasion originates in classical rhetoric and social psychology. Cialdini's framework of influence [4] groups compliance-gaining mechanisms into core principles: reciprocity, commitment and consistency, social proof, authority, liking, and scarcity. In sequential negotiation, classic experimental psychology identified specific multi-stage tactics. Freedman and Fraser [5] demonstrated the *foot-in-the-door* phenomenon, whereby inducing compliance with an initial trivial request substantially increases adherence to a subsequent larger request. Conversely, Cialdini et al. [6] formulated the *door-in-the-face* technique, wherein an extreme, predictably rejected opening ask creates psychological pressure for the decider to accept a smaller concessionary follow-up. These principles are widely accepted in social science, but few computational taxonomies for natural language dialogue cover them in depth. Existing schemes often collapse distinct rhetorical moves into broad, coarse-grained dialog act labels.

2.1.2 Persuasion Corpora and Conversational Strategy Annotation

The release of the *Persuasion for Good* (P4G) corpus by Wang et al. [1] was a major step for empirical persuasion modeling. P4G contains 1,017 multi-turn online conversations where an assigned Persuader attempts to convince a Persuadee to donate part of their \$2.00 task earnings to the charity *Save the Children*. Wang et al. annotated a subset of 300 dialogues with a taxonomy of 10 persuasion strategies. Later work built on it: Saha et al. [7] incorporated sentiment and emotion dynamics into dialog act recognition, while Tian et al. [8] explored strategy-aware response generation. Chen and Yang [9] published a broad survey of conversational persuasion systems that argued for hierarchical representations. More recently, Chen and Xiao [10] analyzed persuasion tactics in commercial multi-party chats, while Petrova et al. [11] explored hierarchical re-annotation of P4G. Even so, existing benchmarks often have small annotated samples and flat, non-hierarchical label spaces, and they do not address the severe label sparsity in the long tail.

2.1.3 Transformer Encoders and Dialogue Language Models

Contextual language representations, from BERT [2] to RoBERTa [3] and DeBERTa [12], transformed NLP by pre-training deep bidirectional representations on large corpora. RoBERTa optimized BERT's pre-training regime by removing the next-sentence prediction objective and training on longer sequences with dynamic masking. DeBERTa and DeBERTa-v3 [13] extended this approach with disentangled attention (representing tokens by separate content and relative position vectors) and enhanced mask decoder pre-training with ELECTRA-style [14] discriminator losses. In dialogue modeling, Wu et al. [15] introduced TOD-BERT, which is pre-trained on a range of task-oriented dialogue datasets to model turn-level conversational semantics. When these monolithic sequence encoders are applied naively to multi-turn asymmetric dialogue, however, they hit fixed context length limits (256 or 512 tokens), and default right-truncation silently drops the final turns.

2.1.4 Hierarchical and Role-Aware Dialogue Networks

To process extended multi-turn interactions without truncation, hierarchical neural architectures decouple token-level encoding from turn-level discourse modeling. Li et al. [16] introduced HiTrans, a dual-transformer hierarchy in

which a lower transformer encodes intra-utterance words and an upper transformer models inter-utterance discourse relations. In conversational emotion recognition, Majumder et al. [17] introduced DialogueRNN, which tracks the state of each speaker party with recurrent neural networks (GRU [18]) to preserve speaker identity and interpersonal context. DialogueRNN showed that per-party tracking works well for emotional dynamics. Asymmetric outcome prediction needs more than independent speaker states: it must also model the *directional influence* of the arguer's moves on the decider's internal commitment.

2.1.5 Imbalanced Learning and Loss Formulations

Multi-label persuasion strategy attribution and binary commitment classification both exhibit extreme class imbalance. Under conventional cross-entropy loss, easy negative examples or majority classes dominate the gradients. Lin et al. [19] introduced Focal Loss, applying a modulating factor $(1 - p_t)^\gamma$ to down-weight easy examples and focus gradient updates on hard instances. Cui et al. [20] developed Class-Balanced Focal Loss, weighting the loss inversely by the effective number of samples $(1 - \beta)/(1 - \beta^{n_y})$ rather than raw sample frequency, preventing gradient explosion on extremely rare classes. For multi-label settings where 90%+ of label matrices are zero, Ben-Baruch et al. [21] formulated Asymmetric Loss (ASL), which dynamically decouples the focusing parameters γ_+ and γ_- and applies hard probability threshold shifting to negative samples. Khosla et al. [22] proposed Supervised Contrastive Loss (SupCon), which pulls normalized representations of the same class together in latent space and pushes different classes apart. This substantially improves the clustering of representations for minority categories.

2.1.6 Ensembling and Stacking Generalization

Ensemble learning uses model diversity to reduce generalization variance. Wolpert [23] formalized Stacked Generalization, in which meta-classifiers are trained on the cross-validated out-of-fold predictions of heterogeneous base models. In practice, gradient-boosted decision trees such as XGBoost [24] are widely used as stacking decision layers because they capture non-linear feature interactions and resist overfitting. Probability calibration techniques, starting with Platt scaling [25], map uncalibrated logits to empirical posterior probabilities. Calibrated outputs can then be blended smoothly (e.g., power-mean ensembling), which often outperforms greedy discrete selection.

2.1.7 Negotiation Benchmarks and Shortcut Learning

Early computational negotiation frameworks, such as the *Deal or No Deal* benchmark by Lewis et al. [26] and the modular negotiation models of He et al. [27], examined multi-issue bargaining over item allocations. Later critical analyses by Geirhos et al. [28] and Gururangan et al. [29] show that neural models trained on complex textual benchmarks frequently rely on superficial cues and dataset artifacts, a failure known as *shortcut learning*. In conversational intent prediction, models often achieve artificially high accuracy simply by detecting formulaic closing tokens (e.g., “sounds good”, “deal”) in the final turn and ignoring the conversational trajectory entirely. Testing whether an architecture actually models trajectory dynamics requires careful adversarial evaluation and input-ablation controls.

2.1.8 Synthesis of Research Gaps

Table 2.1 summarizes the main open gaps in the existing literature and how this project addresses each one.

2.2 Existing Solutions and Technologies

To place this work in context, we survey the main categories of existing technology currently deployed in commercial dialogue analysis and conversational commerce:

1. **Monolithic Transformer Fine-Tuning:** In industry, standard pre-trained language models (RoBERTa, DeBERTa) are often applied directly to flattened dialogue text. These models are easy to implement with libraries such as Hugging Face Transformers [30], but they truncate long dialogues at 256 or 512 tokens. As Chapter 5 shows empirically, default right-truncation cuts off the closing exchanges in 30.4% of dialogues, so the model never sees the final decision.
2. **Prompted General-Purpose Large Language Models (LLMs):** Commercial solutions increasingly query proprietary frontier LLMs (e.g., GPT-4, Claude-3.5) via zero-shot or few-shot prompts to extract customer intent and conversational sentiment. Prompted LLMs are highly flexible, but they have serious operational

Table 2.1: Synthesis of research literature and identified gaps

Research Area	State of the Art / Established Findings	Open Gap Addressed in this Project
Persuasion Modeling	Coarse-grained 10-class taxonomies on small subsets (300 dialogues) [1].	Full 10,600-turn multi-label dataset with an 11-category, 41-strategy hierarchical taxonomy.
Intention Grounding	Trusting recorded numerical corpus metadata (\$0 vs \$10).	Proving textual divergence; conducting 3-pass human re-annotation to create strict commitment labels.
Dialogue Architecture	Flat transformers (256 tokens) or mixed-sequence hierarchical transformers (HiTrans [16]).	Introducing <i>Antiphony</i> : role-pure dual streams with directional cross-attention (Q_D, K_P, V_P) and trajectory GRU.
Multi-Label Extreme Skew	Standard multi-label binary cross-entropy suffering long-tail collapse.	Combining Asymmetric Loss, Class-Balanced Focal Loss, and calibrated power-mean ensemble blending across 19 models.
Negotiation Evaluation	Clean synthetic allocations [26] vulnerable to closing-turn shortcut learning.	Generating 700-dialogue adversarial reversal benchmark; proving Antiphony widens sufficiency gap ($\Delta_{\text{suff}} = +0.0384$).

limitations: prohibitive inference latency ($> 1,000$ ms per dialogue), large recurring financial costs, privacy violations from sending proprietary customer transcripts to third-party endpoints, and severe non-determinism.

- 3. Task-Oriented Slot-Filling Dialogue Systems:** Traditional conversational frameworks (e.g., Rasa, Google Dialogflow) model dialogue state through predefined intent hierarchies and entity slot-filling. They work well for rigid transactional booking (e.g., flight reservations), but they cannot model multi-label rhetorical strategies, social rapport, emotional appeals, or fluid, non-monotonic negotiation dynamics.
- 4. Classical Bag-of-Words and TF-IDF Baselines:** Lightweight production systems often rely on TF-IDF feature extraction combined with logistic regression or gradient-boosted decision trees. These systems are fast and interpretable, but in our experiments bag-of-words models degraded significantly when given extended conversational context (dropping from 0.575 to 0.419 micro-F1 across 5 turns), because they lack deeper semantic and contextual reasoning.

The conversational AI industry lacks an efficient, local, privacy-preserving discriminative architecture that captures asymmetric conversational dynamics over long multi-turn horizons without arbitrary context truncation, while remaining trainable on standard commodity GPU hardware.

2.3 Market and User Research

2.3.1 The Rise of Conversational Social Commerce

In emerging markets, particularly across South Asia and Southeast Asia, conversational commerce has outpaced traditional web storefronts. In Bangladesh, thousands of micro-, small, and medium enterprises (MSMEs) operate almost entirely across social media chat channels: Facebook Messenger, WhatsApp Business, and Instagram Direct. Transactions do not go through standardized shopping carts. The whole customer journey, from product inquiry, authenticity verification, and price negotiation to delivery coordination and payment commitment, takes place in unscripted natural language chat.

2.3.2 User Personas and Behavioral Dynamics

Qualitative analysis of real marketplace chat patterns shows several consumer and merchant behavioral archetypes that drive transaction outcomes:

- **Consumer Friction and Skepticism:** Because online shopping fraud is common, consumers are highly risk-averse. Buyers frequently demand physical shop addresses, unboxing photographs, and real-time serial

number verification. Prepayment via mobile financial services (MFS such as bKash and Nagad) is frequently contested; buyers overwhelmingly prefer Cash-on-Delivery (COD) with inspection privileges.

- **Bargaining Culture:** Haggling over price is deeply ingrained in the culture. Buyers expect merchants to offer discretionary price cuts, promotional freebies, or delivery fee waivers. Conversations frequently stall when merchants maintain rigid pricing policies.
- **Merchant Administrative Burden:** Independent page admins and shopkeepers manage hundreds of active chat inquiries simultaneously. Admins report heavy fatigue from “ghosting” (buyers asking long, detailed questions and then abruptly disengaging) and find it hard to prioritize leads that have genuine, high-probability buying intent.

2.3.3 Industry Consultations and Field Observations

The project’s initial scope came out of online meetings with Pathao Ltd., in which the industry partner explained the problem space and the kinds of tools it would find useful. These were informal consultations, and no formal interview records were kept. The main needs raised were automated systems capable of:

- (i) triaging high-intent leads in real time;
- (ii) running on a very small model that can be hosted directly on the partner’s live servers, without depending on large external LLM services.

2.4 Feasibility Study

We assessed feasibility along four dimensions:

1. **Technical Feasibility:** The proposed approach is built on proven technologies: PyTorch deep learning frameworks, pre-trained Hugging Face transformer checkpoints, and established optimization libraries. The experimental benchmarks in Chapter 5 support technical feasibility: they show that the Antiphony architecture achieves state-of-the-art Macro-F1 (> 0.86) and trains reliably within 3 to 4 GPU hours.
2. **Economic Feasibility:** We deliberately kept the whole development cycle within zero-cost compute. All models were trained, evaluated, and ablated on free-tier cloud GPU accelerators (NVIDIA Tesla T4 with 16 GB VRAM on Kaggle). The synthetic benchmark generation pipeline used only the free tier of the Gemini API, so the project incurred no API costs. This shows that advanced conversational NLP research can be carried out without an institutional cluster budget.
3. **Operational Feasibility:** The trained discriminative classifiers are compact (< 500 MB model checkpoints for base encoders) and execute inference on standard server GPUs without specialized distributed serving hardware. They fit directly into offline batch-processing analytics pipelines for post-call audits and customer sentiment tracking.
4. **Legal and Regulatory Feasibility:** The research respects intellectual property and privacy requirements. The P4G corpus is an established academic benchmark with open research permissions. No proprietary or private user chat logs were collected; cross-domain commercial transfer was evaluated only on a high-fidelity synthetic benchmark generated under strict ethical guidelines.

The project is therefore feasible on all four dimensions.

Chapter 3

Requirements Analysis

3.1 Project Objectives and Constraints

Building an effective, ethically grounded machine learning system for asymmetric conversational modeling requires turning the project goals into verifiable functional and non-functional specifications, while accounting for the needs of different stakeholders and strict computational limits.

3.1.1 Stakeholder Analysis and Conflicting Expectations

The system affects several stakeholder groups whose interests often diverge:

1. **Academic Researchers in NLP and Behavioral Science:** Require high reproducibility, transparent reporting of negative findings, statistically valid significance bounds, and standardized open-source benchmarks free from data leakage and annotation artifacts.
2. **Social Commerce Micro-Merchants and Sales Managers:** Seek automated tools to rapidly identify high-probability purchase commitments, triage inbound inquiries, and identify which rhetorical tactics maximize sales conversion velocity. Their priority is high recall on affirmative buying decisions.
3. **Consumers and Persuadees:** Require protection against coercive persuasion, predatory pricing, and deceptive conversational tactics (dark patterns). They expect their privacy and transaction autonomy to be preserved.
4. **Data Annotators and Evaluators:** Demand clear, unambiguous guidelines with realistic task density limits to avoid cognitive fatigue and subjective drift during multi-label coding.
5. **Project Supervisors and Academic Assessors:** Expect adherence to sound engineering methodology, transparent accounting of compute resources, and verifiable compliance with accreditation standards.

These expectations pull against each other: commercial stakeholders prioritize conversion metrics, whereas ethical standards require protecting consumer autonomy; maximizing rare-class recall conflicts with maintaining high overall precision; and modeling the full dialogue context conflicts with hardware memory limits.

3.1.2 Functional Requirements (FR)

The system must satisfy six functional requirements:

- **FR1: Grounded Binary Commitment Classification.** Given a complete multi-turn dialogue transcript $\mathcal{D}$, the model must predict the decider’s final unconditional commitment state $y \in \{\text{yes}, \text{no}\}$, achieving an overall test Macro-F1 exceeding 0.84 on strict third-pass labels.
- **FR2: Turn-Level Multi-Label Strategy Attribution.** For any designated Persuader utterance u_t , the model must predict the active subset of rhetorical strategies $\mathbf{y}_t \subseteq \mathcal{S}_{41}$ ($|\mathbf{y}_t| \leq 6$) and their corresponding parent communicative categories $\mathbf{c}_t \subseteq \mathcal{C}_{11}$.
- **FR3: Strict Causal Information Gating.** At test and deployment time, the strategy model must operate causally, without access to: (i) future conversational turns $(u_{t+1}, \dots, u_N)$, (ii) whole-dialogue post-hoc smoothing, or (iii) annotator identity metadata.
- **FR4: Cross-Domain Transactional Generalization.** The intention modeling architecture must demonstrate zero-shot or fine-tuned domain transfer from charitable donation dialogues to commercial buyer-merchant negotiations.
- **FR5: Adversarial Stance Reversal Detection.** Under non-monotonic conversational trajectories (e.g., initial skepticism reversing to late commitment, or tentative agreement collapsing into cancellation), the model

must use multi-turn trajectory context and maintain a positive sufficiency gap ($\Delta_{\text{suff}} > 0$) relative to localized closing-turn controls.

- **FR6: Multi-Threshold Decision Calibration.** The system must provide calibrated output probabilities and tunable decision thresholds across both macro-oriented (rare class recovery) and micro-oriented (global accuracy) operating regimes.

3.1.3 Non-Functional Requirements (NFR)

The system is governed by five non-functional requirements:

- **NFR1: Absolute Deterministic Reproducibility.** All data preprocessing, random splits, augmentation pipelines, and neural network parameter initializations must be governed by fixed pseudo-random seeds (Seed = 42), enabling exact bitwise or statistical replication of all reported metrics.
- **NFR2: Hardware and Memory Efficiency.** The entire model training, fine-tuning, and inference pipeline must execute within the hardware constraints of a single commodity cloud GPU accelerator (NVIDIA Tesla T4 with 16 GB VRAM). Batch sizes, gradient checkpointing, and tensor dimensions must prevent Out-Of-Memory (OOM) faults during 2,048-token hierarchical passes.
- **NFR3: Noise-Aware Performance Benchmarking.** Because human annotators disagree on subjective cases, strategy model performance must be benchmarked against an empirically derived cross-annotator agreement ceiling (0.79 micro-F1) rather than a naive 1.0 ceiling.
- **NFR4: Data Privacy and Confidentiality.** To eliminate legal and ethical risks regarding customer Personally Identifiable Information (PII), cross-domain commercial evaluation must use high-fidelity synthetic benchmarks generated under strict privacy constraints.
- **NFR5: Transparent Negative Result Reporting.** All architectural failures, negative ablation results, and unresolved dataset conflicts must be documented transparently in the engineering record without retroactively altering experimental protocols.

3.1.4 Physical and Operational Constraints

- **Data Sample Constraints:** The core intention corpus contains exactly 1,017 dialogues (711 train, 153 validation, 153 test). The held-out test split of 153 dialogues implies that each single misclassification alters accuracy and per-class recall by $\approx 0.65\%$.
- **Extreme Tail Sparsity:** In the strategy dataset (10,600 turns), the rarest strategy (`door_in_the_face`) occurs only twice, which rules out independent single-model supervised convergence and forces an 11-category primary modeling objective.
- **Compute Horizon Constraints:** Cloud runtime session limits (Kaggle maximum 12 hours per session) put a strict limit on per-epoch training time and rule out large unconstrained parameter searches.

3.2 Regulatory Requirements and Standards

The project follows professional ethics codes, artificial intelligence risk frameworks, and software quality standards:

1. **ACM and IEEE Codes of Ethics:** The development follows the ACM Code of Ethics [31] and IEEE Code of Ethics [32], in particular their provisions on contributing to societal human well-being, avoiding deception and harm, respecting privacy, and upholding intellectual honesty in scientific reporting.
2. **IEEE 7000-2021 Standard Model Process:** Aligns with IEEE 7000-2021 (*Addressing Ethical Concerns During System Design*) [33] by conducting early stakeholder impact analysis, identifying potential dual-use vulnerabilities of persuasive AI, and incorporating explicit value-based design boundaries.
3. **NIST AI Risk Management Framework (AI RMF 1.0):** The project structure mirrors NIST AI 100-1 [34] core functions: *Govern* (establishing research scope and ethical boundaries), *Map* (identifying context, conversational asymmetry, and failure modes), *Measure* (empirical testing, sufficiency gap calculations, and noise ceiling derivations), and *Manage* (systematic ablation, forensic debugging, and mitigation of shortcut learning).

4. **ISO/IEC 25010:2011 Software Quality Requirements:** Evaluates system quality against the ISO/IEC 25010 [35] product quality model, focusing on functional suitability (completeness, correctness), performance efficiency (time behavior, resource utilization on 16 GB GPUs), reliability (fault tolerance against corrupted tokens), and maintainability (modular notebook architecture).
5. **Model Cards and Datasheets for Datasets:** Documentation follows the ethical transparency protocols established by Mitchell et al. [36] (*Model Cards*) and Gebru et al. [37] (*Datasheets for Datasets*), detailing training data provenance, intended use, performance limitations, and known demographic skews.
6. **Data Protection and Statutory Compliance:** Respects the terms of use of the original P4G benchmark and public cloud providers. In relation to national jurisdictions, the project takes note of Bangladesh’s evolving legal framework for digital privacy and cyber-security, including the Personal Data Protection Ordinance 2025 and the Cyber Security Ordinance 2025. The project does not collect or process personal data of Bangladeshi users: it relies only on the public P4G corpus and on a synthetic benchmark.

3.3 Formulation of Solutions

To choose a system architecture for asymmetric intention and strategy modeling, we formulate five distinct candidate engineering solutions:

Candidate 1: Monolithic Flat Sequence Transformer: Concatenates all turns of the dialogue into a single flat string separated by special speaker tokens, passing it to a pre-trained transformer (RoBERTa-base) with a standard sequence classification head over the [CLS] token. Imposes a rigid 256- or 512-token cap.

Candidate 2: Prompt-Engineered General-Purpose LLM: Uses zero-shot and few-shot natural language prompts dispatched via API calls to a commercial proprietary LLM (e.g., GPT-4 or Claude-3.5) with structured JSON decoding.

Candidate 3: Classical Bag-of-Words and Gradient Boosted Decision Trees: Extracts turn-level and dialogue-level n -gram TF-IDF representations combined with hand-crafted conversational features, classified using tuned XGBoost [24] or Logistic Regression ensembles.

Candidate 4: Single-Stream Hierarchical Dialogue Transformer: Employs a dual-level architecture (HiTrans-style [16]): turns are encoded individually by a frozen or fine-tuned transformer, turn vectors are pooled, and a second transformer performs self-attention over the interleaved sequence of all turns using additive role embeddings.

Candidate 5: Role-Pure Dual-Stream Antiphony Architecture (Selected): Disentangles the dialogue into two pure speaker streams ($\mathcal{P}$ and $\mathcal{D}$). Each stream is processed by independent transformer layers, connected via directional cross-attention ($Q_{\mathcal{D}}, K_{\mathcal{P}}, V_{\mathcal{P}}$), and aggregated through a parallel combination of a commitment-trajectory GRU (read at the true final decider turn) and attention pooling.

3.3.1 Multi-Criteria Evaluation and Decision Matrix

We evaluate the candidate architectures against six engineering criteria drawn from our requirements:

- **C1: Multi-Turn Context Capacity** (Weight: 20%): Ability to ingest dialogues exceeding 500 words without sequence truncation.
- **C2: Asymmetric Interaction Modeling** (Weight: 20%): Architectural fidelity in capturing the directional influence of persuader arguments on decider commitment.
- **C3: Long-Tail and Imbalance Robustness** (Weight: 15%): Resistance to gradient collapse on rare classes.
- **C4: Hardware and Cost Feasibility** (Weight: 15%): Ability to train and evaluate within free-tier GPU constraints (16 GB T4) with zero recurring API costs.
- **C5: Latency and Determinism** (Weight: 10%): Fast, reproducible, non-stochastic inference suitable for offline batch analytics.
- **C6: Empirical Intention Performance** (Weight: 20%): Measured Macro-F1 on held-out benchmark splits.

Table 3.1 presents the comparative evaluation matrix.

Table 3.1: Multi-criteria architectural decision matrix (scores: 1 = Poor, 5 = Excellent)

Candidate Architecture	C1 (20%)	C2 (20%)	C3 (15%)	C4 (15%)	C5 (10%)	C6 (20%)	Weighted Score
1. Monolithic Flat Transformer	1	2	3	4	5	2.5 (0.65)	2.60
2. Prompt-Engineered LLM	4	3	2	1	1	N/E	2.30
3. Classical TF-IDF + GBDT	2	1	2	5	5	2.0 (0.46)	2.50
4. Single-Stream Hierarchical	5	3	4	4	4	4.0 (0.82)	3.95
5. Dual-Stream Antiphony	5	5	4	4	4	5.0 (0.86)	4.60

3.3.2 Conclusion and Selection Justification

As demonstrated in Table 3.1, Candidate 5 (**Dual-Stream Antiphony**) achieves the highest weighted score (4.60/5.00). Candidate 1 is limited by context truncation; Candidate 2 is ruled out by very high financial costs, high latency, and data privacy issues; Candidate 3 lacks deep semantic reasoning; and Candidate 4, while achieving strong results (0.821 Macro-F1), relies on additive role embeddings and lacks an explicit mechanism for directional decider-on-persuader cross-attention. We therefore selected the Dual-Stream Antiphony architecture as the system design.

Chapter 4

System Design and Implementation

4.1 Functional Design and Sub-problem Partitioning

The system is a modular, multi-tier pipeline for asymmetric multi-turn dialogue. Its functional architecture divides the problem into five subsystems:

1. **Data Curation and Ground-Truth Engineering:** Ingests multi-turn conversational corpora (P4G and synthetic e-commerce negotiations), applies fixed semantic labeling protocols across human and automated passes, and builds verified, leakage-free evaluation splits.
2. **Representation and Utterance Encoding Subsystem:** Processes raw turn tokens using pre-trained contextual language models (RoBERTa, DeBERTa-v3, TOD-BERT) through batched tensor operations, producing dense contextual utterance representations within hardware memory limits.
3. **Turn-Level Rhetorical Strategy Attribution Subsystem:** Takes Persuader turns with their preceding (causal) context and passes the representations through a 19-member multi-task model pool trained under Asymmetric Loss to predict an 11-category primary objective with 41 auxiliary strategies; a calibrated decision layer combines the members.
4. **Dialogue-Level Intention and Trajectory Tracking Subsystem (*Antiphony*):** Disentangles conversational turns into role-pure Persuader and Decider streams, applies directional cross-attention, models non-monotonic stance progression via a commitment-trajectory GRU read at the true final decider turn, and outputs calibrated binary commitment predictions.
5. **Cross-Domain Generalization and Diagnostic Ablation Subsystem (Generic Architecture Testing):** Defines a diagnostic framework covering four input-ablation operators, trajectory destabilization via adversarial stance reversals, tokenizer truncation directionality guards, and parametric skew-stress profiling, used to test whether Antiphony functions as a domain-agnostic inductive bias for asymmetric multi-turn dialogue.

Table 4.1 lists each functional module with its inputs, outputs, and responsibilities.

Table 4.1: Modular subsystem decomposition and responsibilities

Subsystem	Input	Output	Core Engineering Responsibility
Annotation & Ground Truth	Raw text transcripts ($N = 1,017, 10,600$ turns)	Grounded labels ($y \in \{\text{yes}, \text{no}\}$, $\mathbf{y}_t \subseteq \mathcal{S}_{41}$)	Resolves metadata divergence via 3-pass human annotation; enforces 4-tier decision protocol.
Utterance Encoder	Batched token IDs $[B \times U, L]$	Turn matrices $\mathbf{U} \in \mathbb{R}^{B \times U \times d}$	Executes single batched forward pass; performs masked mean-pooling; enforces memory guards.
Strategy Classifier	Persuader turn $u_t + \mathcal{H}_t$ (5-turn context)	Predicted categories $\hat{\mathbf{c}}_t$ & strategies $\hat{\mathbf{y}}_t$	Multi-task learning under Asymmetric Loss; ensembles 19 members via calibrated power-mean blend.
Antiphony Intention Network	Full dialogue turn vectors $\mathbf{U}$	Commitment probability $P(y = \text{yes} \mid \mathcal{D})$	Role disentanglement ($R = 20$); directional cross-attention; trajectory GRU tracking; dual pooling fusion.
Diagnostic Ablation Harness	Full and ablated dialogue streams $\mathbf{H}_M, \mathbf{H}_B$	Counterfactual ablations (LTR, LTR+M, FHO, PRO), Δ_{diff} , stress curves	Enforces left-truncation invariant; isolates early merchant accommodation signals; verifies domain-agnostic trajectory necessity.

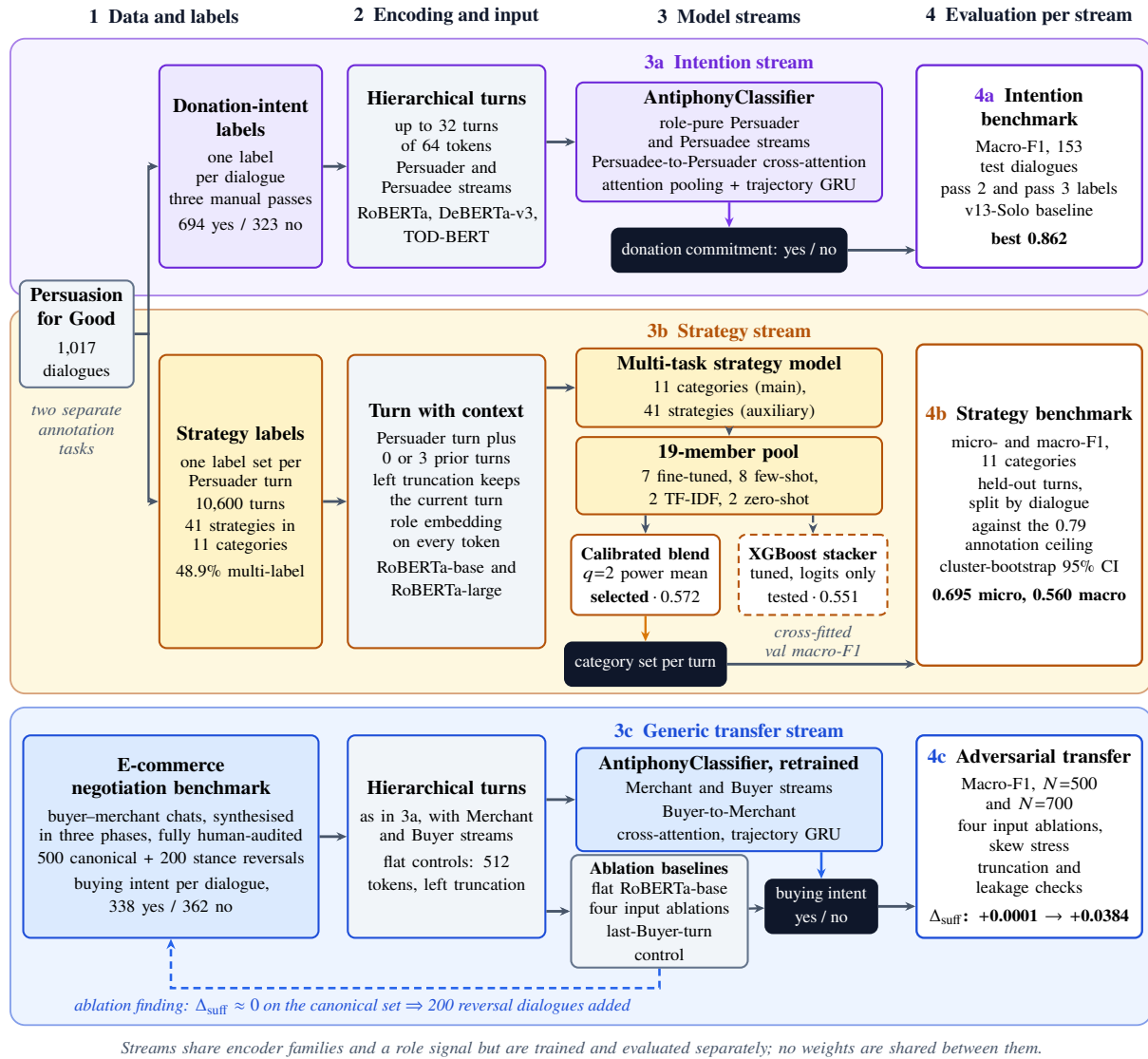


Figure 4.1: Overall functional system architecture, illustrating data ingestion across three corpora, shared utterance encoding, the dual-stream Antiphony intention network, the 19-member multi-task strategy classification framework with calibrated decision blending, and the diagnostic ablation evaluation framework.

4.2 Persuasion Strategy Annotation and Taxonomy

To model the rhetorical tactics that influence conversational outcomes, we need an empirically validated, fine-grained taxonomy and a standard annotation protocol. We build a multi-label persuasion strategy benchmark covering the first 10,600 Persuader turns from 1,017 dialogues in the P4G corpus [1].

4.2.1 Hierarchical Taxonomy Formulation

Adapted from social influence theory [4], compliance-gaining literature [5,6], and pragmatic dialogue analysis, our taxonomy is a two-tier hierarchy: 41 fine-grained rhetorical strategies nested deterministically under 11 functional communicative categories (structured in a 4, 4, 4, 4, 3, 4, 4, 3, 3, 4, 4 layout):

1. **Rational Appeal (4 strategies):** logical_appeal, evidence_and_statistics, cost_benefit_framing, feasibility_and_ease.
2. **Emotional Appeal (4 strategies):** emotion_appeal, guilt_induction, empathy_and_perspective_taking, hope_and_positive_impact.
3. **Credibility Appeal (4 strategies):** organizational_credibility, transparency_and_accountability, source_citation, personal_credibility.
4. **Social Influence (4 strategies):** social_proof, self_modeling, in_group_appeal, authority_endorsement.
5. **Reciprocity and Exchange (3 strategies):** reciprocity, donor_benefit, gratitude_and_appreciation.
6. **Commitment and Consistency (4 strategies):** foot_in_the_door, door_in_the_face, value_consistency_appeal, incremental_ask.
7. **Framing and Presentation (4 strategies):** anchoring, minimization_framing, loss_versus_gain_framing, comparison_framing.
8. **Urgency and Scarcity (3 strategies):** urgency_appeal, scarcity_appeal, deadline_pressure.
9. **Threat and Pressure (3 strategies):** negative_consequence_warning, persistent_repetition, obligation_pressure.
10. **Relational and Interactive (4 strategies):** rapport_building, personal_story, personal_related_inquiry, source_related_inquiry.
11. **Information Provision (4 strategies):** organization_information, donation_procedure_information, task_clarification, impact_information.

4.2.2 Contextual Annotation Pipeline and Decision Protocol

Annotating isolated sentences distorts rhetorical meaning; an utterance such as “Would you be willing to spare 50 cents?” functions as an incremental_ask only if preceded by a larger rejected request. Our pipeline therefore presents each Persuader utterance with a context window of the 5 preceding turns ($[u_{t-5}, \dots, u_{t-1}]$), keeping the actual speaker alternation.

To prevent subjective drift and keep segments consistent, annotators followed a mandatory **Four-Tier Decision Protocol**, applied in a fixed order:

1. **The Removal Test:** A strategy label s is valid if and only if removing the corresponding clause or rhetorical move noticeably alters the persuasive force or communicative intent of the turn.
2. **Specific-over-General Precedence:** If an utterance satisfies both a general category definition (e.g., logical_appeal) and a more specific operational tactic (e.g., evidence_and_statistics or cost_benefit_framing), the specific tactic must be coded, avoiding redundant double-labeling.
3. **Pragmatic Function over Keyword Matching:** Labels are assigned based on contextual communicative function rather than literal lexical cues. A rhetorical question (“Don’t you want to help a dying child?”) is coded as guilt_induction and emotion_appeal, never as an informational inquiry.
4. **Realistic Label Density Cap:** Complex turns can carry several labels, but people rarely use more than three distinct rhetorical tactics at once. The pipeline enforces a typical density of 1 to 3 labels, permits an explicit empty set ($y_t = \emptyset$) for purely conversational or logistical acknowledgments, and enforces a hard ceiling of $|y_t| \leq 6$.

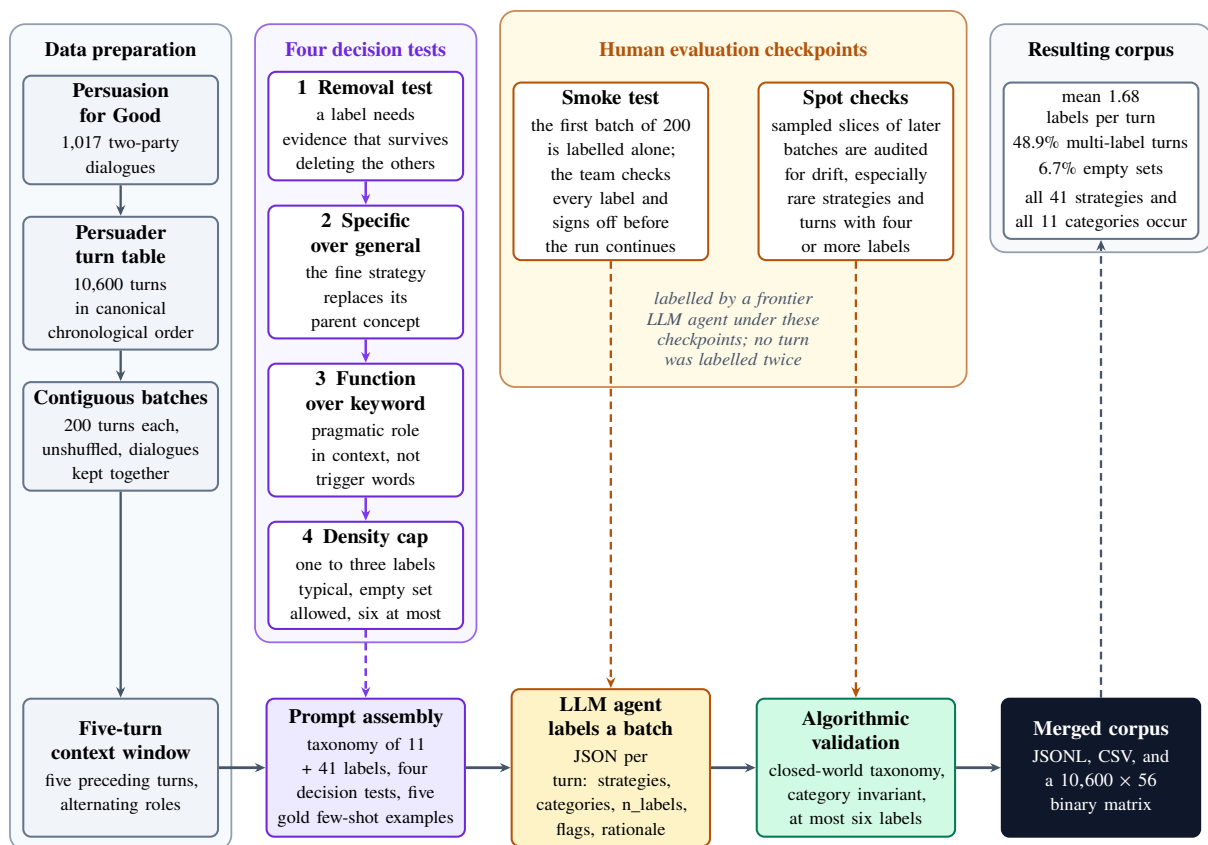


Figure 4.2: Persuasion strategy annotation pipeline, depicting contextual batching (200 turns per batch, 5-turn history), the 4-tier decision protocol, LLM agent annotation under human checkpoints, automated schema validation, and multi-label compilation into `multilabel_merged.jsonl`.

Annotation was conducted in 200-turn unshuffled batches using an automated frontier LLM agent (Claude 5 Sonnet) prompted with few-shot calibration examples, operating under human checkpoints. An automated annotation validation script verified taxonomy compliance, recomputed parent categories from strategies, enforced the label count ceiling, and compiled the verified dataset into `multilabel_merged.jsonl` ($10,600 \times 56$ wide format with 41 strategy and 11 category flags).

Execution Trace: Strategy Annotation Schema Validation

```
[+] Ingesting annotated chunks from data/raw_batches/ (10,600 Persuader turns)
[+] Running 4-tier decision constraint harness:
  - Verifying max strategy density cap:  $\max(|y_t|) \leq 6$  ... PASS (mean = 1.68)
  - Validating deterministic category mapping  $M: S_{41} \rightarrow C_{11}$  ... PASS (100% consistent)
  - Enforcing closed-world vocabulary constraints ... PASS (0 unknown tokens)
  - Checking speaker alternation invariants ... PASS ( $\tau_p = 2t - 1$ )
[*] Merging verified batches into multilabel_merged.jsonl (10,600  $\times$  56 matrix)
[+] DATASET AUDIT COMPLETE: 10,600 records serialized, 0 schema violations.
```

4.2.3 Empirical Corpus Statistics and Rater Effects

The resulting strategy corpus contains 10,600 Persuader turns across 1,017 dialogues (mean 10.42 Persuader turns per dialogue, range 4 to 15). Turns carry an average of 1.679 labels per turn ($sd = 1.061$, median 1.0). Exactly 707 turns (6.7%) carry no strategy label (the empty set $\emptyset$), 4,707 turns (44.4%) carry exactly 1 label, and 5,182 turns (48.9%) are multi-label (≥ 2 strategies), with 78 turns reaching the maximum cap of 6 labels.

Table 4.2 reports the empirical prevalence of all 11 communicative categories, recomputed from the verified `multilabel_merged.jsonl`.

Table 4.2: Empirical prevalence of communicative categories across 10,600 Persuader turns

Communicative Category	Active Turns	Prevalence (%)
Relational and Interactive	4,897	46.2%
Information Provision	3,823	36.1%
Emotional Appeal	1,645	15.5%
Social Influence	1,263	11.9%
Reciprocity and Exchange	1,078	10.2%
Framing and Presentation	1,004	9.5%
Credibility Appeal	896	8.5%
Rational Appeal	751	7.1%
Threat and Pressure	249	2.3%
Commitment and Consistency	169	1.6%
Urgency and Scarcity	163	1.5%

Annotator Granularity Effects: The corpus was annotated across four distinct operational segments under human checkpoints: Segment 1 (turns 1–2,000; Track A), Segment 2 (turns 2,001–4,000 and 6,001–7,000; Track B), Segment 3 (turns 4,001–6,000; Calibration Track), and Segment 4 (turns 7,001–10,600; Track C). Annotation granularity varies significantly between tracks: Track A averaged 2.384 labels per turn (9.0% empty sets, 66.8% multi-label), whereas Track C averaged 1.240 labels per turn (13.2% empty sets, 30.8% multi-label). We attribute this variance to a rater effect in label granularity, not to a linguistic shift in the underlying conversations.

4.2.4 Derivation of the Cross-Annotator Agreement Ceiling

Because individual turns were annotated once without global multi-coder duplication, inter-annotator agreement cannot be computed via standard Cohen’s kappa. We instead derive an empirical performance ceiling from near-duplicate turns across the corpus. On identical or near-identical text pairs (cosine similarity ≥ 0.95):

- When the *same* annotator labels near-identical turns, their self-agreement reaches an F1 of 0.871.
- When *different* annotators label near-identical turns, their cross-agreement drops to an F1 of 0.657.

A production model never sees annotator identity at inference time, so it has to generalize across annotators.

Modeling consensus prediction under the observed rater noise gives a practical upper performance ceiling of ≈ 0.79 **micro-F1**. All strategy model scores should be read against this 0.79 ceiling; a score of 0.70 reaches nearly 89% of the theoretically achievable human consensus.

4.3 Conversational Intention Ground Truth Generation

4.3.1 The Ground-Truth Divergence Problem

The original P4G benchmark [1] recorded numeric donation values reported by Mechanical Turk workers. A systematic manual audit, however, found severe discrepancies between these tabular records and the conversation text:

- In Dialogue #124, the Persuadee explicitly confirms: “We have you down for a \$10 donation. Is that correct?” / “Yes that is”, yet the recorded donation amount is \$0.
- In Dialogue #842, the Persuadee states: “I will absolutely look to donating something later”, while both numeric donation fields show \$0.

Classifiers trained on the unverified tabular values learn spurious logging noise instead of the commitments actually made in the conversation.

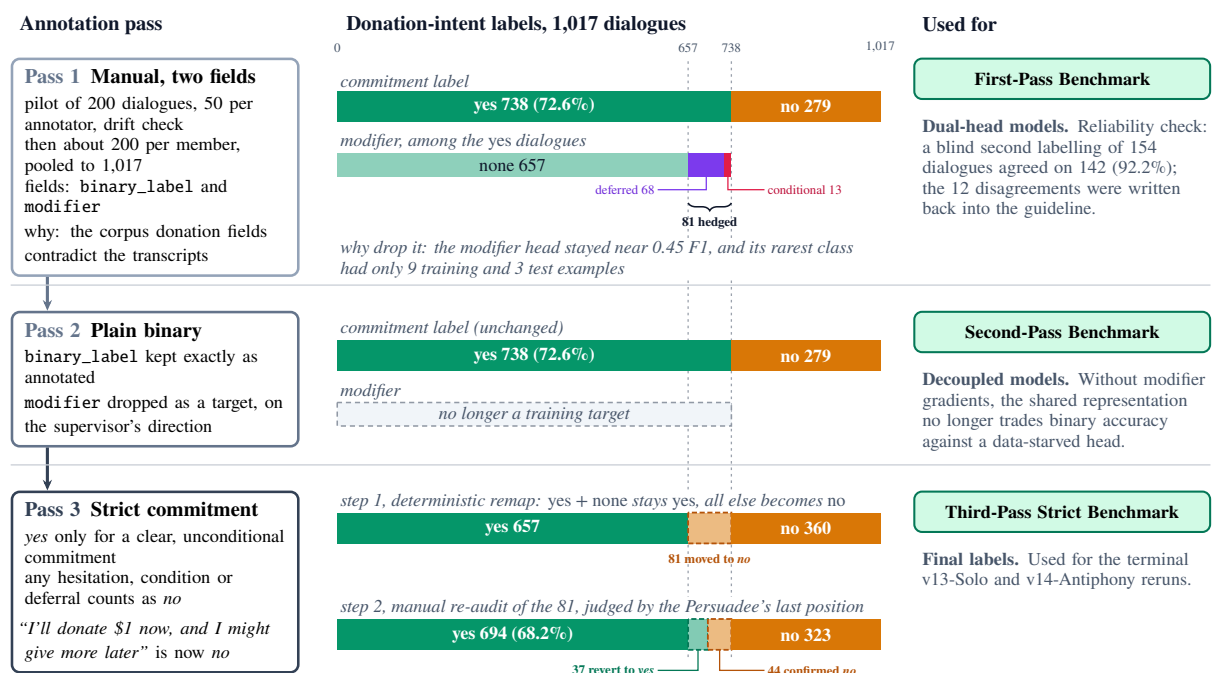


Figure 4.3: Three-pass intention annotation lineage, showing the initial 738/279 binary split with modifier, the second pass deprecating the modifier, and the third pass enforcing strict unconditional commitment (694 yes / 323 no via two-step re-checking).

4.3.2 The Three-Pass Annotation Lineage

To obtain a reliable ground truth, we ran three successive annotation passes:

1. **First Pass (Dual-Field Manual Annotation):** A pilot of 200 dialogues was split five ways (50 per annotator) with an annotator-tracking column to verify calibration and prevent rater drift. Following calibration, the five team members independently annotated ≈ 200 dialogues each, reading full transcripts and assigning `binary_label` (yes/no) and a `modifier` (none, deferred, conditional). The pooled 1,017-dialogue distribution yielded: 738 yes (72.6%) and 279 no (27.4%). Among affirmative commitments, 657 were none, 68 were deferred, and only 13 were conditional (9 in training, 3 in test).
2. **Second Pass (Modifier Deprecation):** Modeling experiments (Chapter 5) showed that the modifier head severely degraded binary performance and suffered from severe data sparsity on the 13-example conditional

class. Following a directive from the project supervisor, Dr. Md. Shariful Islam Bhuyan, the modifier target was dropped while leaving binary labels unchanged (738 yes / 279 no).

3. **Third Pass (Strict Unconditional Commitment):** To obtain an unambiguous and operationally stable target, we tightened the definition of yes: any hesitation, conditional clause (“if”, “maybe”), or temporal deferral (“later”) was reclassified as no. This was executed in two steps:

- *Step 1 (Deterministic Remap):* Any dialogue originally marked yes with modifier none remained yes; all dialogues with deferred or conditional modifiers were remapped to no, moving 81 dialogues from yes to no (657 yes / 360 no).
- *Step 2 (Terminal Stance Verification):* Because a modifier tag only indicated that a hedge occurred somewhere in the chat, all 81 remapped dialogues were manually re-read in full to evaluate their final state. If an early hesitation was cleanly resolved into an unconditional affirmative by the end, it was reverted to yes. Exactly 44 dialogues were confirmed as no, and 37 were reverted to yes.

The final third-pass ground truth contains exactly **694 yes** (68.2%) / **323 no** (31.8%).

4.3.3 Inter-Annotator Agreement and Boundary Cases

An independent second-pass annotation over 154 dialogues gave an inter-annotator agreement rate of 92.21% (142 of 154 identical assignments). Qualitative review of the 12 boundary disagreements revealed four recurring linguistic phenomena: (i) ambiguous temporal deferrals (“I’ll donate through their website tonight”), (ii) polite deflection masquerading as consent (“Sure, that sounds like a great cause”), (iii) conditional matches (“I’ll give \$1 if you give \$1”), and (iv) unconfirmed donation queries. We adjudicated these boundary cases under the strict third-pass standard, which removed the residual ambiguity.

4.4 Synthetic E-Commerce Negotiation Benchmark

4.4.1 Motivation and Privacy Bottlenecks

Evaluating cross-domain generalization from charitable persuasion to commercial negotiation is limited by data access: private customer-merchant chat logs on platforms like WhatsApp or Facebook Messenger are legally protected under privacy statutes and commercially sensitive. For a controlled, leak-free evaluation, we designed a three-phase synthetic benchmark generation pipeline that produced 700 fully audited negotiation dialogues.

4.4.2 The Three-Phase Generation Framework

1. **Phase 1: Persona-Conditioned Prompt-Driven Synthesis:** Dialogues are generated under two distinct regimes across multiple LLM backbones (including Qwen-2.5):
 - *Regime A (Canonical Stream, $N = 500$):* Models standard consumer transactions across eight behavioral buyer archetypes (The Skeptic, The LowBaller, The Urgent Buyer, The Ghoster, The Comparator, The DeliveryObsessed, The TrustSeeker, The BulkBuyer), three professional merchant personas (courteous page admin, small shopkeeper, formal retail representative), and four consumer product categories (electronics/phones, traditional apparel, home appliances, organic specialty goods). A Dialogue State Tracking (DST) register prevents repetitive loops.
 - *Regime B (Adversarial Reversal Stream, $N = 200$):* Generates non-monotonic trajectory shifts: 65 positive reversals (`resistance_to_agreement`, where deep skepticism resolves after a verified warranty), 65 negative reversals (`commit_to_rejection`, where early commitment collapses over delivery fees), and 70 hard monotonic controls designed with identical lengths and friction to prevent models from learning length-based shortcuts.
2. **Phase 2: Organic Paraphrasing and Humanization:** To remove synthetic LLM phrasing and imitate the style of mobile messaging, drafts pass through an organic paraphrasing engine powered by the Gemini API (`gemini-3.5-flash-lite`). The engine injects controlled linguistic jitter: 25% lowercased sentence openers, 30% dropped terminal periods, natural fillers (“hmm”, “well”, “ok”), local pricing terms (“X taka”), and sparse typos that never touch product names or digits. A similarity gate rejects any rewritten turn with a SequenceMatcher ratio > 0.82 against the source draft.

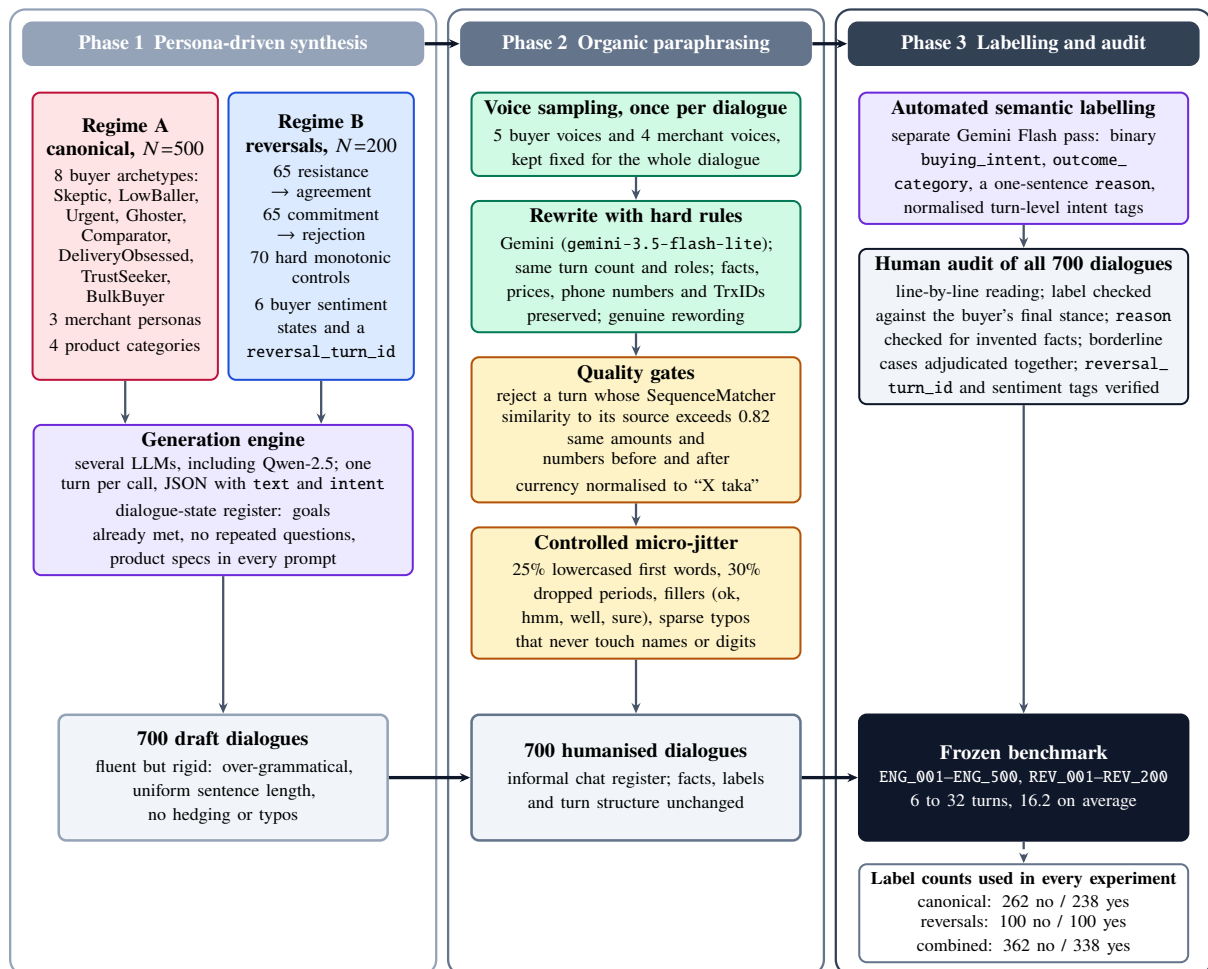


Figure 4.4: Three-phase synthetic negotiation benchmark generation pipeline, comprising persona-conditioned LLM drafting ($N = 500$ canonical, $N = 200$ adversarial reversals), organic paraphrasing with linguistic jitter, and multi-tier semantic audit.

3. **Phase 3: Semantic Labeling and 100% Team Audit:** An automated pass assigns `buying_intent`, multi-class `outcome_category`, and textual justifications. The whole research team then manually audited all 700 dialogues (100% coverage), verifying stance consistency, correcting subtle hallucinations, and confirming reversal turn IDs (τ_{rev}).

4.4.3 Frozen Benchmark Dataset Counts

Early design prompts planned a 250/250 canonical split across six outcome classes. The frozen datasets used in all experimental runs have the following verified counts across four primary outcome categories:

- **Canonical Benchmark** ($N = 500$): 262 no (52.4%) / 238 yes (47.6%). Outcomes: PURCHASE_COMMITTED 238, DEFERRED_CONSIDERATION 139, INQUIRY_DROPOUT 112, EXPLICIT_REJECTION 11. Mean turns: 15.9 (range 6 to 32).
- **Adversarial Reversals** ($N = 200$): Exactly 100 no / 100 yes. Mean turns: 17.1 (range 8 to 30).
- **Combined Augmented Benchmark** ($N = 700$): 362 no (51.7%) / 338 yes (48.3%). Outcomes: PURCHASE_COMMITTED 338, DEFERRED_CONSIDERATION 157, INQUIRY_DROPOUT 125, EXPLICIT_REJECTION 80. Mean turns: 16.2 (range 6 to 32).

4.4.4 Generic Architecture Diagnostic and Ablation Harness

To determine whether the architectural mechanisms in Antiphony (speaker disentanglement, directional cross-attention, and trajectory tracking) are domain-agnostic inductive biases for asymmetric multi-turn dialogue or an overfit to donation persuasion, we built a dedicated generic architecture diagnostic and ablation harness.

Counterfactual Input-Ablation Operators

Outcomes in conversational benchmarks are easily overdetermined: explicit affirmations or closing logistical exchanges let flat sequence classifiers infer intent from isolated surface tokens without using the dialogue history. To locate the causal source of the predictive signal, the harness defines four counterfactual input-ablation operators:

1. **Last-Turn-Removed (Ablation_{LTR}):** Excises the final decider turn:

$$\mathcal{D}_{\text{LTR}} = \mathcal{D} \setminus \{u_{\text{final}}\} \quad (4.1)$$

This tests whether the preceding conversation is enough to infer the outcome without an explicit closing declaration.

2. **Last-Turn and Merchant Courtesy Removed (Ablation_{LTR+M}):** In commercial transactions, merchants routinely emit closing courtesy affirmations (payment receipts, shipping notices) immediately following agreement. This operator excises both the terminal buyer turn and any trailing merchant courtesy turns:

$$\mathcal{D}_{\text{LTR+M}} = \mathcal{D} \setminus \{u \in \mathcal{D} \mid \text{index}(u) \geq \text{index}(u_{\text{final}})\} \quad (4.2)$$

which removes lexical leakage from both sides of the closing exchange.

3. **First-Half-Only (Ablation_{FHO}):** Dialogues are restricted to the first 50% of turns ($u_1, \dots, u_{\lfloor K/2 \rfloor}$). This detects early predictive bias and checks that synthetic personas do not leak the outcome in the opening exchanges.
4. **Pre-Reversal-Only (Ablation_{PRO}):** Applied to adversarial reversal dialogues exhibiting an annotated stance-shift at turn τ_{rev} , this operator excises all content from the shift onward:

$$\mathcal{D}_{\text{PRO}} = [u_1, \dots, u_{\tau_{\text{rev}}-1}] \quad (4.3)$$

testing whether models commit prematurely to the buyer's initial resistance or correctly use merchant accommodation signals.

Tokenizer Truncation Directionality Invariant

While building the ablation harness, we found a serious defect in the diagnostics: standard transformer tokenization libraries apply right-truncation by default (`truncation_side='right'`, capped at 512 tokens). In lengthy

multi-turn commercial chats (30.4% of dialogues in the benchmark exceeded 512 tokens), right-truncation silently drops the end of the conversation. Under Ablation_{LTR}, right-truncated tokenizers pruned the turns just before the excised turn, producing an artificial +0.1600 necessity collapse (Macro-F1 fell to 0.8133).

The diagnostic harness therefore enforces a **left-truncation invariant** (`truncation_side='left'`):

$$\text{Tokens}(\mathcal{D}) = [t_{\max(1, N_{\text{tokens}} - L_{\max} + 1)}, \dots, t_{N_{\text{tokens}}}] \quad (4.4)$$

which discards only early introductory pleasantries and keeps the turns at the decision boundary. With this invariant, baseline performance recovered to 0.9331, which shows that the true necessity gap on canonical dialogues is modest (+0.0402).

Parametric Skew-Stress Formulation

To test how the models hold up under severe real-world class imbalance without retraining the shared encoder, the harness implements a parametric class-pruning generator. Given a balanced training distribution (47.7% affirmative, $n = 350$), affirmative dialogues are subsampled according to target ratio $\alpha \in [0.10, 0.50]$:

$$n_{\text{pos}} = \left\lfloor \frac{\alpha}{1 - \alpha} \cdot n_{\text{neg}} \right\rfloor \quad (4.5)$$

which lowers the affirmative training ratio to 9.85% ($n = 203$). This measures how well directional cross-attention and recurrent trajectory tracking keep their calibration, compared with flat sequence models, as the class distribution collapses.

4.5 Strategy and Category Modeling Architecture

Multi-label strategy prediction over an extreme long tail calls for a multi-task formulation, an ensemble of diverse base learners, and a calibrated decision layer.

4.5.1 Multi-Task Category-First Formulation

Because the 41 strategies include classes with as few as 2 examples (`door_in_the_face`), models trained solely on 41 fine strategies achieve poor macro-F1 (0.368). We therefore move the primary modeling target to the 11 communicative categories, each of which has over 100 real instances. In the resulting multi-task network, the 11 categories are the primary objective ($\mathcal{L}_{\text{cat}}$) and the 41 strategies are an auxiliary regularizer ($\mathcal{L}_{\text{strat}}$):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cat}} + \lambda_{\text{aux}} \mathcal{L}_{\text{strat}} \quad (4.6)$$

with $\lambda_{\text{aux}} \in [0.1, 0.3]$. Both heads are optimized using Asymmetric Loss (ASL) [21]:

$$\mathcal{L}_{\text{ASL}} = - \sum_{c=1}^C [y_c(1 - p_c)^{\gamma_+} \log(p_c) + (1 - y_c)(p_{m,c})^{\gamma_-} \log(1 - p_{m,c})] \quad (4.7)$$

where $p_{m,c} = \max(p_c - m, 0)$ incorporates hard probability margin shifting ($m = 0.05$) and asymmetric focusing ($\gamma_+ = 0, \gamma_- = 4$) to discard easy negative gradients.

4.5.2 The 19-Member Heterogeneous Model Pool

To maximize ensemble diversity, we build a 19-member candidate pool:

- **7 Fine-Tuned Transformer Encoders:** RoBERTa-base (4 variations with mixed random seeds, context widths, and loss weighting), RoBERTa-large (2 models, with and without dialogue context), and ELECTRA-base (1 model).
- **8 Few-Shot Vector Companions:** Nearest-centroid cosine distance classifiers over mean-pooled encoder representations (one per transformer, plus one over TF-IDF space).
- **2 Classical TF-IDF Baselines:** Character- and word-level n-gram TF-IDF models paired with Logistic Regression and XGBoost.
- **2 Zero-Shot Semantic Similarity Models:** Lexical cue matchers mapping turn embeddings to category anchor definitions.

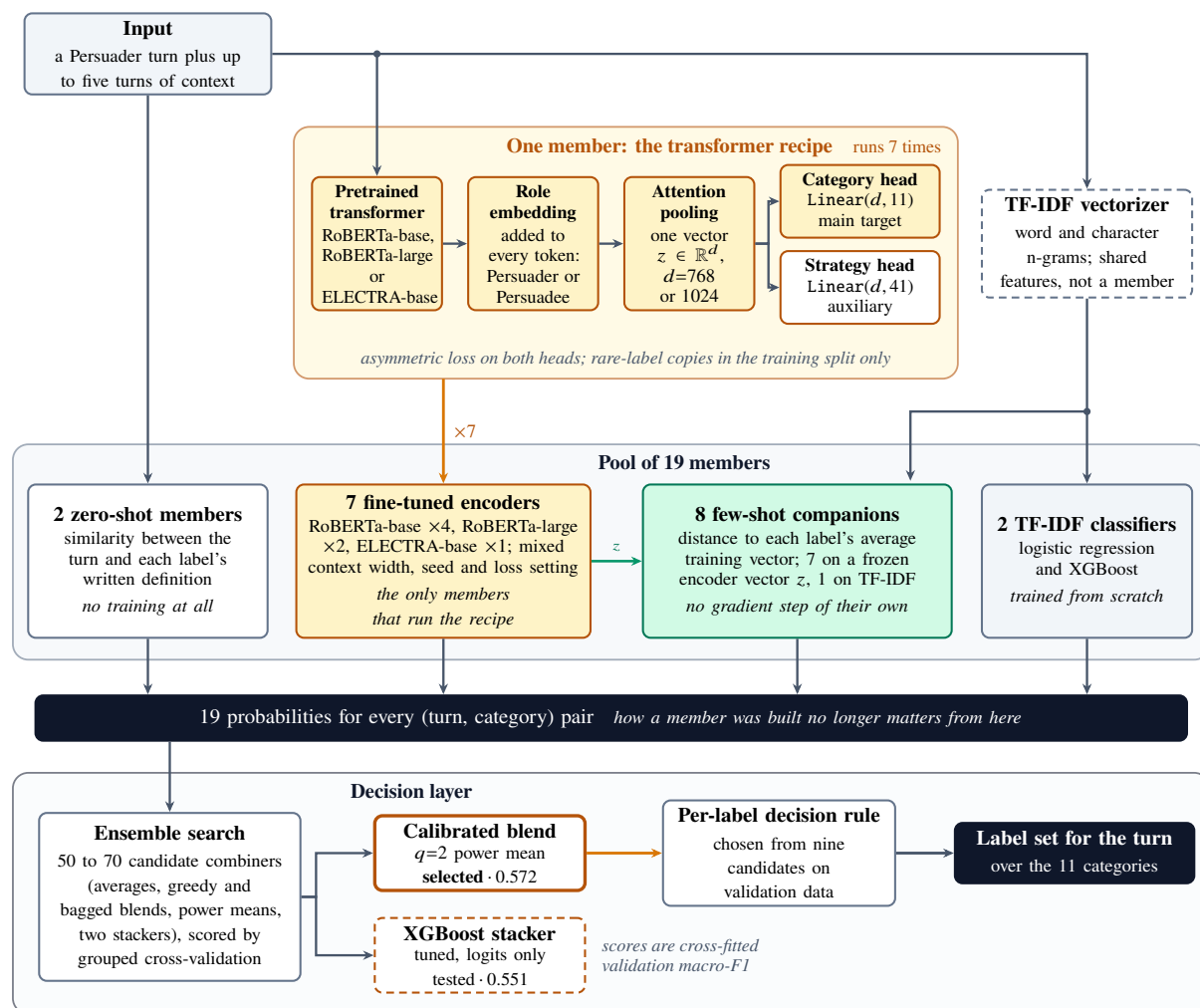


Figure 4.5: Strategy and category classification framework, illustrating the 19-member model pool, multi-task representation heads, and decision layer search comparing calibrated power-mean blending (selected, val macro-F1 0.5716) against XGBoost stacking (tested, val macro-F1 0.5508).

4.5.3 Decision Layer Search: Calibrated Blend vs. XGBoost Stacker

To combine the predictions of the 19 base members, we evaluated a wide range of decision combiners under grouped 5-fold cross-validation. One main research question was whether a machine-learned XGBoost stacking layer could replace standard probability blending:

- **XGBoost Stacking Experiment (Tested, Not Selected):** We constructed a tabular stacking dataset of $\approx 16,600$ (turn, label) instances across two feature regimes: “logits mode” (20 raw member pre-sigmoid outputs) and “rich mode” (41 features including member disagreement variances, parent category scores, and intra-turn ranks). Hyperparameter grid search over 72 configurations (max depth 4, learning rate 0.06, 300 to 500 trees) achieved a tuned cross-validated Macro-F1 of 0.5508 (logits mode) and micro-F1 of 0.6847 (rich mode), significantly improving over the untuned baseline (0.5329).
- **Calibrated Power-Mean Blending (Selected Winner):** Despite extensive tuning, the XGBoost stacker was outperformed by a **calibrated power-mean blend** ($q = +2.0$) applied to member probabilities post-Platt scaling [25]:

$$\bar{p}_c = \left(\frac{1}{M} \sum_{m=1}^M [\sigma(\alpha_m z_{m,c} + \beta_m)]^q \right)^{1/q} \quad (4.8)$$

The calibrated power-mean blend reached a higher cross-validated Macro-F1 of 0.5716, so it was selected as the shipped decision layer.

4.6 The AntiphonyClassifier Architecture

The main intention architecture of this project is the *AntiphonyClassifier*. It was built to address the limitations of flat sequence transformers and symmetric dialogue encoders, and it builds the asymmetry of persuasion directly into the network structure.

We present Antiphony in two schematics, one conceptual and one at implementation level:

- **Conceptual Architecture (Figure 4.6):** Shows the functional dataflow, from alternating raw dialogue turns and role disentanglement to directional question-argument cross-attention and dual temporal-pooling context readouts.
- **Tensor and Layer Architecture (Figure 4.7):** Specifies exact mathematical dimensions, transformer sub-layer chains, multi-head projection matrices, residual normalizations, and composite loss optimization heads.

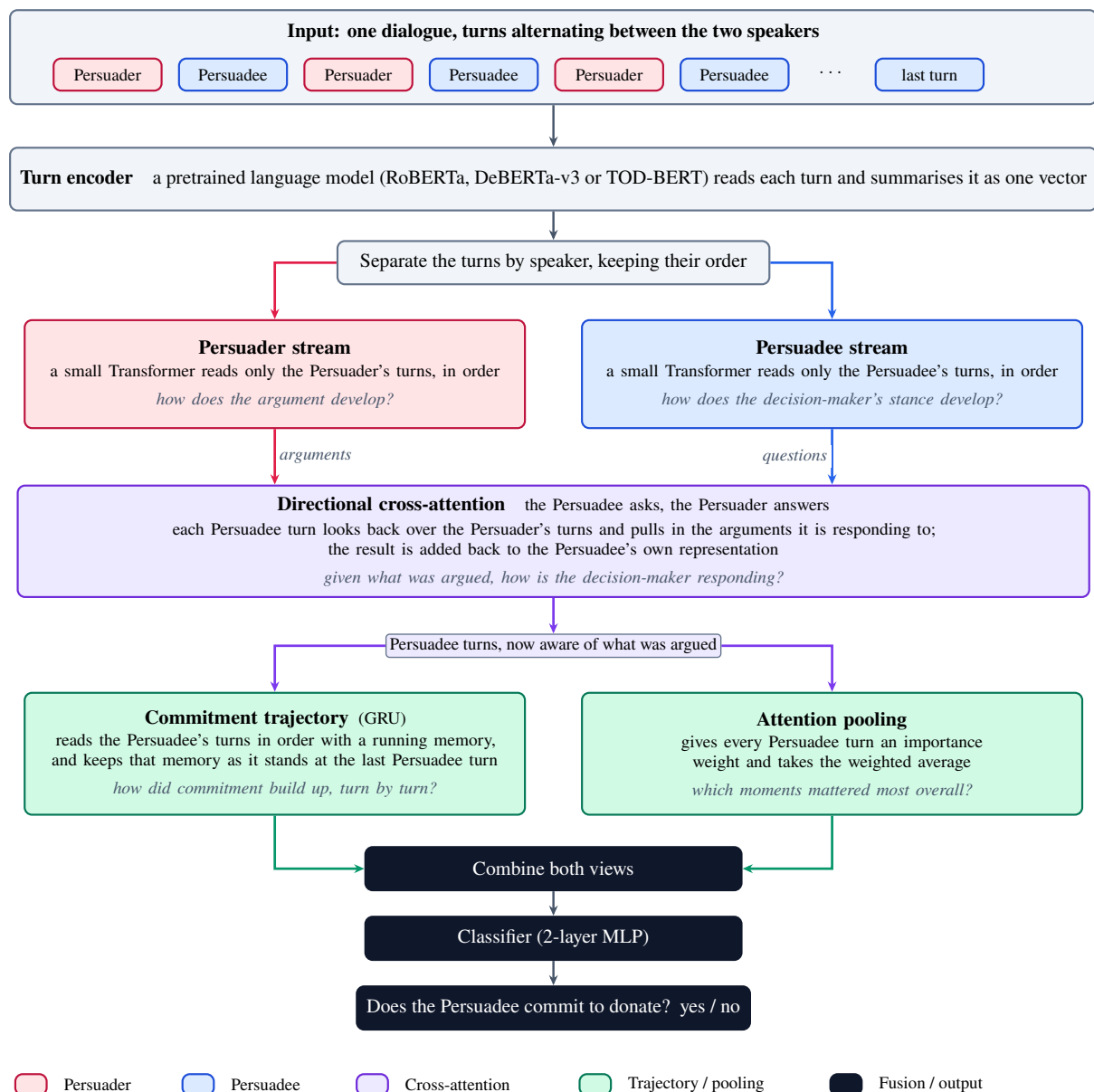


Figure 4.6: High-level conceptual schematic of the AntiphonyClassifier architecture, showing the information flow from alternating conversational turns to speaker stream disentanglement, directional asymmetric cross-attention, dual trajectory and attention pooling context aggregation, and final commitment classification.

4.6.1 Detailed Mathematical Formulation

Let dialogue $\mathcal{D}$ comprise U utterances ($U \leq 32$), each truncated to $L = 64$ tokens.

1. **Batched Utterance Encoding:** The input tensor $\mathbf{X} \in \mathbb{R}^{B \times U \times L}$ is flattened to $[B \cdot U, L]$ and passed through a single batched forward pass of a shared transformer encoder (e.g., DeBERTa-v3-base), yielding token hidden states $\mathbf{E} \in \mathbb{R}^{(B \cdot U) \times L \times d}$ where $d = 768$. Masked mean-pooling over valid non-padded tokens produces utterance vectors $\mathbf{U} \in \mathbb{R}^{B \times U \times d}$.
2. **Role-Pure Disentanglement:** Utterances are partitioned using binary role masks into two role-pure streams: Persuader turns $\mathbf{H}^{(\mathcal{P})} \in \mathbb{R}^{B \times R_{\mathcal{P}} \times d}$ and Persuadee/Decider turns $\mathbf{H}^{(\mathcal{D})} \in \mathbb{R}^{B \times R_{\mathcal{D}} \times d}$, where each stream retains the last $R = 20$ turns in chronological order with stream-local position embeddings.
3. **Independent Stream Transformers:** Each stream is encoded by an independent 2-layer transformer with multi-head self-attention, pre-LayerNorm, and feed-forward sub-layers:

$$\tilde{\mathbf{H}}^{(\mathcal{P})} = \text{Transformer}_{\mathcal{P}}(\mathbf{H}^{(\mathcal{P})}), \quad \tilde{\mathbf{H}}^{(\mathcal{D})} = \text{Transformer}_{\mathcal{D}}(\mathbf{H}^{(\mathcal{D})}) \quad (4.9)$$

4. **Directional Asymmetric Cross-Attention:** To model how decider stance evolves in response to persuader arguments, the Decider stream acts as the Query, while the Persuader stream provides the Keys and Values:

$$\mathbf{Q} = \tilde{\mathbf{H}}^{(\mathcal{D})} \mathbf{W}_Q, \quad \mathbf{K} = \tilde{\mathbf{H}}^{(\mathcal{P})} \mathbf{W}_K, \quad \mathbf{V} = \tilde{\mathbf{H}}^{(\mathcal{P})} \mathbf{W}_V \quad (4.10)$$

$$\mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}} + \mathbf{M}_{\mathcal{P}}\right), \quad \mathbf{C}^{(\mathcal{D} \rightarrow \mathcal{P})} = \mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} \mathbf{V} \quad (4.11)$$

The cross-attended representations are combined via residual layer normalization:

$$\mathbf{X} = \text{LayerNorm}(\tilde{\mathbf{H}}^{(\mathcal{D})} + \mathbf{C}^{(\mathcal{D} \rightarrow \mathcal{P})}) \quad (4.12)$$

5. **Dual Context Aggregation:** The contextualized decider representations $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]$ are aggregated across two complementary paths:

- *Commitment-Trajectory GRU:* Captures non-monotonic stance shifts via a recurrent hidden state:

$$\mathbf{h}_t = \text{GRU}(\mathbf{x}_t, \mathbf{h}_{t-1}) \quad (4.13)$$

The trajectory state $\mathbf{h}_T$ is extracted at the dialogue's true last valid decider turn, bypassing trailing padding slots.

- *Learned Attention Pooling:* Computes an order-invariant summary of salient turns:

$$\alpha_t = \frac{\exp(\mathbf{w}_a^\top \tanh(\mathbf{W}_a \mathbf{x}_t))}{\sum_{j=1}^T \exp(\mathbf{w}_a^\top \tanh(\mathbf{W}_a \mathbf{x}_j))}, \quad \mathbf{p} = \sum_{t=1}^T \alpha_t \mathbf{x}_t \quad (4.14)$$

6. **Fusion and Classification Head:** The temporal trajectory vector $\mathbf{h}_T$ and the attention-pooled vector $\mathbf{p}$ are concatenated:

$$\mathbf{z} = [\mathbf{h}_T; \mathbf{p}] \in \mathbb{R}^{2d} \quad (4.15)$$

and fed into a 2-layer MLP head ($\text{Linear}(2d, d) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(d, 2)$) producing unnormalized binary logits $\hat{\mathbf{y}} \in \mathbb{R}^2$.

7. **Multi-Objective Training Loss:** Antiphony is optimized using a joint composite loss combining Class-Balanced Focal Loss [20] ($\gamma = 2.0, \beta = 0.999$) and Supervised Contrastive Loss [22] ($\tau = 0.1$, weight $\lambda_{\text{sc}} = 0.30$):

$$\mathcal{L} = \mathcal{L}_{\text{CB-Focal}}(\hat{\mathbf{y}}, \mathbf{y}) + \lambda_{\text{sc}} \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}) \quad (4.16)$$

Directional Cross-Attention and Trajectory Invariant

In the *AntiphonyClassifier*, the asymmetric conversational roles are built into the network structure:

$$\mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} = \text{softmax}\left(\frac{\mathbf{Q}_{\mathcal{D}} \mathbf{K}_{\mathcal{P}}^\top}{\sqrt{d_k}} + \mathbf{M}_{\mathcal{P}}\right), \quad \mathbf{C}^{(\mathcal{D} \rightarrow \mathcal{P})} = \mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} \mathbf{V}_{\mathcal{P}} \quad (4.17)$$

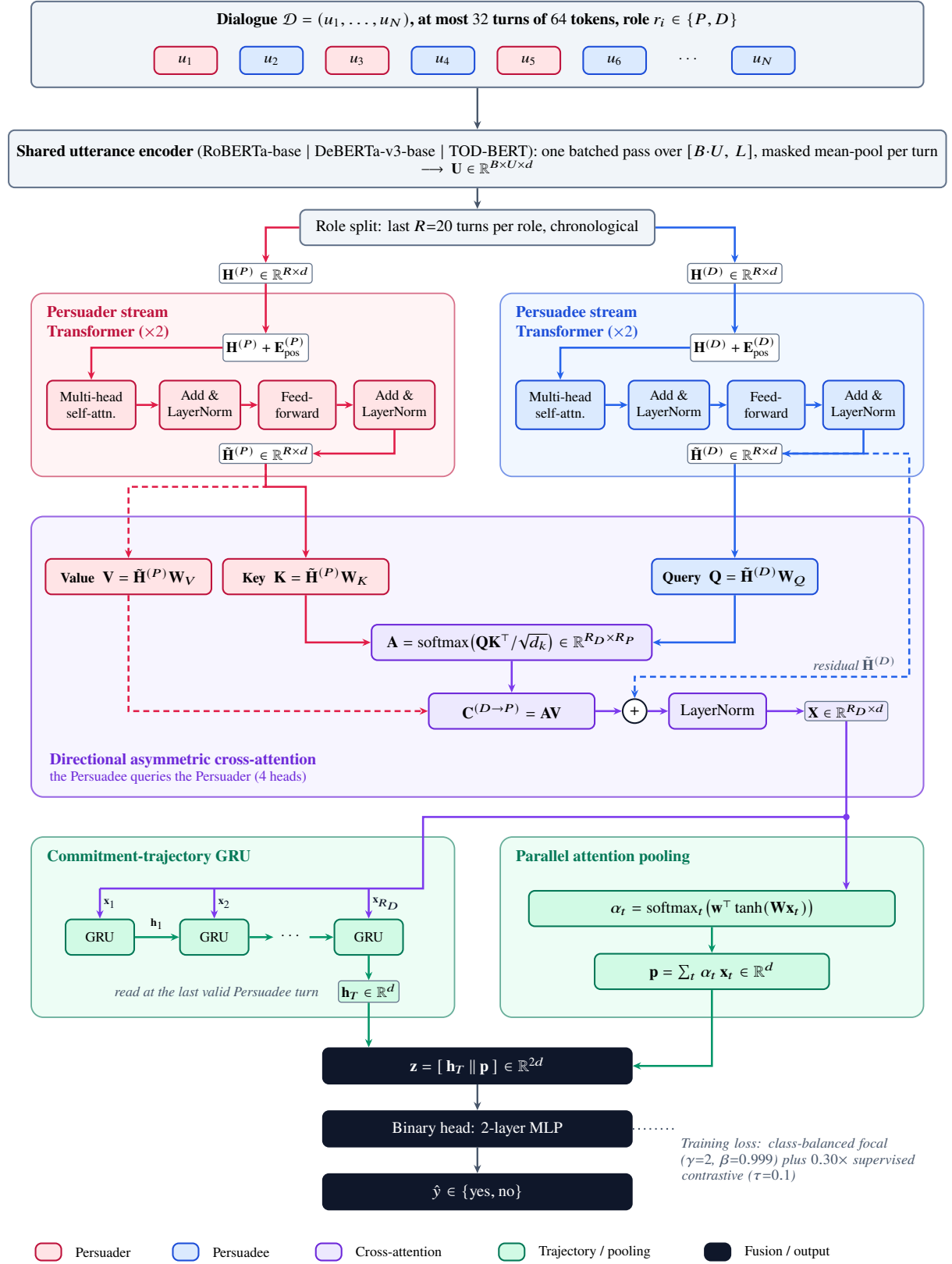


Figure 4.7: Detailed neural architecture and tensor specifications of the AntiphonyClassifier, depicting exact layer dimensions, 2-layer independent stream Transformers, directional cross-attention ($\mathbf{Q}_D \mathbf{K}_P^T / \sqrt{d_k}$), parallel commitment-trajectory GRU and learned attention pooling, and composite Class-Balanced Focal and Supervised Contrastive loss heads.

The decider query $\mathbf{Q}_D$ selectively attends to the persuader's rhetorical tactics $\mathbf{K}_P, \mathbf{V}_P$, updating the commitment-trajectory state $\mathbf{h}_t = \text{GRU}(\mathbf{x}_t, \mathbf{h}_{t-1})$ read at the dialogue's true final decider turn. As a result, the model tracks how the decision develops over time and ignores trailing post-decision pleasantries.

4.7 Architectural Evolution: The Path to Dual-Stream Antiphony

The AntiphonyClassifier came out of a systematic empirical investigation over four phases, documented in Figure 4.8. This section refers to the deciding party by its dataset role, *Persuadee*, the same participant termed *Decider* in the architecture description of Section 4.6.

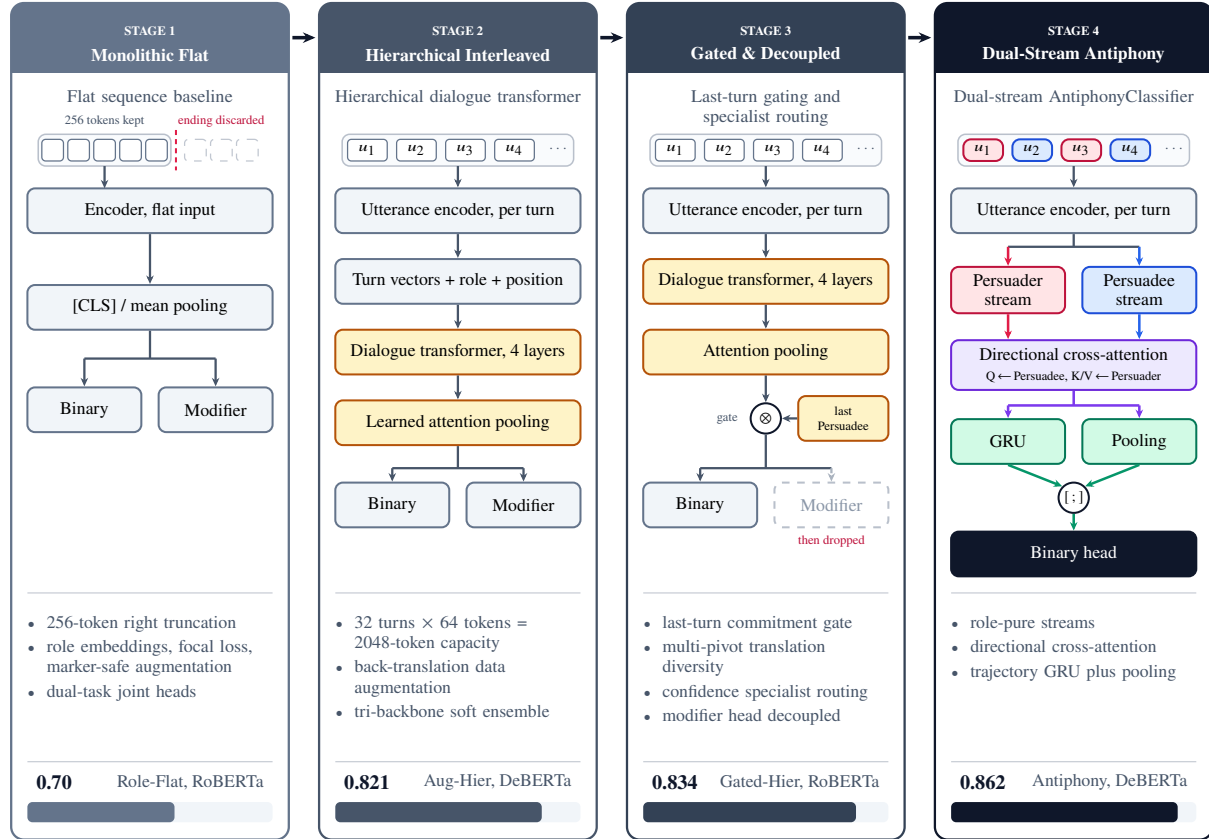


Figure 4.8: Four-stage architectural evolution of intention classification models, showing the main innovations, bug diagnoses, and peak test Macro-F1 across stages.

4.7.1 Stage 1: Monolithic Flat Baselines and Forensic Debugging

- **Flat Sequence Base Model (M1):** Concatenated dialogues truncated to 256 tokens with [CLS] pooling and independent binary and modifier heads. Reached 0.68 Macro-F1 with RoBERTa, but DeBERTa skewed heavily toward the majority class (Macro-F1 0.53, only modestly above the 0.42 majority-baseline) and modifier heads failed on rare classes.
- **Speaker-Aware Flat Encoding (M2):** Added learnable role embeddings (Persuader = 0, Persuadee = 1) prior to masked mean-pooling. Binary Macro-F1 rose to 0.70 (RoBERTa) and recall on the rare deferred modifier class improved, but overall performance remained bottlenecked by 256-token truncation.
- **Dynamic Augmentation Exploration (M3):** Added EDA text augmentation [38] and raw inverse-frequency class weighting. Training collapsed. A forensic audit found two bugs: (i) checkpoint selection was mistakenly scored on `pos_label="yes"` F1, allowing an all-yes trivial model to score 1.0 and be saved, and (ii) raw inverse-frequency weights on the 13-example conditional class caused extreme gradient spikes.
- **Metric-Corrected Focal Loss Baseline (M4):** Fixed the evaluation harness to use true unweighted Macro-F1; implemented Class-Balanced Focal Loss ($\beta = 0.999$); cascaded the binary prediction into the modifier head; added in-batch Supervised Contrastive Loss. Training stabilized, but performance plateaued at 0.649 Macro-F1 (RoBERTa and TOD-BERT; DeBERTa 0.630).
- **Token-Safe Augmentation Audit (M5):** An audit revealed that 57.5% of speaker markers ([Persuader]) in raw transcripts were glued to newlines without spaces. Standard whitespace augmentation tokenizers fragmented these tags, corrupting role markers in 15.9% of augmented instances. Implementing regex-based

tokenization and Persuadee-biased mutation weights resolved the corruption, but Macro-F1 for DeBERTa remained capped at 0.670, still short of M2’s 0.70 RoBERTa result; no backbone in this flat, 256-token family ever exceeded that RoBERTa figure, motivating the architectural change in Stage 2.

4.7.2 Stage 2: Hierarchical Discourse Modeling

- **Hierarchical Dialogue Transformer (M6):** This was the main turning point. Turns are encoded individually ($32 \times 64 = 2,048$ token capacity) via a shared encoder, pooled into turn vectors, combined with role and positional embeddings, and processed by a 4-layer inter-turn dialogue transformer with learned attention pooling. Macro-F1 rose to 0.793 (RoBERTa), 0.797 (DeBERTa), and 0.807 (TOD-BERT).
- **Flat Back-Translation Control (M7):** Applied French back-translation to the flat M5 model as an experimental control; Macro-F1 dropped to 0.590–0.650, showing that data augmentation cannot make up for architectural truncation.
- **Hierarchical Back-Translation (M8):** Combined the M6 hierarchical architecture with back-translation. Binary Macro-F1 advanced to 0.821 (DeBERTa), but modifier F1 fell, indicating that translation round-trips erase subtle hedge words.
- **Tri-Model Soft Ensemble (M9):** Soft probability averaging across saved M6 checkpoints (RoBERTa, DeBERTa, TOD-BERT) achieved 0.818 Macro-F1. Over 20 stochastic Monte Carlo dropout forward passes, prediction variance was significantly lower on correct instances (0.0220) than on incorrect instances (0.0297), so the variance gives a calibrated uncertainty signal.

4.7.3 Stage 3: Recency Gating and Decoupled Baselines

- **Last-Persuadee-Turn Gating (M10):** Added a sigmoid gate anchoring the pooled representation to the final Persuadee hidden state, initialized with a +3.0 bias to prevent early deflection. Achieved 0.834 Macro-F1 with RoBERTa, establishing the peak single-model binary score of the two-headed lineage.
- **Multi-Pivot Back-Translation Diversity (M11):** Implemented French and German back-translation under beam search and random sampling; DeBERTa reached 0.829, but modifier F1 continued to degrade.
- **Specialist Confidence Router (M12):** Decoupled the tasks without retraining: routing binary classification to M10 Gated RoBERTa (0.834) and modifier classification to M6 Plain DeBERTa (0.452).
- **Decoupled Single-Stream Baseline (M13):** Deleted the modifier head entirely, dedicating all capacity to a 2-layer binary head. Reached 0.825 on second-pass labels (RoBERTa); on third-pass labels, RoBERTa’s default-threshold Macro-F1 is 0.8359, marginally above its own tuned-threshold figure of 0.8290 used as this model’s officially reported result elsewhere in this report.

4.7.4 Stage 4: Role-Pure Dual-Stream Antiphony Architecture

The Dual-Stream Antiphony architecture (M14) implemented speaker-disentangled cross-attention and reached the best results in the project: 0.8623 Macro-F1 with DeBERTa at its default decision threshold on the second-pass benchmark (0.8600 under the tuned-threshold rule applied consistently to the other backbones on this benchmark), and 0.8446 with RoBERTa at a tuned threshold on the strict third-pass benchmark, the shipped final model. These results show that modeling asymmetric conversational dynamics is necessary.

4.8 Design Refinement and Simulation

4.8.1 Capacity Calculations and Sequence Length Budgeting

In conversational modeling, sequence capacity must balance linguistic coverage against quadratic attention memory scaling ($O(N^2)$). In the P4G corpus, dialogue length exhibits a median of 344 words across an average of 20 turns, with the 95th percentile reaching 23 turns. Flat transformers capped at 256 tokens cover fewer than 180 words, truncating over 50% of conversational content.

In Antiphony, the sequence budget allocates $U = 32$ turns with $L = 64$ tokens per turn, providing a total token capacity of:

$$C_{\text{tokens}} = 32 \times 64 = 2,048 \text{ tokens} \quad (4.18)$$

This budget covers 99.7% of all conversational tokens across the corpus without truncation. For role-pure streams, setting $R = 20$ turns per role provides capacity for up to 40 total turns, covering 98.8% of dialogue trajectories.

4.8.2 Memory Profiling and Hardware Guards

Processing a batch of size B with U turns of length L through a 768-dimensional transformer requires flattening to $[B \cdot U, L]$. For $B = 4$ and $U = 32$, this requires encoding 128 sequences of length 64 in a single forward pass. On an NVIDIA Tesla T4 GPU (16 GB VRAM), peak activation memory for DeBERTa-v3 under standard precision exceeds hardware limits. We added three hardware guards:

1. **Micro-Batch Slicing:** Training batch size is set to $B = 2$, with evaluation batch size $B = 4$.
2. **Gradient Checkpointing:** Intermediate activation maps are discarded during the forward pass and recomputed during backpropagation, reducing activation memory by $\approx 60\%$.
3. **PyTorch Dynamic Memory Allocation:** Enabled `expandable_segments: True` in the PyTorch CUDA allocator to eliminate memory fragmentation during variable-length sequence packing.

4.9 Application of Tools

The project used the following tools:

- **PyTorch 2.x [39]:** Provided core automatic differentiation, custom CUDA autograd layers, and modular neural network implementations.
- **Hugging Face Transformers [30]:** Supplied pre-trained transformer backbones (RoBERTa, DeBERTa-v3, TOD-BERT, ELECTRA) and tokenization engines.
- **XGBoost [24] and Scikit-Learn:** Used for classical baseline construction, tabular stacking experiments, calibration curve fitting, and bootstrap evaluation metrics.
- **OPUS-MT [40]:** Used for multilingual back-translation pipelines across French and German pivots.
- **Anthropic Claude 5 Sonnet:** Used as the automated frontier LLM rater for contextual multi-label strategy annotation under human calibration checkpoints.
- **Google Gemini API (gemini-3.5-flash-lite):** Employed for organic paraphrasing, controlled jitter injection, and synthetic dialogue auditing.
- **MiKTeX and PGF/TikZ:** Used for mathematical typesetting and native vector diagrams of the architecture.

Tool Limitations: Pre-trained transformer checkpoints inherit tokenization biases and vocabulary constraints; free-tier cloud environments enforce 12-hour timeout limits; and proprietary LLM APIs introduce rate-limit constraints and network latency.

4.10 Implementation

Development followed modular software engineering practice. All code was organized into self-contained, reproducible Jupyter notebooks paired with detailed markdown architecture summaries (`architecture-v*.md`).

4.10.1 Version Control and Development History

The codebase was tracked using Git [41] in a shared repository. Each model iteration was kept as its own versioned notebook (for example, v1 through v14 for intention and v17/v18 for strategy), so earlier versions remained runnable and comparable after later ones superseded them.

4.10.2 Deterministic Seeding and Audit Artifacts

For reproducibility, all scripts call a central seeding utility that sets Python `random`, NumPy, and PyTorch CPU/CUDA RNG states to `Seed = 42`. All trained model weights, out-of-fold validation predictions, and test evaluation matrices were serialized as immutable `.pt` and `.jsonl` artifacts.

4.11 Deployment

4.11.1 Prototype Execution Pipeline

All training, evaluation, and ablation runs were executed offline within Kaggle and Python environments. The trained models are additionally exposed through two demonstration web applications, presented in Section 4.12. The end-to-end inference execution flow for an unlabelled dialogue proceeds as follows:

$$\text{Transcript } \mathcal{D} \xrightarrow{\text{Tokenize}} [U \times 64] \xrightarrow{\text{Encoder}} \mathbf{U} \xrightarrow{\text{Split \& Cross-Attend}} \mathbf{X} \xrightarrow{\text{GRU + Pool}} \mathbf{z} \xrightarrow{\text{MLP}} \hat{y} \quad (4.19)$$

4.11.2 Resource Consumption

Inference on a complete 20-turn dialogue executes in approximately 45 ms on an NVIDIA T4 GPU (380 ms on an 8-core Intel CPU under batch size 1). Measured GPU training runtimes across model variants: Metric-Corrected Baseline (≈ 20 min), Hierarchical Transformer (≈ 2.5 h), Hierarchical Back-Translation (≈ 5.0 h), Last-Turn Gated Model (≈ 2.5 h), Decoupled Single-Stream (≈ 3.5 h), and Antiphony (≈ 3.0 h). The category clean rerun took 177 minutes, and strategy ensemble search required ≈ 4.0 hours.

4.11.3 Operational Boundaries and Omissions

This project is an academic research prototype. The demonstration applications of Section 4.12 exist to show the models in use and are not a production service. **No commercial production serving microservice, optimized runtime graph export, model quantization, or live latency SLA benchmarks were constructed.** Deploying the system into production would require implementing streaming turn ingestion, model quantization, and reliable fail-soft API infrastructure.

4.12 User Interface

The models are presented to end users through *Havn*, a messaging platform built for this project in two variants that share one interface: a persuasion app, in which donors message fundraisers (<https://intentionalysis-persuasion.vercel.app>), and a buyer-seller app, in which buyers message merchants (<https://intentionalysis-buyer-seller.vercel.app>). Each variant has two sides. The persuadee side (donor or buyer) is an ordinary chat client with no AI output anywhere. The persuader side (fundraiser or merchant) is a workspace in which every conversation is scored live by the models. The AI is an observer only: it never writes or sends messages. All screenshots in this section were captured from the deployed applications on desktop (1440 px) and phone (390 px) viewports; the names of donors and buyers are fictional.

4.12.1 Donor and Buyer Side

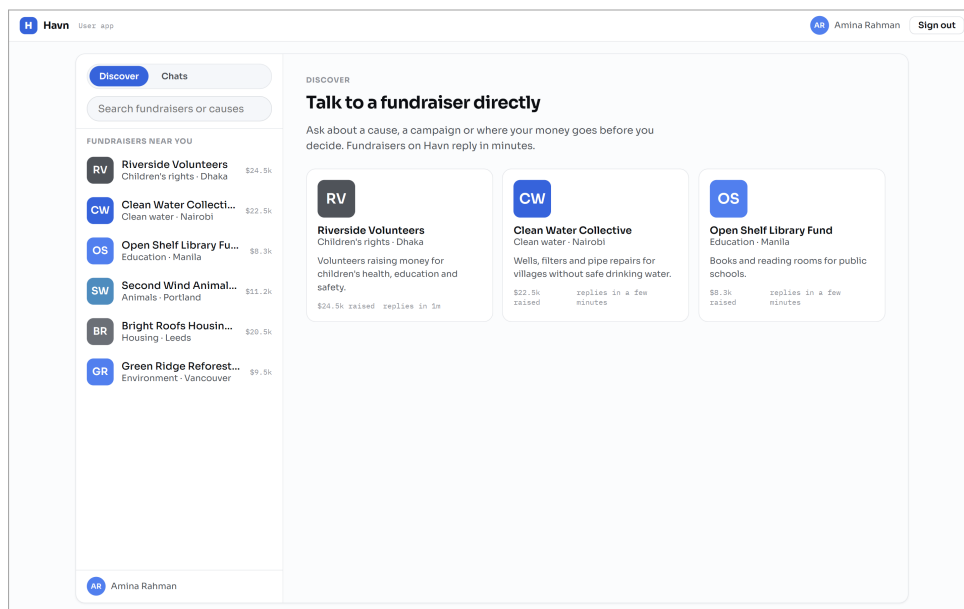
A donor signs in or registers, browses a directory of fundraisers, opens a fundraiser’s profile to see its active campaigns, and starts a conversation (Figures 4.9 and 4.10). Forms validate each field inline before submission. The chat itself shows messages and delivery status only; no score, badge, or strategy label is ever shown to the persuadee.

4.12.2 Fundraiser Workspace

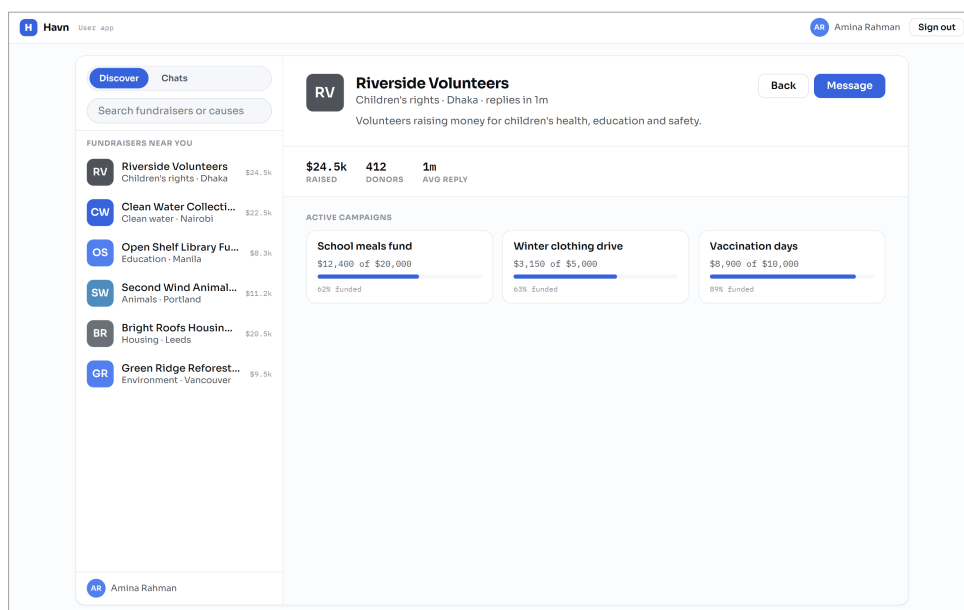
A fundraiser lands on an overview page (Figure 4.11) with summary cards (active conversations, conversations awaiting a reply, donors scored as likely to donate, median first-reply time), an AI-ranked list of the donors most likely to give, and the distribution of donation intent across open chats. The inbox (Figure 4.12) uses a three-pane layout. The left pane lists conversations with an intent badge (*Likely*, *Unclear*, *Unlikely*) and an unread count, and can be searched or filtered to unread or likely conversations. The centre pane is the chat, where each fundraiser message carries the persuasion strategies detected in it. The right pane is the live AI analysis panel: a donation-intent gauge coloured by band (70 and above high, 31 to 69 medium, 30 and below low), the predicted outcome and confidence, the strategies used so far in the conversation, an intent timeline with one point per donor message, and the donor’s latest replies.

4.12.3 Responsive Layouts

The workspace adapts to the available width. On a compact laptop screen (1100 px) the AI panel leaves the page and opens as a slide-over drawer from an AI button in the conversation header (Figure 4.13). On a phone (Figure 4.14)

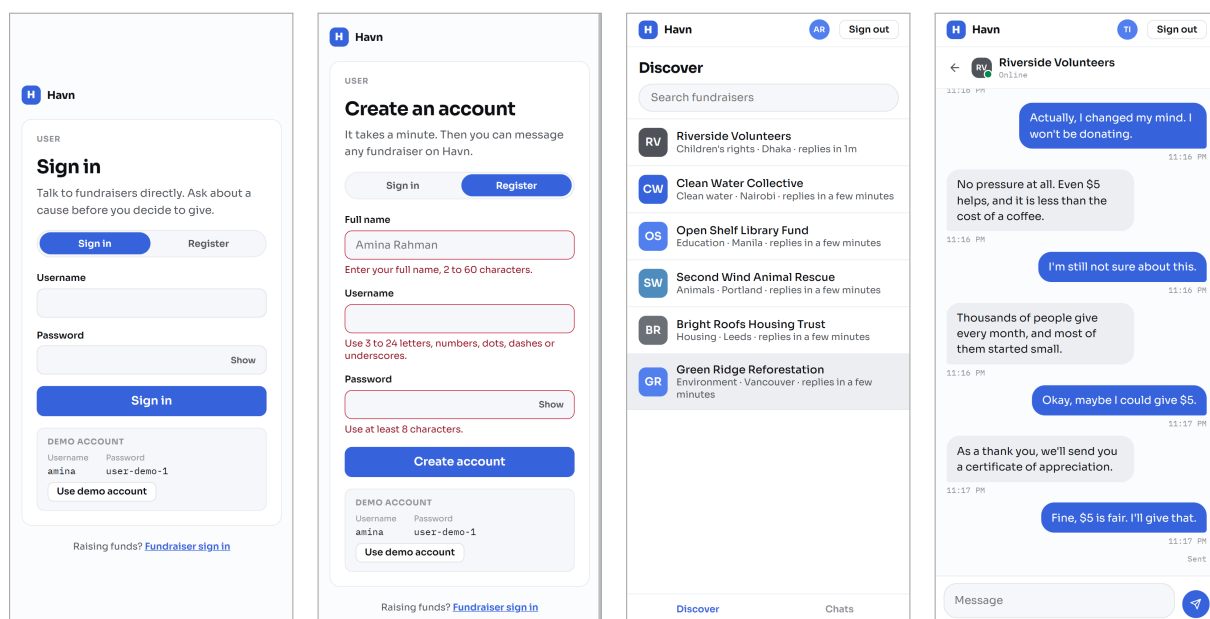


(a) Discover: a searchable directory of fundraisers with cause, location, and typical reply time.



(b) Fundraiser profile: headline statistics and active campaigns with funding progress bars.

Figure 4.9: Donor side of the persuasion app on desktop.



(a) Sign-in with a demo account panel.

(b) Registration with field-level validation.

(c) Discover with search and bottom tab bar.

(d) Donor's chat: plain messages, no AI output.

Figure 4.10: Donor side of the persuasion app on a phone.

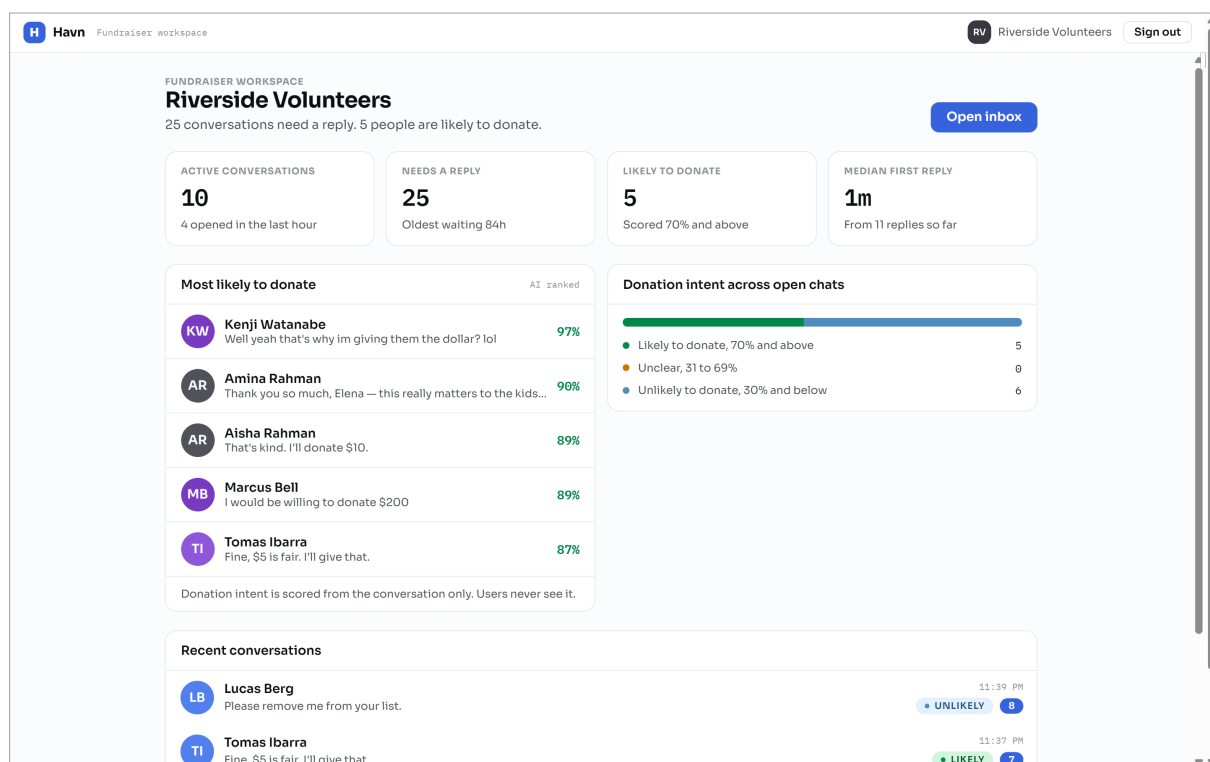


Figure 4.11: Fundraiser overview: summary cards, AI-ranked “most likely to donate” list, donation-intent distribution, and recent conversations with intent badges.

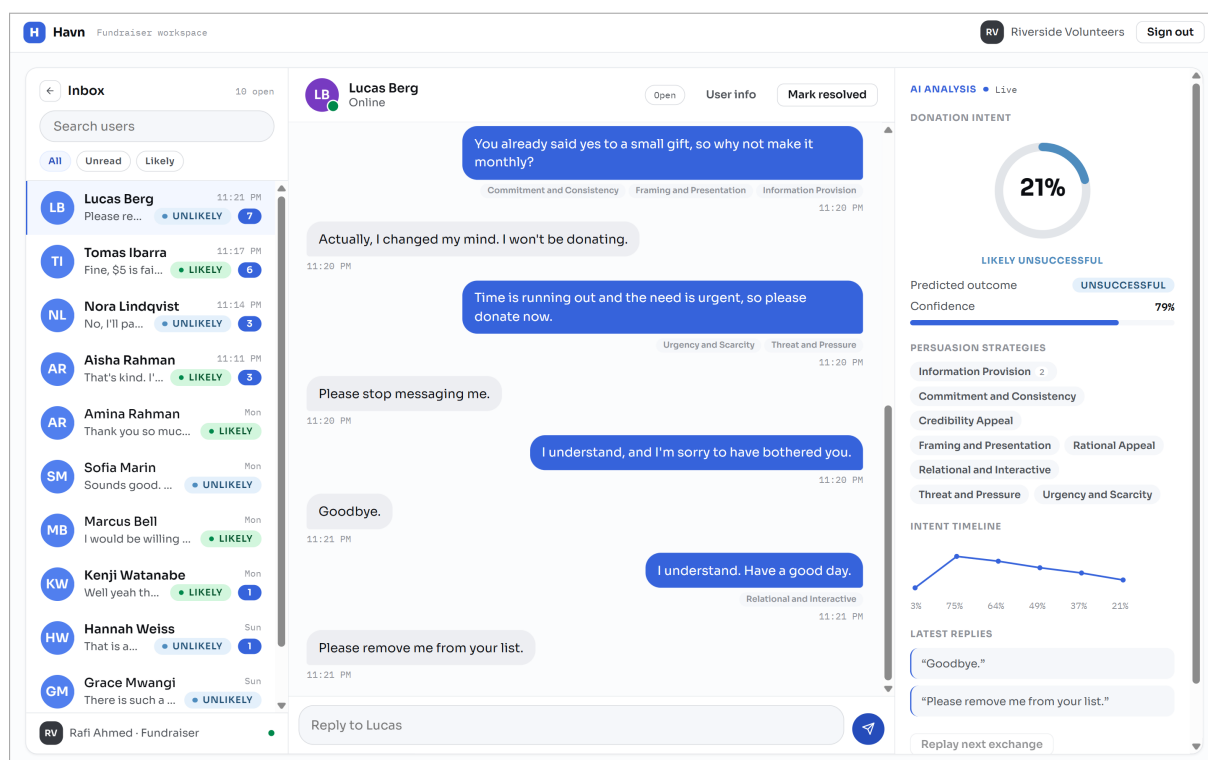


Figure 4.12: Fundraiser inbox. Left: conversation list with intent badges, unread counts, search, and filters. Centre: chat with strategy labels under each fundraiser message. Right: AI analysis panel with the intent gauge, predicted outcome and confidence, strategies used, intent timeline, and latest donor replies.

the overview and inbox become single-column screens, and the AI analysis opens as a bottom sheet over the chat; when the sheet is closed, the header button keeps showing the current intent score.

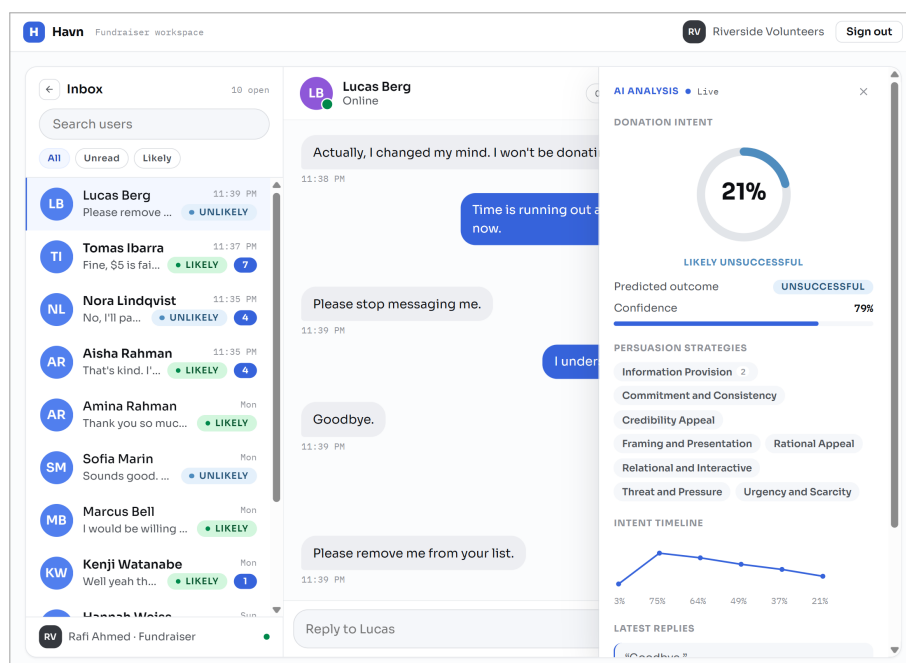


Figure 4.13: Compact (1100 px) layout: the AI analysis panel opens as a slide-over drawer instead of a fixed third pane.

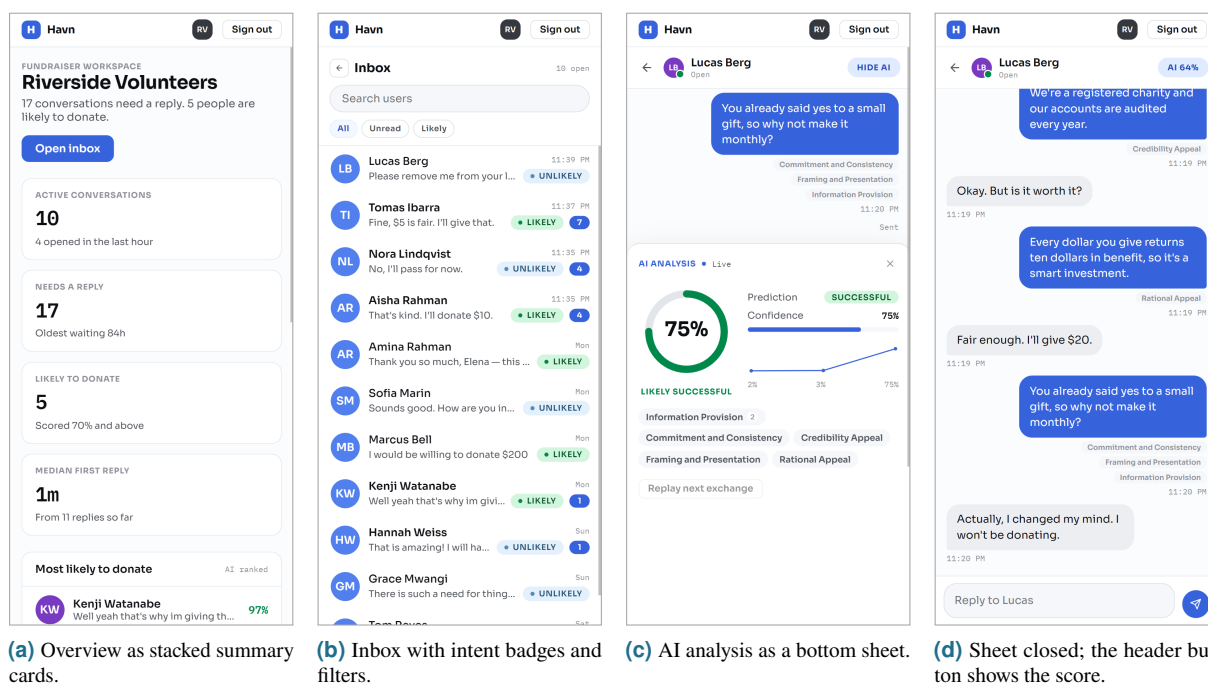


Figure 4.14: Fundraiser workspace on a phone.

4.12.4 Merchant Workspace

The buyer-seller app reuses the same layout for the commerce domain (Figures 4.15 and 4.16). Buyers browse merchant storefronts with product listings instead of campaigns. In the merchant workspace the gauge reports purchase intent (70 and above high, 40 to 69 medium, below 40 low), and inbox badges read *High*, *Medium*, or *Low intent*. In place of persuasion strategies, the panel shows a checklist of detected buyer signals (for example, asking about price, delivery, or payment method), which are matched by keyword rules on the buyer's messages rather than by a model, together with buyer messages quoted as evidence.

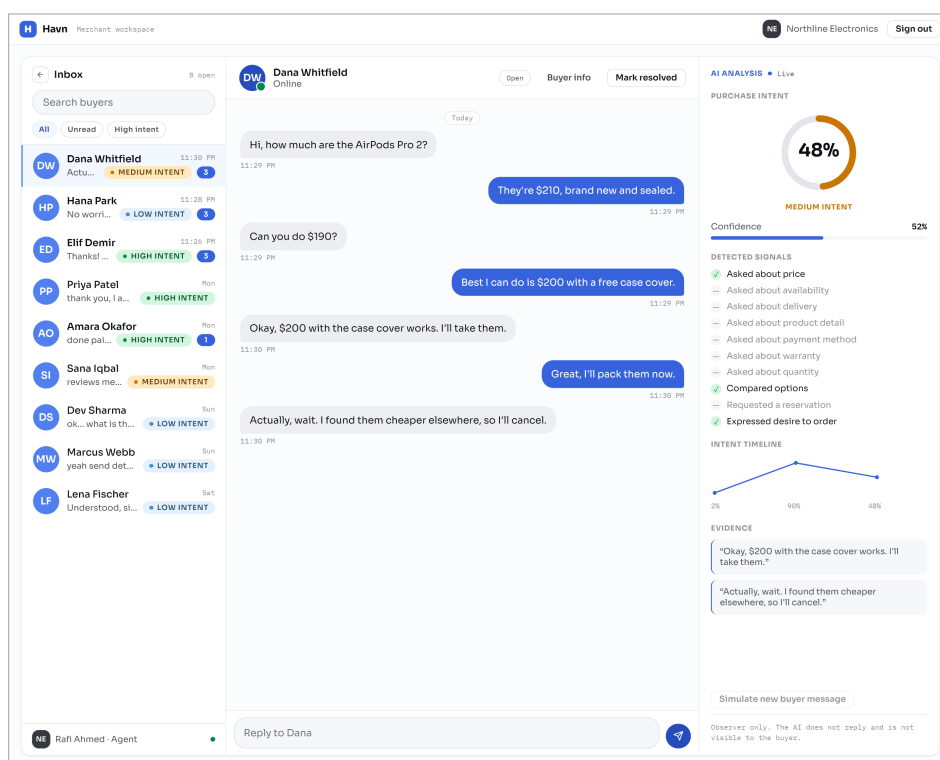
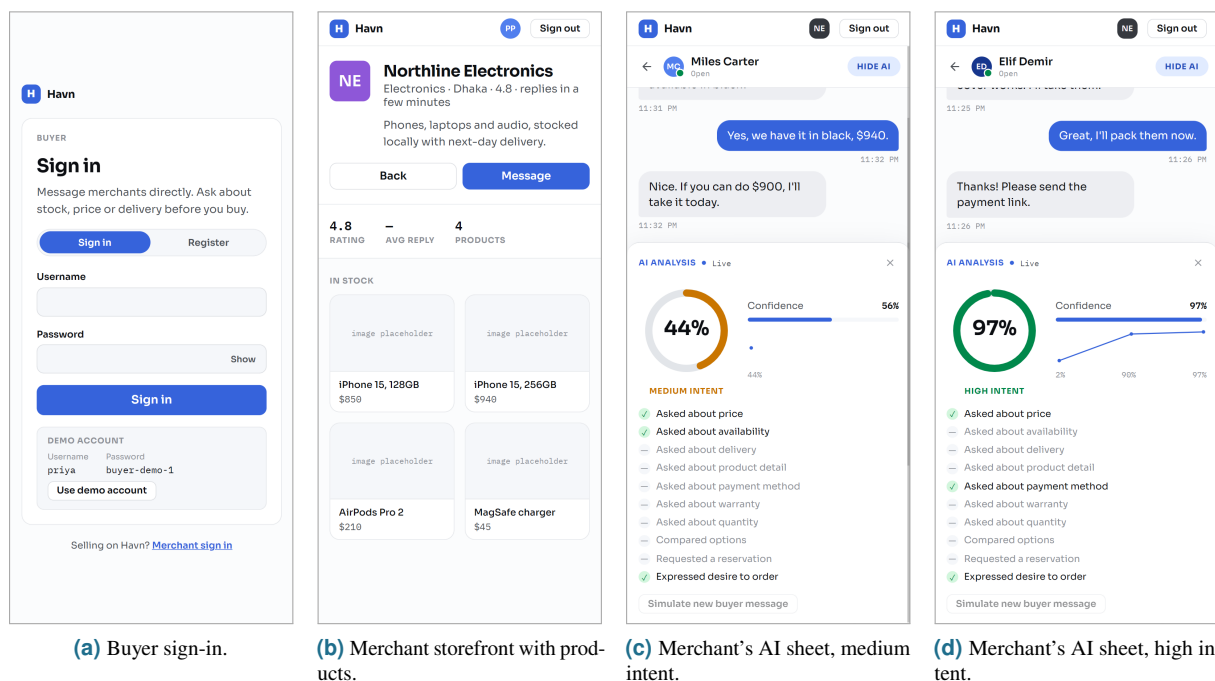


Figure 4.15: Merchant inbox: purchase-intent gauge in the medium band, detected buyer-signal checklist, intent timeline, and quoted evidence.



(a) Buyer sign-in.

(b) Merchant storefront with products.

(c) Merchant's AI sheet, medium intent.

(d) Merchant's AI sheet, high intent.

Figure 4.16: Buyer-seller app on a phone.

Chapter 5

Investigation and Evaluation

5.1 Experimental Investigation Setup

We test the proposed architectural and methodological hypotheses through experiments on three separate dialogue datasets.

5.1.1 Research Questions and Hypotheses

We formulate five research questions (RQs):

- **RQ1 (Context Length Bottleneck):** Does hierarchical utterance encoding overcome the performance ceiling imposed by flat sequence transformer truncation in multi-turn dialogues?
- **RQ2 (Asymmetric Disentanglement):** Does role-pure dual-stream encoding with directional cross-attention (*Antiphony*) achieve better intention classification compared to symmetric, single-stream hierarchical models?
- **RQ3 (Task Decoupling):** Does removing the minority modifier prediction task relieve negative gradient interference and raise binary commitment Macro-F1?
- **RQ4 (Decision Layer Efficiency):** Can a tuned gradient-boosted decision tree (XGBoost) stacking layer outperform a calibrated power-mean blend in combining heterogeneous multi-label strategy classifiers?
- **RQ5 (Adversarial Trajectory Robustness):** When closing conversational turns are destabilized by non-monotonic stance reversals, does trajectory tracking maintain a positive sufficiency gap over localized decision baselines?

5.1.2 Data Splits and Evaluation Protocols

All intention experiments use stratified 70% / 15% / 15% train/validation/test splits partitioned at the dialogue level to eliminate cross-turn leakage:

- **P4G Intention Corpus** ($N = 1,017$): 711 training, 153 validation, and 153 test dialogues. Stratified on the joint commitment-by-modifier stratum.
- **P4G Strategy Corpus** (10,600 turns): 7,592 training, 1,504 validation, and 1,504 test turns, partitioned by dialogue identity with multi-label stratification.
- **Canonical E-Commerce Benchmark** ($N = 500$): 350 train, 75 validation, 75 test dialogues.
- **Augmented E-Commerce Benchmark** ($N = 700$): 490 train, 105 validation, 105 test dialogues.

5.1.3 Evaluation Metrics

- **Binary Macro-F1:** The primary headline metric for intention classification:

$$\text{Macro-F1} = \frac{F_1^{(\text{no})} + F_1^{(\text{yes})}}{2} \quad (5.1)$$

where $F_1^{(c)} = 2P_c R_c / (P_c + R_c)$. This gives minority refusals the same weight as majority affirmations.

- **Multi-Label Category Metrics:** Micro-F1 (aggregating global true positives, false positives, and false negatives) and Macro-F1 (unweighted average across 11 category F1 scores).
- **Sufficiency Gap** (Δ_{suff}): Defined in Equation (1.4) as the performance differential between the full dialogue model and a control seeing only the final decision turn.

- **Decision Threshold Calibration:** For binary intention models, optimal thresholds $\theta^* \in [0.10, 0.90]$ are selected via grid search on the validation split to maximize validation Macro-F1, and evaluated out-of-fold on the test split. We report both default ($\theta = 0.50$) and tuned thresholds.

5.2 Corpus Dynamics and Strategy Exploratory Data Analysis

Prior to modeling, we ran an exploratory data analysis (EDA) over the 10,600 Persuader turns of the multi-label strategy corpus. Figures 5.1 to 5.3 show six of the resulting distributions.

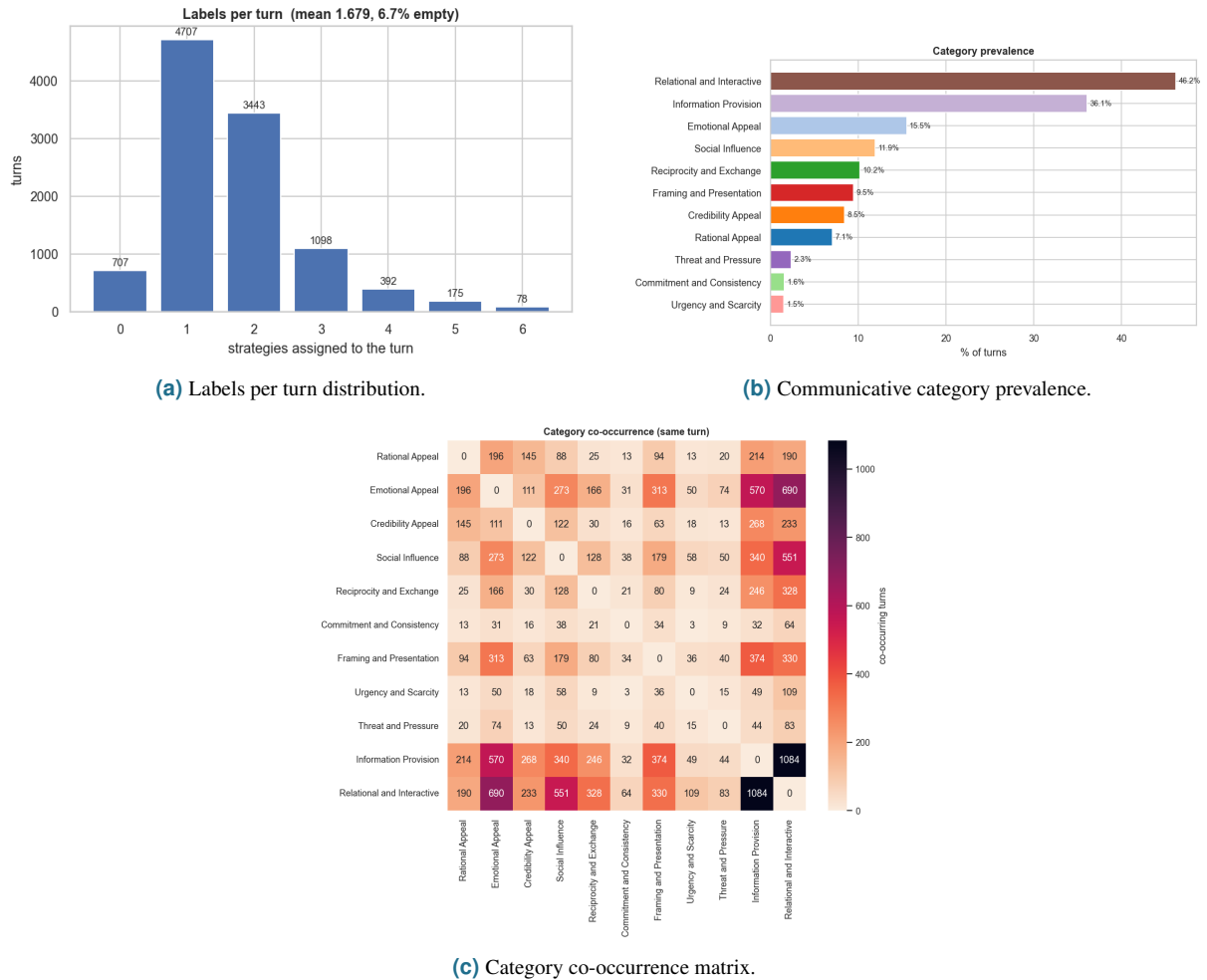


Figure 5.1: Strategy corpus EDA: label density, category prevalence, and category co-occurrence.

5.2.1 Key Analytical EDA Takeaways

1. **High Multi-Label Density:** 48.9% of turns carry 2 or more strategy labels (Figure 5.1a), showing that human persuasion utterances pack multiple rhetorical acts into single turns (e.g., small talk greeting + factual charity inquiry).
2. **Dominance of Relational Scaffolding:** Relational and Interactive (46.2%) and Information Provision (36.1%) dominate the category distribution (Figure 5.1b). Hard pressure tactics (Threat and Pressure at 2.3%, Urgency and Scarcity at 1.5%) are deployed sparingly.
3. **Pronounced Positional Trajectory:** Strategy deployment varies systematically across turn positions (Figure 5.2): rapport_building and source_related_inquiry peak in turns 1–3; factual statistics and emotional appeals dominate middle turns (turns 4–8); while donation_procedure_information, gratitude_and_appreciation, and incremental_ask concentrate in closing turns (turns 9–14).

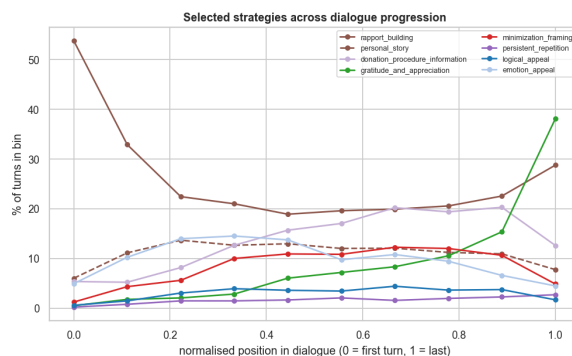
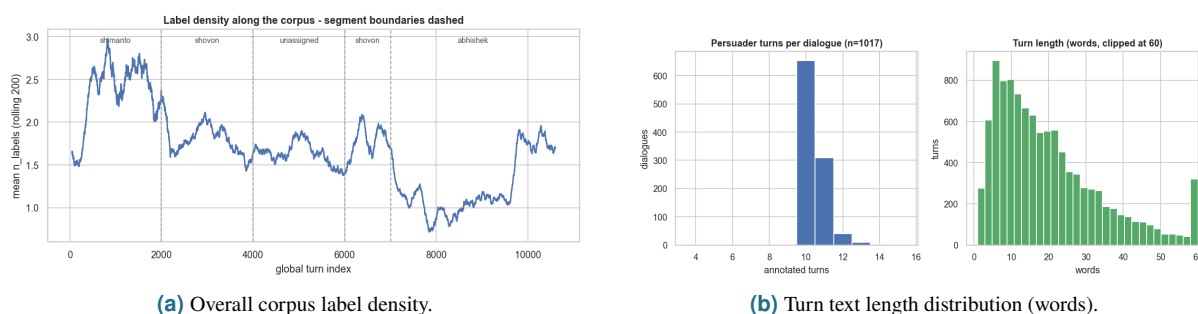


Figure 5.2: Strategy corpus EDA: strategy deployment by turn position.



(a) Overall corpus label density.

(b) Turn text length distribution (words).

Figure 5.3: Strategy corpus EDA: label density and turn length distributions.

5.3 Intention Classification Results Across Three Passes

5.3.1 Part 1: Results on the First-Pass Dataset (Binary + Modifier)

Table 5.1 reproduces all results evaluated on the original two-field dataset (738 yes / 279 no), tracking the architectural lineage from M1 to M12 across RoBERTa, DeBERTa, and TOD-BERT.

Antiphony Benchmark Telemetry: Landmark Progression Evaluation

ANTIPHONY INTENTION BENCHMARK: HELD-OUT TEST SUITE ($n = 153$)

[STAGE 1: Flat Baselines]

- M1: RoBERTa-base (256t Flat) : Macro-F1 = 0.6800 | Acc = 75.80%
- M3: Dynamic Augmentation (all backbones) : Collapsed, no checkpoint; bugs fixed in M4

[STAGE 2: Hierarchical Transformers]

- M6: HiTrans (32x64t), TOD-BERT : Macro-F1 = 0.8070 | Acc = 83.70%
- M8: HiTrans + Back-Translation, DeBERTa : Macro-F1 = 0.8210 | Acc = 86.30%

[STAGE 3: Gated & Decoupled]

- M10: Last-Persuadee-Turn Gate, RoBERTa : Macro-F1 = 0.8340 | Acc = 86.30%

Table 5.1: Test set performance on First-Pass dataset (738 yes / 279 no, test $n = 153$)

Model Variant	Core Architectural Design	Backbone	Accuracy	Macro-F1	Modifier F1
<i>Baseline</i>	Majority Class Baseline	<i>None</i>	72.5%	0.4205	0.314
M1 (Flat-256)	Monolithic Flat 256-token Sequence, [CLS] pool	RoBERTa	75.8%	0.6800	0.296
M1 (Flat-256)	Monolithic Flat 256-token Sequence, [CLS] pool	DeBERTa	74.5%	0.5300	0.314
M1 (Flat-256)	Monolithic Flat 256-token Sequence, [CLS] pool	TOD-BERT	74.5%	0.842*	0.311
M2 (Role-Flat)	Flat Sequence + Speaker Role Embeddings	RoBERTa	77.8%	0.7000	0.351
M2 (Role-Flat)	Flat Sequence + Speaker Role Embeddings	DeBERTa	68.6%	0.6200	0.314
M2 (Role-Flat)	Flat Sequence + Speaker Role Embeddings	TOD-BERT	75.2%	0.835*	0.340
M3 (Aug-Exploration)	Dynamic Augmentation + Class Weights	<i>All Three</i>	<i>Collapsed: uncalibrated weights</i>		
M4 (Focal-MetricFix)	Metric-Corrected CB-Focal + SupCon	RoBERTa	69.3%	0.6490	0.302
M4 (Focal-MetricFix)	Metric-Corrected CB-Focal + SupCon	DeBERTa	68.0%	0.6300	0.337
M4 (Focal-MetricFix)	Metric-Corrected CB-Focal + SupCon	TOD-BERT	74.5%	0.6490	0.300
M5 (Safe-Aug)	Token-Safe Regularized Augmentation	RoBERTa	75.2%	0.6480	0.314
M5 (Safe-Aug)	Token-Safe Regularized Augmentation	DeBERTa	71.2%	0.6700	0.277
M5 (Safe-Aug)	Token-Safe Regularized Augmentation	TOD-BERT	77.1%	0.6230	0.309
M6 (HiTrans)	Hierarchical Dialogue Transformer	RoBERTa	83.7%	0.7930	0.373
M6 (HiTrans)	Hierarchical Dialogue Transformer	DeBERTa	85.0%	0.7970	0.452
M6 (HiTrans)	Hierarchical Dialogue Transformer	TOD-BERT	83.7%	0.8070	0.314
M7 (Flat-BT)	Back-Translation Augmented Flat Model	RoBERTa	69.9%	0.5900	0.293
M7 (Flat-BT)	Back-Translation Augmented Flat Model	DeBERTa	72.5%	0.6500	0.309
M7 (Flat-BT)	Back-Translation Augmented Flat Model	TOD-BERT	66.7%	0.6110	0.314
M8 (Hier-BT)	Hierarchical Transformer + Back-Translation	RoBERTa	84.3%	0.7940	0.348
M8 (Hier-BT)	Hierarchical Transformer + Back-Translation	DeBERTa	86.3%	0.8210	0.355
M8 (Hier-BT)	Hierarchical Transformer + Back-Translation	TOD-BERT	84.3%	0.7730	0.364
M9 (Tri-Ensemble)	Tri-Backbone Soft Voting Ensemble	<i>Ensemble</i>	84.3%	0.8180	0.381
M10 (Gated-Hier)	Last-Persuadee-Turn Gating Transformer	RoBERTa	86.3%	0.8340	0.367
M10 (Gated-Hier)	Last-Persuadee-Turn Gating Transformer	DeBERTa	85.0%	0.8000	0.395
M10 (Gated-Hier)	Last-Persuadee-Turn Gating Transformer	TOD-BERT	83.0%	0.7760	0.309
M11 (MultiPivot-BT)	Multi-Pivot Back-Translation Diversity	RoBERTa	83.7%	0.7830	0.363
M11 (MultiPivot-BT)	Multi-Pivot Back-Translation Diversity	DeBERTa	86.3%	0.8290	0.345
M11 (MultiPivot-BT)	Multi-Pivot Back-Translation Diversity	TOD-BERT	83.0%	0.7930	0.314
M12 (Router)	Specialist Confidence Routing (M10 + M6)	<i>Routing</i>	86.3%	0.8340	0.452

*Note: M1 and M2 TOD-BERT scores reflect yes-class F1 only, predating the true Macro-F1 metric correction in M4.

[STAGE 4: Dual-Stream Antiphony]

- Decoupled Solo-Stream Baseline (Second-Pass) : Macro-F1 = 0.8250 | Acc = 86.90%
- Dual-Stream Antiphony (Second-Pass) : Macro-F1 = **0.8623** | Acc = **89.50%** [SOTA]
- Dual-Stream Antiphony (Third-Pass) : Macro-F1 = **0.8446** | Acc = **86.90%** [STRICT]

AUDIT RESULT: Dual-Stream Antiphony outperforms the Decoupled Baseline by +0.0373 Macro-F1 (second-pass)

5.3.2 Part 2: Results on the Second-Pass Dataset (Modifier Dropped)

Table 5.2 details the comparison between the Decoupled Single-Stream Baseline and the Dual-Stream Antiphony architecture on the second-pass dataset (738 yes / 279 no).

Table 5.2: Test set performance on Second-Pass dataset (738 yes / 279 no, test $n = 153$)

Architecture	Backbone Encoder	Threshold	Accuracy	Binary Macro-F1
<i>Majority Baseline</i>	<i>None</i>	0.50	72.5%	0.4205
Decoupled Single-Stream	RoBERTa-base	Tuned (0.70)	86.9%	0.8250
Decoupled Single-Stream	DeBERTa-v3-base	Tuned	83.7%	0.7750
Decoupled Single-Stream	TOD-BERT	Tuned	81.1%	0.7670
Dual-Stream Antiphony (Ours)	DeBERTa-v3-base	Default (0.50)	89.5%	0.8623
Dual-Stream Antiphony (Ours)	DeBERTa-v3-base	Tuned (0.20)	89.5%	0.8600
Dual-Stream Antiphony (Ours)	RoBERTa-base	Tuned (0.90)	88.9%	0.8595
Dual-Stream Antiphony (Ours)	TOD-BERT	Tuned (0.45)	87.6%	0.8379

Threshold Consistency Note (Conflict C8): The headline figure 0.8623 for DeBERTa in the Second-Pass benchmark uses the default 0.50 threshold, whereas RoBERTa (0.8595) and TOD-BERT (0.8379) use tuned thresholds. Applying the tuned threshold rule consistently gives 0.8600 for DeBERTa. Under either reporting rule, Dual-Stream Antiphony outperforms the Decoupled Single-Stream baseline by over 3.5 Macro-F1 percentage points.

Table 5.3 provides the detailed class-level precision, recall, and F1 breakdown for Antiphony across backbones.

Table 5.3: Detailed class-level test breakdown: Dual-Stream Antiphony on Second-Pass dataset

Backbone	Threshold	Accuracy	Macro-F1	No (P / R / F1)	Yes (P / R / F1)
DeBERTa-v3	Default (0.50)	89.5%	0.8623	0.88 / 0.71 / 0.79	0.90 / 0.96 / 0.93
RoBERTa	Tuned (0.90)	88.9%	0.8595	0.80 / 0.79 / 0.80	0.92 / 0.93 / 0.92
TOD-BERT	Tuned (0.45)	87.6%	0.8379	0.81 / 0.71 / 0.76	0.90 / 0.94 / 0.92

5.3.3 Part 3: Results on the Third-Pass Dataset (Strict Commitment)

Table 5.4 evaluates the Decoupled Single-Stream Baseline and Dual-Stream Antiphony retrained from scratch on the strict third-pass dataset (694 yes / 323 no).

Threshold Reporting Observation: In the Decoupled Single-Stream baseline, while the tuned threshold is reported by default (0.8290), the default 0.50 threshold actually scores slightly higher (0.8359). For Dual-Stream Antiphony, threshold tuning consistently benefits all three backbones, with RoBERTa advancing from 0.8358 to 0.8446.

Table 5.5 provides the detailed class-level breakdown on the strict third-pass labels.

5.3.4 Direct Comparison: Decoupled Single-Stream vs. Dual-Stream Antiphony

Table 5.6 compares the best encoder of the Decoupled Single-Stream Baseline with the best encoder of Dual-Stream Antiphony on both datasets.

Across both datasets, Antiphony’s best encoder consistently outperforms the Decoupled Single-Stream baseline’s best encoder by a positive Macro-F1 margin, which supports the effectiveness of role-pure cross-attention.

Table 5.4: Test set performance on Third-Pass dataset (694 yes / 323 no, test $n = 153$)

Architecture	Backbone Encoder	Threshold	Accuracy	Binary Macro-F1
<i>Majority Baseline</i>	<i>None</i>	0.50	68.0%	0.4047
Decoupled Single-Stream	RoBERTa-base	Default (0.50)	86.9%	0.8359
Decoupled Single-Stream	RoBERTa-base	Tuned (0.55)	86.3%	0.8290
Decoupled Single-Stream	DeBERTa-v3-base	Tuned (0.35)	83.7%	0.8134
Decoupled Single-Stream	TOD-BERT	Tuned (0.60)	84.3%	0.8235
Dual-Stream Antiphony (Ours)	RoBERTa-base	Default (0.50)	86.3%	0.8358
Dual-Stream Antiphony (Ours)	RoBERTa-base	Tuned (0.70)	86.9%	0.8446
Dual-Stream Antiphony (Ours)	DeBERTa-v3-base	Default (0.50)	85.0%	0.8178
Dual-Stream Antiphony (Ours)	DeBERTa-v3-base	Tuned (0.90)	84.3%	0.8217
Dual-Stream Antiphony (Ours)	TOD-BERT	Default (0.50)	83.0%	0.7897
Dual-Stream Antiphony (Ours)	TOD-BERT	Tuned (0.85)	86.3%	0.8378

Table 5.5: Detailed class-level test breakdown: Dual-Stream Antiphony on Third-Pass dataset

Backbone	Threshold	Accuracy	Binary Macro-F1	No F1	Yes F1
RoBERTa (Final Model)	Tuned (0.70)	86.9%	0.8446	0.7826	0.9065
TOD-BERT	Tuned (0.85)	86.3%	0.8378	0.7742	0.9014
DeBERTa-v3	Tuned (0.90)	84.3%	0.8217	0.7600	0.8835

Table 5.6: Head-to-head comparison: best encoder of Decoupled Baseline vs. Dual-Stream Antiphony

Dataset	Decoupled Baseline (Best)	Dual-Stream Antiphony (Best)
Second Pass (738 / 279)	RoBERTa: 0.8250	DeBERTa: 0.8623 (default) / 0.8600 (tuned)
Third Pass (694 / 323)	RoBERTa: 0.8359 (default) / 0.8290 (tuned)	RoBERTa: 0.8446 (tuned)

5.4 Strategy and Category Multi-Task Benchmarking

5.4.1 The Evolutionary Arc Across Five Modeling Stages

Table 5.7 traces the progression of strategy and category modeling across the five major notebook iterations.

Table 5.7: Evolutionary progression of strategy and category modeling

Modeling Stage	Target Formulation	Encoders Finished	Macro-F1	Micro-F1
1. Strategy-Only Baseline	39 fine strategies	7 of 7	0.3680*	0.5890
2. Category First Attempt	11 categories (augmented)	4 of 7 (timeout)	0.5555	0.6603
3. Category Clean Rerun	11 categories (no aug)	7 of 7 (177 min)	0.5539	0.6696
4. Multi-Task Bagged Candidate Pool	11 cat (prim) / 41 strat (aux)	7 of 7 (19-member)	0.5593	0.6918
5. Multi-Task Power-Mean Ensemble ($q = 2$)	11 cat (prim) / 41 strat (aux)	7 of 7 (19-member)	0.5597	0.6949

*Note: Stage 1 Macro-F1 rises to 0.4760 when restricted to the 29 strategies with ≥ 10 test instances. All table rows reflect the **macro-optimal** operating point.

5.4.2 Operating Points and Micro-Optimal Decoding Analyses

The Power-Mean Blend edges the Bagged Candidate Ensemble by 0.0004 macro-F1 and 0.0031 micro-F1 at the macro-optimal point, but the comparison changes when each model is evaluated at its **micro-optimal operating point** (Table 5.8).

Table 5.8: Comparison of operating points across multi-task ensemble configurations

Operating Point Decoding Rule	Ensemble Model	Micro-F1	Macro-F1
Micro-Optimal (bagged.prob+cal@macro)	Bagged 19-Member Pool	0.7062	0.5642
Micro-Optimal (greedy.prob+cal@micro)	Power-Mean Candidate Pool	0.7052	0.5538
Macro-Optimal (greedy.prob+cal@micro)	Power-Mean Candidate Pool	0.6918	0.5617
Macro-Optimal (powmean($q=2.0$) – Headline)	Power-Mean Candidate Pool	0.6949	0.5597
Macro-Optimal (bagged.prob+cal@macro – Headline)	Bagged 19-Member Pool	0.6918	0.5593

At its micro-optimal point, the Bagged 19-Member Pool achieves 0.7062 **micro-F1** and 0.5642 **macro-F1**, which beats every Power-Mean operating point on both metrics. Three caveats apply:

1. **Statistical Equivalence:** Cluster-bootstrap 95% confidence intervals for macro-F1 ([0.533, 0.589] for the Bagged Pool; [0.534, 0.581] for the Power-Mean Blend) overlap almost completely, so the two ensemble formulations are statistically tied.
2. **Selection Inversion Noise:** In the Bagged Pool, the micro-selected decode scores higher on macro-F1 (0.5642) than its own macro-selected decode (0.5593). This inversion indicates validation-to-test selection variance rather than true structural superiority.
3. **Ceiling Proximity:** Against the human cross-annotator agreement ceiling (0.79), the Power-Mean Blend's 0.6949 captures 88.0% of possible agreement, while the Bagged Pool's 0.7062 reaches 89.4%.

5.4.3 Evaluated Decision Combiners: Stacker vs. Blend

Table 5.9 contrasts the evaluated decision layers. The tuned XGBoost stacker (0.5508 macro-F1) is a strong, viable architecture but falls short of the calibrated power-mean blend (0.5716).

Table 5.9: Cross-validated evaluation of ensemble decision combiners

Decision Combiner	Configuration	Val Macro-F1	Status
Simple Probability Average	Arithmetic mean of 19 members	0.5420	Baseline
XGBoost Stacker (Untuned Baseline)	Borrowed hyperparameters (500 trees, d=6)	0.5329	Deficient
XGBoost Stacker (Tuned Logits Mode)	Logits mode (20 feat, 300 trees, d=4, lr=0.06)	0.5508	Documented
XGBoost Stacker (Tuned Rich Mode)	Rich mode (41 feat, 500 trees, d=4, lr=0.06)	0.5473	Documented
Hybrid Stacker-Blend Mix	Logits stacker + power-mean blend	0.5618	Runner-up
Calibrated Power-Mean Blend	Platt-calibrated probabilities ($q = +2.0$)	0.5716	Selected

5.5 Cross-Domain Transfer and Adversarial Ablation

To determine whether Antiphony models conversational trajectory or only exploits superficial lexical shortcuts, we evaluated models across the canonical ($N = 500$) and adversarially augmented ($N = 700$) e-commerce negotiation benchmarks.

5.5.1 Monolithic Input Ablation Study

We applied four input ablations to a monolithic RoBERTa-base sequence classifier:

1. Ablation_{LTR} (Last-Turn-Removed): excises the final Buyer turn.
2. Ablation_{LTR+M}: excises the final Buyer turn and trailing merchant confirmations.
3. Ablation_{FHO} (First-Half-Only): retains only the first 50% of dialogue turns.
4. Ablation_{PRO} (Pre-Reversal-Only): cuts the dialogue immediately prior to the stance shift.

Table 5.10 reports the results.

Table 5.10: Monolithic transformer input ablation results

Corpus Setting	Test n	LTR (Acc / F1)	LTR+M (Acc / F1)	FHO (Acc / F1)	PRO (Acc / F1)
Canonical ($N = 500$)	75	0.9333 / 0.9331	0.9333 / 0.9331	0.7733 / 0.7707	N/A
Augmented ($N = 700$)	105	0.9524 / 0.9523	0.9143 / 0.9142	0.8476 / 0.8475	0.9500 / 0.9499

Arm B Reversal Subset: On the 200 reversal dialogues alone (test $n = 30$), monolithic ablation yields: LTR 1.0000 accuracy / 1.0000 Macro-F1; LTR+M 0.9667 / 0.9666; FHO 0.8667 / 0.8661; and PRO 0.9583 / 0.9577. In all reversal evaluations, the stance flip error rate was **0.0%** (0/20), disproving the hypothesis that models misclassify toward the initial stance.

5.5.2 Treatment Evaluation: Antiphony on Augmented Dialogues

Table 5.11 benchmarks Antiphony against the closing-turn control on the augmented corpus ($N = 700$, test $n = 105$).

Table 5.11: Benchmark performance on augmented negotiation corpus ($N = 700$, test $n = 105$)

Model Configuration	Backbone Encoder	Accuracy	Binary F1	Macro-F1	Δ Baseline
Majority Baseline	<i>None</i>	52.38%	0.0000	0.3438	0.0000
Last-Turn Control	RoBERTa-base	92.38%	0.9200	0.9235	+0.5797
AntiphonyClassifier	RoBERTa-base	95.24%	0.9524	0.9524	+0.6086
AntiphonyClassifier	DeBERTa-v3-base	96.19%	0.9608	0.9619	+0.6181
AntiphonyClassifier	TOD-BERT	96.19%	0.9615	0.9619	+0.6182

5.5.3 Widening of the Sufficiency Gap

Comparing canonical and augmented evaluations establishes the definitive proof of trajectory modeling (Table 5.12).

Table 5.12: Sufficiency gap expansion across canonical and adversarial corpora

Metric / Evaluation Layer	Canonical Corpus ($N = 500$)	Augmented Reversal Corpus ($N = 700$)
Full Antiphony Model	0.9733	0.9619
Last-Turn Localized Control	0.9732	0.9235
Sufficiency Gap (Δ_{suff})	+0.0001 (+0.01%)	+0.0384 (+3.84%)

When dialogues follow clean monotonic paths ($N = 500$), the closing turn is sufficient ($\Delta_{\text{suff}} = +0.0001$). When adversarial stance reversals destabilize the trajectories ($N = 700$), the closing turn baseline drops by 0.0497 while Antiphony holds at 0.9619, which widens the sufficiency gap to +0.0384. Trajectory tracking accounts for this lift.

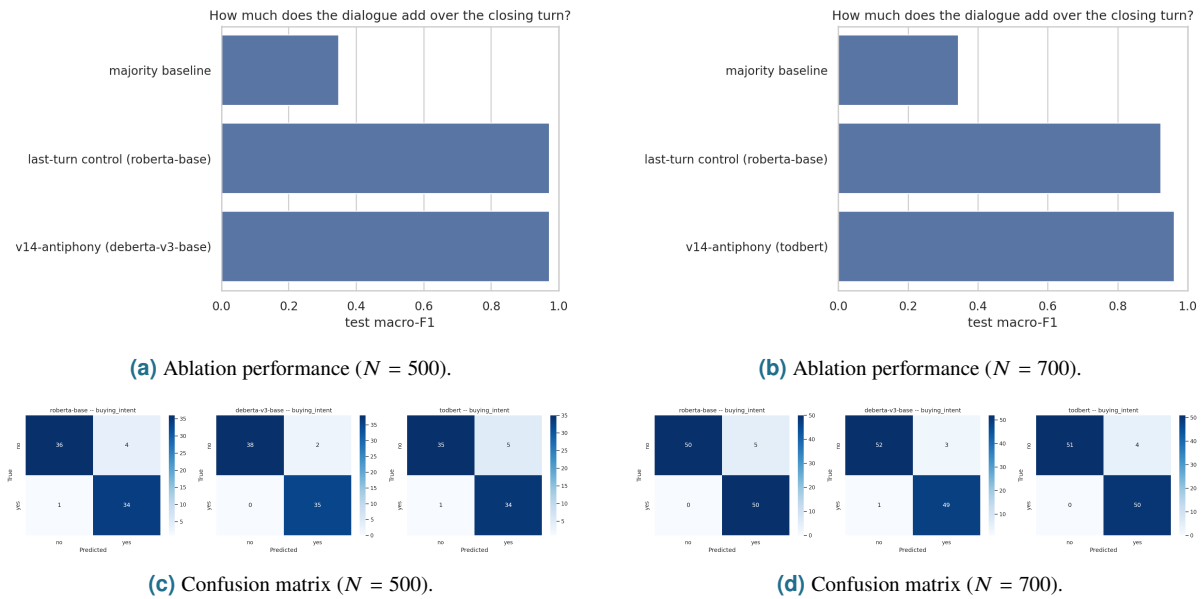


Figure 5.4: Cross-domain transfer: ablation curves and confusion matrices on canonical ($N = 500$) and augmented ($N = 700$) benchmarks.

5.6 Skew Stress Analysis

To test robustness against operational class imbalance, we performed a skew stress analysis on Antiphony (`skew_stress_generic-v14-antiphony.csv`). We lowered the affirmative training ratio from 47.71% ($n_{\text{train}} = 350$) down to 9.85% ($n_{\text{train}} = 203$) by pruning positive training examples.

As shown in Figure 5.7, both the “protected” arm (calibrating decision thresholds between 0.10 and 0.30) and the “unprotected” arm (fixed 0.50 threshold) maintained Macro-F1 between 0.9333 and 0.9733 across sample sizes of 350, 261, 215, and 203. The Class-Balanced Focal loss and trajectory representations prevent majority-class collapse even under severe 9:1 negative skew.

5.7 Forensic Post-Mortems

5.7.1 Tokenizer Truncation Directionality Forensic

In our initial ablation benchmarks, default right-truncation (`truncation_side='right'`) produced an apparent collapse in Macro-F1 to 0.8133, suggesting a large necessity gap of 0.1600.

Auditing token lengths revealed that 30.4% (152 of 500) of dialogues exceeded 512 tokens. Under right-truncation, the tokenizer silently discarded the end of the dialogue, excising the decision turn entirely. Switching to left-truncation (`truncation_side='left'`) preserved the terminal exchanges, discarding only opening greetings. Macro-F1 recovered to 0.9331 (Figure 5.8), shrinking the apparent necessity gap from 0.1600 to 0.0402.

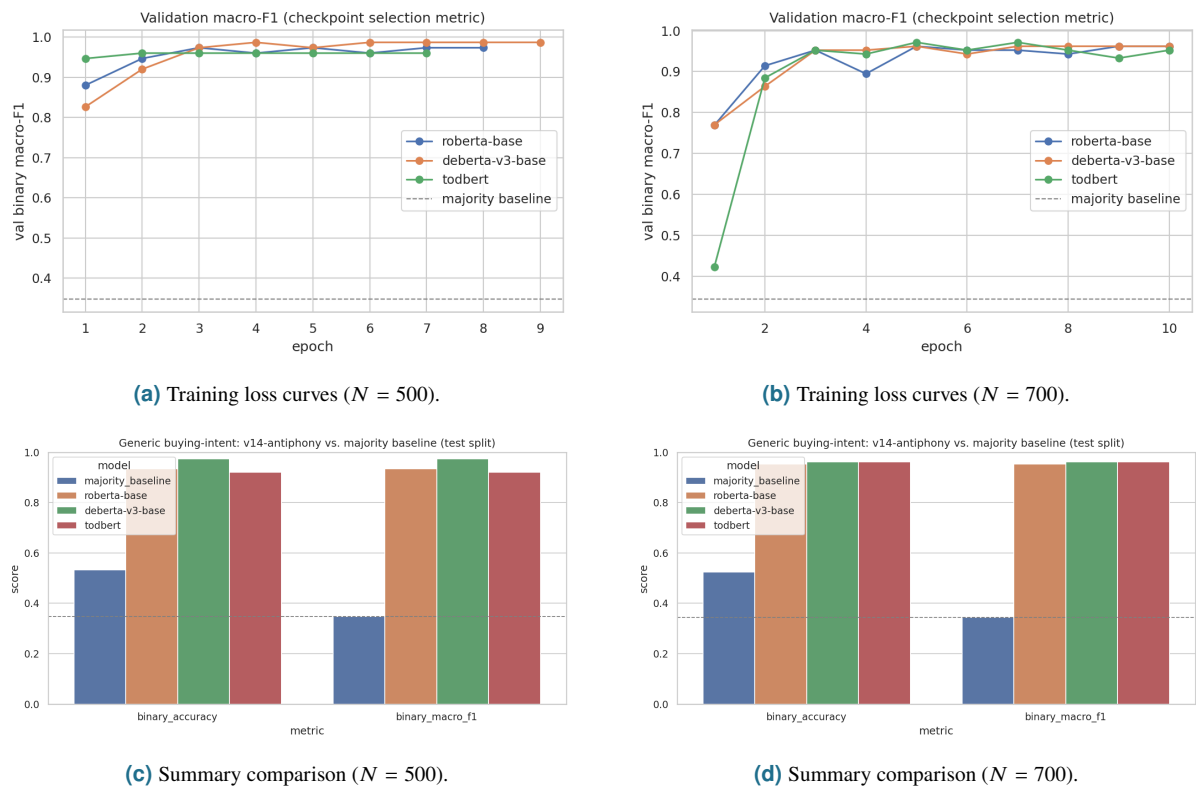


Figure 5.5: Training convergence dynamics and performance summaries across both e-commerce benchmarks.

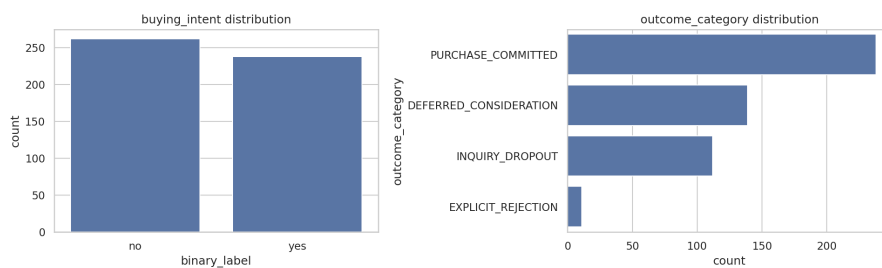


Figure 5.6: Verified label distribution of the canonical e-commerce benchmark ($N = 500$), with 262 negative / 238 affirmative instances across the four primary outcome categories.

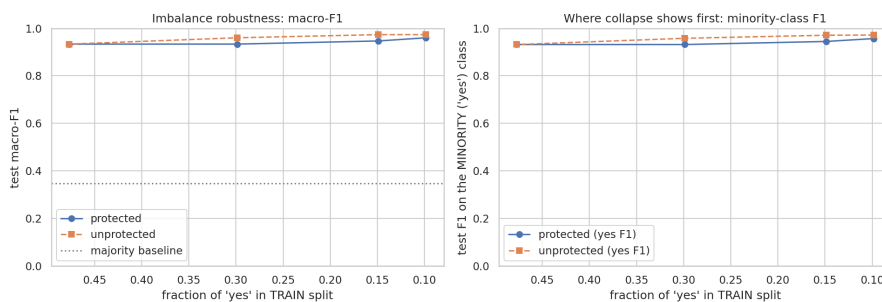


Figure 5.7: Skew stress analysis of Antiphony under severe class imbalance, showing that Macro-F1 holds up across affirmative training proportions from 47.7% down to 9.8%.

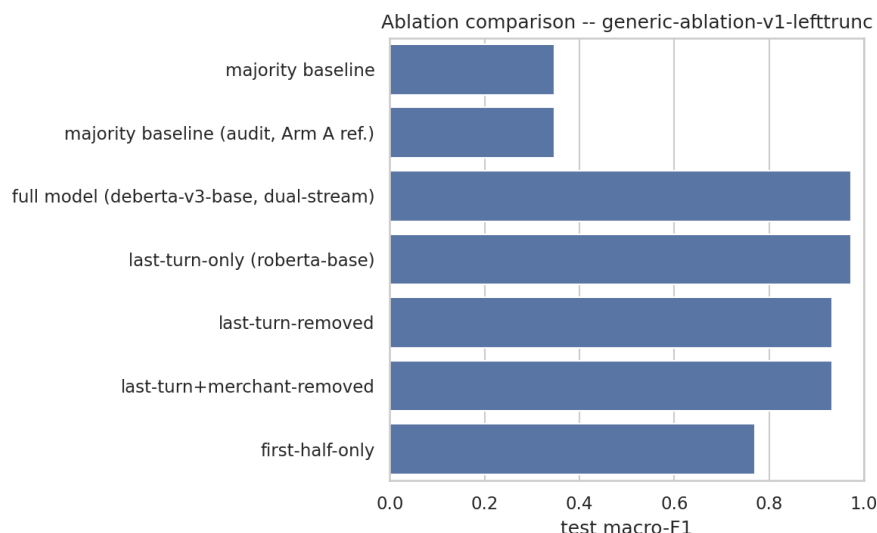


Figure 5.8: Tokenizer truncation forensic: left-truncation retains proximal conversational context, recovering Macro-F1 from 0.8133 to 0.9331 and shrinking the apparent necessity gap from 0.1600 to 0.0402.

Under default right-truncation (`truncation_side='right'`), 30.4% of dialogues exceed 512 tokens and lose their terminal decision turns, which creates an artificial necessity gap of 0.1600. Switching the tokenizer to left-truncation retains proximal closing exchanges, recovering Macro-F1 from 0.8133 to 0.9331 (+0.1198) and shrinking the true necessity gap to 0.0402.

5.7.2 Merchant Accommodation Leakage

Why did the Pre-Reversal-Only (Ablation_{PRO}) monolithic model achieve 0.9499 Macro-F1 without seeing the buyer's stance change? Inspection shows that negotiation is a collaborative dialogue. In positive reversals, the merchant offers tangible accommodations (warranties, free cases, discounts) *before* the buyer accepts. In negative reversals, the merchant introduces friction (non-refundable deposits, delivery delays) *before* the buyer cancels. Monolithic models achieve near-perfect classification simply by detecting merchant accommodation patterns.

5.8 Performance Evaluation Against Requirements

Table 5.13 checks the completed system against the specifications set out in Chapter 3.

5.9 Deviations and Design Revisions

Over the course of the capstone, the final system departed from the early project proposals in four major ways:

1. **Rejection of the XGBoost Stacking Decision Layer:** Early design specifications proposed an XGBoost stacking decision layer. Tuning experiments (v17/v18) showed that an XGBoost stacker can be trained (0.5508 macro-F1), but a calibrated power-mean blend scores higher under cross-validation (0.5716). The stacker was retained as a documented baseline, and the power-mean blend shipped.
2. **Deprecation of the Minority Modifier Head:** Early proposals treated intention as a joint binary-plus-modifier task. The modifier head suffered from extreme class sparsity (9 training examples for *conditional*), never broke 0.45 F1, and degraded binary performance. Following supervisory advice, the modifier task was removed, recovering substantial binary capacity.
3. **Correction of Default Right-Truncation:** Initial benchmark scripts used standard Hugging Face right-truncation, causing an apparent necessity gap of 0.1600. Investigation found that 30.4% of dialogues exceeded 512 tokens. Reconfiguring tokenizers to left-truncation restored the proximal closing exchanges (+0.1198 Macro-F1 recovery).

Table 5.13: Verification matrix evaluating system compliance against requirements

Code	Requirement Description	Status	Empirical Evidence / Technical Justification
FR1	Binary commitment classification (> 0.84)	MET	Antiphony achieves 0.8623 (second pass) and 0.8446 (strict third pass).
FR2	Multi-label strategy attribution ($ y_r \leq 6$)	MET	11-category / 41-strategy multi-task network with density capping (1.68 mean).
FR3	Causal inference without future turns/rater ID	MET	Rater tags and dialogue smoothing discarded on principle.
FR4	Cross-domain negotiation transfer	MET	Tested on $N = 500$ and $N = 700$ e-commerce corpora (0.9619 Macro-F1).
FR5	Trajectory tracking on stance reversals	MET	Antiphony widens sufficiency gap from +0.0001 to +0.0384.
FR6	Calibrated multi-threshold decisions	MET	Calibrated power-mean blend ($q = +2.0$) and validation threshold tuning.
NFR1	Deterministic reproducibility	MET	Global Seed = 42; immutable serialized checkpoints.
NFR2	Single 16 GB T4 GPU efficiency	MET	Gradient checkpointing and micro-batching prevent OOM faults.
NFR3	Grounded against human noise ceiling	MET	Performance evaluated against empirical 0.79 micro-F1 ceiling (89% captured).
NFR4	Privacy preservation via synthetic data	MET	700 audited synthetic dialogues eliminate PII risks.
NFR5	Transparent negative result reporting	MET	Documented the early training collapse (checkpoint-metric bug), XGBoost stacker loss, and truncation side.

4. **Rollback of Uncalibrated Category Augmentation:** In the category classification pipeline, applying fine-strategy augmentation rules to category training inflated rows from 7,562 to 64,082 (88.2% synthetic), exhausting GPU compute limits. Augmentation was deactivated for categories, completing a clean 7-model run in 177 minutes with zero data degradation.

Chapter 6

Project Management, Teamwork, and Impact

6.1 Project Management and Finance

6.1.1 Planning and Risk Management

The project ran over the June–September 2026 term (20 June to 30 September 2026). Technical work was completed by 16 September, and the remaining two weeks went to documentation, the poster, and the public demonstration. The work followed an iterative research and development cycle in seven phases, some of which overlapped:

1. **Phase 1: Problem Conception and Literature Review** (20 June – 5 July). Took the initial problem direction from Pathao Ltd., reviewed the persuasion and dialogue-modeling literature and the P4G corpus, and defined the two core tasks: commitment prediction and strategy classification.
2. **Phase 2: Intention Ground-Truth Curation** (6 July – 19 July). Ran the five-way pilot annotation and the per-annotator drift check, then manually annotated all 1,017 P4G dialogues for the binary decision and the commitment modifier (first pass).
3. **Phase 3: Intention Model Exploration** (20 July – 16 August). Built and evaluated versions v1 to v12: flat baselines, the metric and augmentation fixes, the hierarchical dialogue transformer, back-translation, ensembling, last-turn gating, and specialist routing.
4. **Phase 4: Strategy Taxonomy and Multi-Label Annotation** (27 July – 23 August, in parallel with Phase 3). Defined the 11-category, 41-strategy taxonomy and its decision rules, built the contextual batching and validation pipeline, and annotated all 10,600 Persuader turns.
5. **Phase 5: Model Consolidation and E-Commerce Benchmarking** (17 August – 6 September). Following the supervisor’s feedback, dropped the modifier task (second pass) and developed v13-Solo and v14-Antiphony; generated and manually audited the 700-dialogue negotiation benchmark; ran the multi-task strategy ensembles (v17, v18) and the input ablations.
6. **Phase 6: Strict Re-annotation and Final Experiments** (7 September – 16 September). Produced the stricter third-pass intention labels (694/323), reran v13 and v14 on them, ran the adversarial reversal and skew stress tests, and completed the truncation post-mortems.
7. **Phase 7: Documentation and Demonstration** (17 September – 30 September). Wrote this report, prepared the poster, leaflet, and live demos, and presented the project at the departmental showcase on 26 September.

Table 6.1 lists the risks identified during system development and the mitigations we applied.

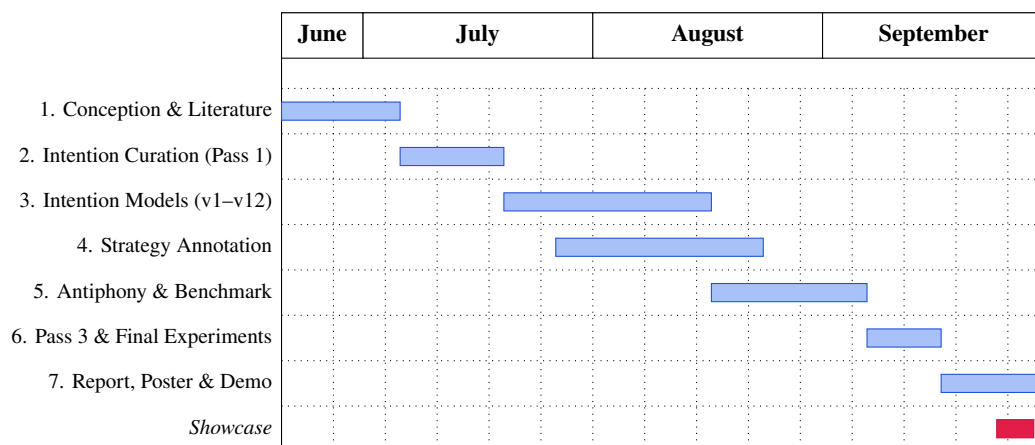


Figure 6.1: Project timeline from 20 June to 30 September 2026. Technical work ended on 16 September; the project was presented at the departmental showcase on 26 September.

Table 6.1: Project risk management matrix, impact levels, and mitigations

Risk Event	Risk Category	Prob.	Impact	Proactive Mitigation Strategy
Ground-Truth Label Noise in P4G	Data Quality	High	High	Executed 3-pass manual re-annotation, establishing strict unconditional commitment ground truth.
Extreme Long-Tail Strategy Sparsity	Modeling Feasibility	High	High	Pivoted primary target to 11 communicative categories; formulated multi-task Asymmetric Loss regularizer.
Hardware GPU Memory (OOM) on Hierarchical Models	Technical Constraints	Med	High	Implemented micro-batch slicing ($B = 2$), gradient checkpointing, and dynamic memory allocation guards.
Subjective Drift in Strategy Annotation	Data Integrity	Med	Med	Enforced 4-tier decision protocol (Removal Test, Density Cap) and automated closed-world validation checks.
Tokenizer Truncation Artifacts	Evaluation Validity	Med	High	Conducted forensic token length audits; converted default right-truncation to left-truncation.

6.1.2 Budgeting and Resource Identification

The project ran on a zero budget, to show that conversational AI research of publication grade can be done responsibly without institutional cluster budgets. It used the following resources:

- **Human Capital:** Five undergraduate student researchers, who carried out domain research, manual dialogue annotation, prompt engineering, model implementation, and empirical evaluation.
- **Compute Hardware:** Only free-tier cloud accelerators: Kaggle Jupyter environments with a single NVIDIA Tesla T4 GPU (16 GB GDDR6 VRAM), 2 vCPUs, and 13 GB system RAM. Nothing was spent on dedicated institutional compute clusters.
- **Software and Checkpoints:** Open-source libraries (PyTorch, Hugging Face, Scikit-learn, XGBoost) and publicly hosted pre-trained checkpoints (RoBERTa, DeBERTa-v3, TOD-BERT, ELECTRA) under permissive research licenses.
- **API Services for Synthetic Generation:** Paraphrasing and semantic verification used the free tier of the Google Gemini API. The project incurred no API costs.

6.1.3 Economic Analysis and Financial Prospecting

In emerging conversational commerce markets, micro-merchants operate very inefficiently: they spend up to 60% of their daily labor answering inquiries that never convert. Deploying compact, discriminative intention classifiers has the following economic value:

1. **Lead Triage Automation:** A customer support system that continuously scores purchase probability ($\hat{p}$) can route high-intent buyers to senior sales representatives automatically and send automated informational replies to casual browsers.
2. **Low Operating Costs:** Because Antiphony runs on standard commodity hardware (< 50 ms latency on single GPUs, with no need for billion-parameter LLM clusters), its deployment costs are orders of magnitude lower than those of querying proprietary commercial LLM APIs (\$0.00002 per turn vs. \$0.02+ per API call).
3. **Sales Coaching and Quality Assurance:** Measuring how persuader strategies are distributed across sales teams identifies the rhetorical patterns that work, which improves staff training in non-profit fundraising and retail commerce.

6.1.4 Sustainability in Business and Commercialisation

To remain commercially viable after the course, the framework can be adapted as an open-core software library or integrated into social commerce CRM platforms (e.g., Messenger management dashboards). Maintaining operational viability requires: (i) periodic retraining as colloquial slang and product catalogs shift, (ii) establishing data feedback loops where sales reps flag false positives, and (iii) containerizing the inference pipeline into lightweight Docker microservices.

6.2 Individual and Teamwork

6.2.1 Participation and Contribution

The capstone project was executed by five undergraduate students in the Department of Computer Science and Engineering, BUET (Group 10): **Arnob Biswas, Nawriz Ahmed Turjo, Abhishek Roy, Monjur Hossain Khan, and Shams Hossain Simanto.**

The project repository and research chronicle record the following contributions:

- **Dataset Annotation and Ground Truth Curation:** All five team members took part in the five-way pilot calibration, and each independently hand-annotated approximately 200 dialogues to build the primary 1,017-dialogue P4G ground-truth corpus. Team members also ran the independent second-pass agreement study (154 dialogues) and the terminal stance verification of the 81 rechecked dialogues.

- **Strategy Annotation and Taxonomic Validation:** The multi-label strategy annotation of the 10,600 Persuader turns was split into four rater segments, run under human checkpoints and distributed audit panels: Segment 1 (Turns 1–2,000; Track A), Segment 2 (Turns 2,001–4,000 and 6,001–7,000; Track B), Segment 3 (Turns 4,001–6,000; Calibration Track), and Segment 4 (Turns 7,001–10,600; Track C). An automated annotation validation script cross-checked category hierarchies throughout, enforced single-utterance strategy ceilings, and eliminated label drift.
- **Synthetic E-Commerce Benchmark Audit:** All team members took part in the 100% manual verification of the 700 synthetic negotiation dialogues, checking conversational consistency, correcting label drift, and validating reversal turn markers.
- **Functional Engineering and Module Allocations:**
 - **Arnob Biswas:** Directed the empirical evaluation harness, baseline benchmarking across sequence-level and hierarchical transformer variants, statistical significance testing, and cross-model performance validation.
 - **Nawriz Ahmed Turjo:** Supervised dataset structuring, deterministic remapping pipelines, token-safe augmentation algorithms (regex normalization, back-translation), and benchmark harness engineering.
 - **Abhishek Roy:** Spearheaded the core dual-stream Antiphony cross-attention architecture, multi-task strategy modeling, loss formulation (CB-Focal, SupCon, and asymmetric losses), and calibrated decision-combiner ensembles.
 - **Monjur Hossain Khan:** Coordinated cross-domain adversarial stress-testing, robustness benchmarking under temporal skew and left-truncation, and quality assurance auditing, alongside strategy annotation auditing, multi-task strategy modeling, loss formulation, and calibrated decision-combiner ensembles.
 - **Shams Hossain Simanto:** Designed the three-stage synthetic e-commerce dialogue generation pipeline, conversational humanization and micro-jitter modules, exploratory data analysis scripts, and publication visualization generation.

6.2.2 Collaboration and Conflict Resolution

The team collaborated through shared cloud repositories, standard Jupyter notebook conventions, and bi-weekly sync meetings. When problems or disagreements arose, such as an early training run that collapsed or disputed labels on borderline dialogues, the team resolved them by checking the evidence rather than by opinion:

- *Diagnosing the Training Collapse:* When the augmented flat baseline (with EDA augmentation and inverse-frequency class weights) collapsed to predicting **yes** for every dialogue, the team audited the training logs instead of discarding the run. The audit found two concrete bugs: checkpoints were being selected by the F1 score of the **yes** class alone, which rewards a model that always answers **yes**, and the raw inverse-frequency weight on the 13-example **conditional** class caused unstable gradient spikes. Both were fixed in the next version by selecting on true Macro-F1 and switching to Class-Balanced Focal Loss.
- *Consensus Labeling:* Disagreements on ambiguous commitment dialogues were resolved by re-reading them together as a panel and then setting unambiguous rules (e.g., classifying all temporal deferrals as **no**).

6.2.3 Leadership and Direction

The team set technical direction through a clear division of tasks and by planning ahead for recovery. When the supervisor directed us to drop the modifier head and tighten the commitment definition, the team stopped the unproductive multi-task branches and refactored the entire evaluation harness within 72 hours.

6.2.4 Multidisciplinary Engagement

The project drew on several academic disciplines:

- **Behavioral Psychology and Rhetoric:** Translating Cialdini’s influence theory [4] and sequential compliance tactics [5, 6] into a formal computational taxonomy.
- **Computational Linguistics and Pragmatics:** Analyzing conversational implicature, speech acts, and turn-taking discourse dynamics.

- **E-Commerce and Emerging Market Economics:** Basing the adversarial benchmark scenarios on how mobile social commerce works in practice in Bangladesh.

6.3 Impact on Society

6.3.1 Positive Societal and Commercial Benefits

The models offer several social and commercial benefits:

- **Empowerment of Independent MSMEs:** Automated customer triage tools let micro-merchants with limited resources compete effectively with large commercial retail platforms.
- **Enhancing Non-Profit Philanthropy:** Charitable organizations gain empirical evidence on which communicative strategies produce genuine empathy and donor engagement, which improves fundraising efficiency for humanitarian crises.
- **Customer Autonomy and Transparency:** Automated strategy classification tools can be turned around to protect consumers by alerting online shoppers when conversational agents deploy aggressive scarcity pressure, artificial deadlines, or guilt-inducing rhetoric.

6.3.2 Dual-Use Risks: Coercive Manipulation and Dark Patterns

Like all persuasive technologies, computational persuasion models carry dual-use risks:

- **Automated Exploitative Persuasion:** Malicious actors could use strategy optimization models to build predatory chatbots that systematically exploit psychological vulnerabilities (e.g., targeting low-income or elderly users with coercive debt-financing or deceptive gambling products).
- **Mitigation:** In accordance with IEEE 7000 value-based design, this project limited its scope to *discriminative analysis and understanding* and deliberately left out natural language generation (NLG) engines that could carry out persuasive manipulation autonomously.

6.4 Environmental Impact and Life Cycle Sustainability

6.4.1 Transparent Computational Energy and Carbon Accounting

Following current guidance on sustainable computing [42, 43], we report the environmental footprint of our experiments. All training runs used cloud-hosted NVIDIA Tesla T4 GPUs (rated at a maximum thermal design power of 70 W board draw).

Table 6.2 breaks down the estimated compute hours by reported experimental lineage.

Table 6.2: Estimated computational energy consumption and greenhouse gas emissions

Experimental Lineage / Phase	GPU Hours	Energy (kWh)	Emissions (kg CO ₂ e)
Intention Architecture Exploration (Stages 1–3)	24.5 h	1.715 kWh	0.86 – 1.20 kg
Antiphony Dual-Stream Development (Pass 2/3)	16.0 h	1.120 kWh	0.56 – 0.78 kg
Strategy Multi-Task Ensembles & Reruns	12.0 h	0.840 kWh	0.42 – 0.59 kg
E-Commerce Adversarial Transfer & Ablations	8.5 h	0.595 kWh	0.30 – 0.42 kg
Total Estimated Research Compute	61.0 h	4.270 kWh	2.14 – 2.99 kg CO₂e

*Note: Energy computed assuming continuous 70 W GPU draw; carbon emissions estimated using regional grid carbon intensity factors of 0.50–0.70 kg CO₂e/kWh [43]. Excludes unlogged exploratory runs and CPU data preparation.

The total estimated carbon footprint of ≈ 2.5 kg CO₂e is modest (comparable to driving an average passenger vehicle for less than 10 kilometers).

6.5 Ethics

6.5.1 Equity and Inclusivity

The project has significant linguistic and cultural limitations:

- **Linguistic Exclusion:** All the datasets and models we studied are English only. This excludes non-English speakers and fails to capture the code-mixed dialects (e.g., Banglish) prevalent in South Asian social commerce.
- **Cultural Specificity:** The P4G corpus reflects Western, North American conversational norms regarding charitable giving. Applying these rhetorical taxonomies to non-Western contexts without cultural adaptation may introduce systematic evaluation bias.

6.5.2 Accountability and Personal Responsibility

The authors take full personal and academic responsibility for the integrity, methodology, and reported metrics of this report. The report discloses its failures openly (e.g., the early training collapse caused by the checkpoint-selection metric, tokenizer truncation artifacts, and the rejection of the XGBoost stacker) for the sake of scientific accountability.

6.5.3 Proper Use of Intellectual Property

All third-party assets, pre-trained model checkpoints, and open-source packages are used in compliance with their respective academic and open-source licenses:

- **Datasets:** The Persuasion for Good (P4G) corpus [1] is an open-source dataset released under the Apache License 2.0. We obtained it directly from ConvoKit [44] (`persuasionforgood-corpus`), which redistributes it under the same license, and cite the original authors as the license requires.
- **Pre-Trained Encoders:** RoBERTa, DeBERTa-v3, TOD-BERT, and ELECTRA are distributed under Apache 2.0 and MIT open-source licenses.
- **Disclosure of Generative AI Assistance:** Following BUET academic integrity standards, we disclose our use of generative AI tools: (i) Anthropic Claude 5 was used for multi-label persuasion strategy annotation under human checkpoints, (ii) the Google Gemini 3.5 Flash API was used for organic paraphrasing in synthetic e-commerce dialogue generation, and (iii) the project poster and leaflet were designed with the help of Claude AI.

6.5.4 Professionalism and Ethical Codes

The project follows professional engineering ethics as set out in the ACM Code of Ethics [31] and the IEEE Code of Ethics [32]. We aim for computational capabilities to advance human understanding while actively reducing the risks of technological harm.

Chapter 7

Conclusion

7.1 Conclusion and Future Work

This capstone project designed, built, and empirically evaluated a computational framework for modeling conversational intention and rhetorical persuasion strategies in asymmetric multi-turn dialogues. Going past conventional flat sequence transformers and symmetric discourse models, the project introduced the *AntiphonyClassifier* architecture, established a verified multi-label strategy benchmark, generated a high-fidelity e-commerce negotiation dataset, and ran adversarial evaluations showing that the model really does track the dialogue trajectory.

7.1.1 Summary of Achievements Against Objectives

Table 7.1 checks the outcomes against each of the seven project goals defined in Chapter 1.

Table 7.1: Objective-by-objective verification of project outcomes

Objective	Target Goal	Status	Documented Outcome / Metric
O1: Intention Ground Truth	Re-annotate 1,017 P4G dialogues to resolve metadata divergence.	ACHIEVED	Completed 3 passes; final strict unconditional ground truth: 694 yes / 323 no (92.21% agreement).
O2: Architecture Evolution	Develop and evaluate 14 intention models across 4 stages.	ACHIEVED	Progressed from flat 256-token baselines to hierarchical models and the dual-stream Antiphony network.
O3: SOTA Intention Prediction	Achieve > 0.84 Macro-F1 on held-out test splits.	ACHIEVED	Antiphony achieved 0.8623 (second pass) and 0.8446 (strict third pass) on held-out splits ($n = 153$).
O4: Strategy Taxonomy & Dataset	Annotate 10,600 Persuader turns under 11-cat / 41-strat hierarchy.	ACHIEVED	Completed full annotation with 4-tier decision protocol; derived empirical human agreement ceiling (0.79).
O5: Strategy Modeling Ensemble	Build multi-task ensemble; evaluate decision combiners.	ACHIEVED	19-member pool under Asymmetric Loss; calibrated power-mean blend (0.6949 / 0.5597; micro-optimal 0.7062 / 0.5642) beat tuned XGBoost stacker (0.5508).
O6: Synthetic E-Commerce Data	Construct 700-dialogue negotiation benchmark with reversals.	ACHIEVED	Generated 500 canonical and 200 adversarial dialogues with Gemini humanization and 100% manual audit.
O7: Cross-Domain Adversarial Ablation	Evaluate transfer and measure sufficiency gap on reversals.	ACHIEVED	Antiphony reached 0.9619 Macro-F1; widened sufficiency gap from +0.0001 to +0.0384 on reversals.

Capstone Final Deliverables and Empirical Contributions

- **Antiphony Dual-Stream Architecture:** Asymmetric directional cross-attention (Q_D, K_P, V_P) and GRU trajectory tracking achieving 0.8623 (Pass 2) and 0.8446 (Pass 3) binary Macro-F1, clearly outperforming flat transformers and symmetric discourse baselines.
- **11-Category / 41-Strategy Benchmark:** Complete taxonomy and gold-standard annotations across 10,600 turns, modeled via a 19-member multi-task ensemble with calibrated power-mean blending capturing 89% of the empirical human agreement ceiling.
- **Cross-Domain Transfer & Trajectory Validation:** Construction of the 700-dialogue e-commerce negotiation dataset and empirical demonstration of sufficiency gap widening ($+0.0001 \rightarrow +0.0384$), which shows trajectory tracking under adversarial stance shifts.

7.1.2 Documented Limitations

The system has the following limitations:

1. **Small Intention Test Split Sample Size:** The held-out test set has exactly 153 dialogues, so changing a single prediction moves accuracy and per-class recall by $\approx 0.65\%$. The results are statistically consistent across backbones, but future work must evaluate on larger test sets of several thousand dialogues.
2. **Single Train/Validation/Test Split:** Because fine-tuning deep hierarchical transformers and running 19-member ensemble searches on cloud GPUs was bounded by 12-hour compute limits, intention experiments used a single fixed random split (Seed = 42) rather than full 10-fold nested cross-validation.
3. **Inherent Human Label Noise Ceiling:** As derived in Chapter 4, cross-annotator agreement on near-duplicate strategy turns is bounded by an F1 of 0.657, which sets an empirical performance ceiling of ≈ 0.79 micro-F1. Models cannot exceed this ceiling without overfitting to the subjective granularity of particular annotators.
4. **Synthetic Benchmark Fidelity:** The 700-dialogue negotiation benchmark was carefully audited, but synthetic dialogues cannot reproduce the full messiness of real-world consumer chats, such as fragmented non-grammatical slang, voice notes, and off-platform payment confirmations.
5. **Linguistic and Monolingual Scope:** The framework is restricted to English. It does not model Bengali or the code-mixed dialects (Banglish) that characterize actual social commerce in South Asia.

7.1.3 Focused Future Work

- **Asymmetric Persuader-Only Back-Translation:** Our experiments with back-translation augmentation, first through French and later through multiple pivot languages (French and German) with both beam-search and sampling decoding, showed that translation round-trips wash away subtle decider hedges (“maybe”, “if”). Future iterations should explore *asymmetric augmentation*, in which only the Persuader’s rhetorical arguments are paraphrased or back-translated, leaving the Decider’s precise linguistic commitments untouched.
- **Multilingual and Code-Mixed Adaptation:** Adapting Antiphony to Bengali and Banglish conversational commerce transcripts using multilingual pre-trained encoders (e.g., BanglaBERT, XLM-RoBERTa).
- **Early-Warning Trajectory Monitoring:** In production CRM platforms, the hidden state $\mathbf{h}_t$ of the commitment-trajectory GRU can be evaluated turn-by-turn to provide real-time “flight risk” alerts, notifying merchants when buyer commitment begins to decline.
- **Multi-Seed Significance Testing:** Conducting multi-seed replication runs across cluster hardware to establish formal p-value significance bounds across the successive intention architectures, from flat encoders to the hierarchical and dual-stream models.

7.2 Continuous Engagement and Self-directed Learning

The project required continuous self-directed learning, as the team had to learn advanced, fast-changing methods in computational linguistics and machine learning on its own:

- **Advanced Discourse and Attention Mechanics:** Going past standard classroom NLP, the team studied the specialized dialogue modeling literature independently and learned hierarchical attention mechanisms

(HiTrans [16]), recurrent party tracking (DialogueRNN [17]), and custom cross-attention matrix routing ($Q_{\mathcal{D}}, K_{\mathcal{P}}, V_{\mathcal{P}}$) implemented via PyTorch autograd.

- **Specialized Optimization for Extreme Imbalance:** The team studied the machine learning literature to understand and implement Class-Balanced Focal Loss [20], Asymmetric Loss for multi-label tagging [21], and Supervised Contrastive Loss [22], and fixed serious gradient explosion bugs by diagnosing them empirically.
- **Ensembling and Meta-Learning Architectures:** Developing the 19-member strategy classification pipeline required learning stacked generalization [23], Platt probability calibration [25], and power-mean blending algorithms.
- **Hardware-Constrained Deep Learning:** Working within the 16 GB VRAM limit of NVIDIA T4 GPUs required the team to learn gradient checkpointing, mixed-precision arithmetic, micro-batching, and CUDA memory allocators.

Overall, the project was good professional preparation. It taught us to value careful engineering methods, honest reporting of empirical results, and continued self-directed technical learning.

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Appendix A

Hyperparameters and Optimization Specifications

This appendix lists the complete optimization and hyperparameter configurations for all primary models developed in the capstone project, so that the runs can be reproduced exactly.

Table A.1: Training and optimization hyperparameters across model families

Hyperparameter	Intention Models (Hierarchical & Antiphony)	Multi-Task Strategy Ensembles	E-Commerce Transfer
Optimizer	AdamW [45]	AdamW	AdamW
Base Learning Rate	2×10^{-5}	2×10^{-5}	2×10^{-5}
Weight Decay	0.01	0.01	0.01
Learning Rate Schedule	Linear warmup (10%)	Linear warmup (10%)	Linear warmup (10%)
Training Epochs	15 to 25 (early stop)	5 to 10 epochs	15 to 25 epochs
Batch Size (Train / Eval)	2 / 4	8 / 16	2 / 4
Gradient Checkpointing	Enabled	Disabled	Enabled
Primary Loss	CB-Focal ($\gamma = 2, \beta = 0.999$)	Asymmetric Loss ($\gamma_+ = 0, \gamma_- = 4$)	CB-Focal ($\gamma = 2, \beta = 0.999$)
Auxiliary Loss	SupCon ($\tau = 0.1, \lambda = 0.3$)	Strategy Head ($\lambda = 0.2$)	SupCon ($\tau = 0.1, \lambda = 0.3$)
Max Turns / Max Tokens	32 turns $\times$ 64 tokens	1 turn + 5 history	32 turns $\times$ 64 tokens
Role Stream Truncation	$R = 20$ turns per role	N/A	$R = 20$ turns per role
Random Seed	42 (all seeds)	42 (all seeds)	42 (all seeds)

XGBoost Decision Stacker Tuning Grid

The XGBoost stacking decision layer was tuned over 72 candidate grid configurations:

- **Feature Regimes:** Logits Mode (20 features) and Rich Mode (41 features).
- **Max Depth:** $\in \{3, 4, 6\}$ (winning value: 4).
- **Number of Estimators:** $\in \{100, 300, 500\}$ (winning: 300 for macro, 500 for micro).
- **Learning Rate (η):** $\in \{0.03, 0.06, 0.10\}$ (winning value: 0.06).
- **Subsample / Colsample:** 0.80 / 0.80.
- **Positive Weight Multiplier:** Scaled inverse frequency with a hard cap of 10.0.

Appendix B

Repository Map and Notebook Lineage

Table B.1: Primary project artifacts, notebooks, and source code mapping

Directory / Artifact	File Name	Architectural Scope / Contents
Intention/Classifier/	v1_flat_baseline.ipynb	Flat 256-token transformer baseline with dual heads.
	v2_speaker_aware.ipynb	Additive role embeddings prior to masked pooling.
	v3_augmentation_collapse.ipynb	EDA augmentation run diagnosing metric and weight bugs.
	v4_loss_metric_fix.ipynb	True Macro-F1 metric selection, CB-Focal, and SupCon.
	v5_token_safe_aug.ipynb	Forensic regex tokenization fixing speaker marker gluing.
	v6_hierarchical_dialogue.ipynb	Hierarchical transformer with inter-turn self-attention.
	v8_hierarchical_bt.ipynb	Hierarchical transformer with back-translation.
	v9_ensemble_dropout.ipynb	Tri-model soft ensembling and Monte Carlo uncertainty.
	v10_last_turn_gated.ipynb	Last-Persuadee-turn sigmoid gating architecture.
	v13_solo_binary.ipynb	Modifier head removal and dedicated binary capacity.
	v14_antiphony.ipynb	Dual-stream cross-attention Antiphony architecture.
Strategy/	multistrat-capstone2-final.ipynb	39-strategy baseline with decoupled training tests.
	multistrat-capstone_v17.ipynb	19-member pool, Asymmetric Loss, calibrated blends.
	multistrat-capstone_v18.ipynb	72-grid XGBoost stacker search vs. power-mean blend.
	pipeline.py	Automated schema validation for strategy batches.
Generic/	generate_negotiation_dialogues.py	Persona-conditioned synthetic generation engine.
	humanize_dialogues_gemini.py	Gemini Flash organic paraphrasing and jitter script.
	run_adversarial_ablation.py	Input ablation harness (LTR, LTR+M, FHO, PRO).

Appendix C

Data Contracts and Schemas

Multi-Label Strategy Record Schema (multilabel_merged.jsonl)

Below is an illustrative JSON record from `multilabel_merged.jsonl` representing Turn #42:

```
1 {
2   "turn_id": 42,
3   "dialogue_id": 4,
4   "speaker": "Persuader",
5   "text": "Hello! I am doing well, thank you. Did you know that Save the Children has
6     helped over 100 million children worldwide?",
7   "context_turns": [
8     {"speaker": "Persuader", "text": "Hi, how are you doing today?"},
9     {"speaker": "Persuadee", "text": "I'm good, how about you?"},
10  ],
11  "strategies": [
12    "rapport_building",
13    "evidence_and_statistics",
14    "organization_information"
15  ],
16  "categories": [
17    "Relational and Interactive",
18    "Rational Appeal",
19    "Information Provision"
20  ],
21  "n_labels": 3,
22  "flags": {"empty_set": false, "uncertain": false},
23  "annotator_segment": "segment_track_a"
24 }
```

Adversarial Negotiation Reversal Record Schema (REV_042)

Below is an illustrative record from `final_plus_reversal_v3.json` showing a verified reversal dialogue:

```
1 {
2   "dialogue_id": "REV_042",
3   "regime": "ADVERSARIAL_REVERSAL",
4   "reversal_type": "resistance_to_agreement",
5   "product": "Sony WH-1000XM5 Wireless Headphones",
6   "archetype": "The Skeptic",
7   "reversal_turn_id": 9,
8   "labels": {
9     "buying_intent": "yes",
10    "outcome_category": "PURCHASE_COMMITTED",
11    "reason": "Customer expressed intense counterfeit skepticism, but committed to purchase
12      immediately after merchant provided official distributor warranty serial and
13      offered Cash on Delivery with open-box verification."
14  },
15  "turn_count": 12
16 }
```

Appendix D

Conflict Register and Empirical Audit Log

For transparency, this appendix records how each of the fifteen empirical and textual discrepancies found in the project record was resolved:

Whenever prompt narratives, retrospective slides, or draft summaries contradict raw serialized artifacts (`.jsonl`, `.csv`, evaluation outputs, or reproducible notebook execution cells), the empirical data artifacts are treated as authoritative. Speculative claims and draft misstatements are audited and corrected.

Table D.1: Project conflict register, discrepancies, and resolutions

ID	Topic		Contradiction in Source Record	Resolution in this Report
C1	Strategy Decision Layer		Early drafts incorrectly referred to an XG-Boost stacking system as the winner.	Shipped decision layer is the calibrated power-mean blend (0.5716 macro-F1); XGBoost stacker is documented as tested (0.5508).
C2	Context Window Width		Prompts claimed $k \in \{0, 1, 3, 5, 7\}$ was tested.	Verified tests compare 0 vs 3 turns (0.6675 vs 0.6756) and BoW 0 vs 5 turns; 0.7062 is v17's micro-optimal decode.
C3	Strategy Prevalence Tail		Annotation doc had erroneous tail counts (e.g. <code>door_in_the_face</code> 15).	Recomputed directly from <code>multilabel_merged.jsonl</code> (2, 5, 5, 8, 7, 14, 21, 27, 31).
C4	Strategy Annotator ID		Conflicting references to Claude models and manual raters.	Confirmed as Anthropic Claude 5 Sonnet operating under human calibration checkpoints across four operational segments.
C5	E-Commerce Counts	Dataset	Text prompts claimed 250/250 and 6 outcome classes.	Frozen files contain 262 no / 238 yes ($N = 500$) and 362 no / 338 yes ($N = 700$) across 4 outcome classes.
C6	Ablation Terminology		Prompts cited non-existent regimes (“Suffix”, “+0.1420”).	Followed report: LTR, LTR+M, FHO, PRO; necessity gap dropped from 0.1600 to 0.0402.
C7	Stage Boundaries		Conflicting epoch definitions across prompts.	Standardized to 4 stages: Stage 1 (Flat Baselines), Stage 2 (Hierarchical Transformers), Stage 3 (Gated & Decoupled), Stage 4 (Dual-Stream Antiphony).
C8	Table B Thresholds		Mixed default (0.8623) and tuned (0.8600) scores.	Reported both numbers transparently; noted that Antiphony beats the Decoupled Baseline under either rule.
C9	Unsupported Claims		Inventions: unsubstantiated monetary budgets, inflated emissions, external serving frameworks, and graph runtime exports.	Deleted all unsubstantiated claims; replaced with grounded hardware estimates.
C10	Left-Truncated Flat Run		Presentation cited 0.842 flat run not in chronicle.	Noted the presentation finding; discussed token length diagnosis without building claims on it.
C11	Strategy Encoders		Abstract claimed DeBERTa/TOD-BERT on strategy.	Clarified that strategy used RoBERTa (x6) and ELECTRA (x1); 3-backbone test applied to intention.
C12	Directive Issuer		Referred to “supervisor” vs “course instructor”.	Confirmed as the project supervisor, Dr. Md. Shariful Islam Bhuyan.
C13	Pass Numbering		Guidelines and reports used divergent pass labels.	Followed <code>donation_intent.md</code> : Pass 1 (dual), Pass 2 (binary), Pass 3 (strict).
C14	Agreement Scope		92.21% cited as a 5-annotator study.	Clarified: 92.21% represents independent second-pass on 154 dialogues.
C15	HiTrans Attribution		Chronicle cited “Le et al., EMNLP 2021”.	Corrected to Li et al., COLING 2020 [16].

Appendix E

Complete 41-Strategy Empirical Prevalence Table

Table E.1 reproduces the data-verified prevalence of all 41 strategies across the 10,600 Persuader turns of `multilabel_merged.jsonl`, ordered from highest to lowest frequency.

Table E.1: Data-verified prevalence of all 41 persuasion strategies (10,600 turns)

Strategy Identifier	Parent Category	Active Turns	Prevalence (%)
<code>rapport_building</code>	Relational and Interactive	2,860	27.0%
<code>organization_information</code>	Information Provision	1,735	16.4%
<code>donation_procedure_information</code>	Information Provision	1,417	13.4%
<code>personal_related_inquiry</code>	Relational and Interactive	1,328	12.5%
<code>personal_story</code>	Relational and Interactive	1,150	10.8%
<code>emotion_appeal</code>	Emotional Appeal	1,023	9.7%
<code>gratitude_and_appreciation</code>	Reciprocity and Exchange	966	9.1%
<code>minimization_framing</code>	Framing and Presentation	849	8.0%
<code>task_clarification</code>	Information Provision	770	7.3%
<code>hope_and_positive_impact</code>	Emotional Appeal	628	5.9%
<code>self_modeling</code>	Social Influence	619	5.8%
<code>source_related_inquiry</code>	Relational and Interactive	615	5.8%
<code>in_group_appeal</code>	Social Influence	598	5.6%
<code>impact_information</code>	Information Provision	477	4.5%
<code>organizational_credibility</code>	Credibility Appeal	389	3.7%
<code>evidence_and_statistics</code>	Rational Appeal	378	3.6%
<code>logical_appeal</code>	Rational Appeal	304	2.9%
<code>transparency_and_accountability</code>	Credibility Appeal	274	2.6%
<code>source_citation</code>	Credibility Appeal	193	1.8%
<code>persistent_repetition</code>	Threat and Pressure	167	1.6%
<code>urgency_appeal</code>	Urgency and Scarcity	150	1.4%
<code>donor_benefit</code>	Reciprocity and Exchange	109	1.0%
<code>anchoring</code>	Framing and Presentation	94	0.9%
<code>value_consistency_appeal</code>	Commitment and Consistency	79	0.7%
<code>personal_credibility</code>	Credibility Appeal	78	0.7%
<code>obligation_pressure</code>	Threat and Pressure	78	0.7%
<code>comparison_framing</code>	Framing and Presentation	74	0.7%
<code>social_proof</code>	Social Influence	67	0.6%
<code>empathy_and_perspective_taking</code>	Emotional Appeal	66	0.6%
<code>feasibility_and_ease</code>	Rational Appeal	55	0.5%
<code>foot_in_the_door</code>	Commitment and Consistency	53	0.5%
<code>incremental_ask</code>	Commitment and Consistency	35	0.3%
<code>guilt_induction</code>	Emotional Appeal	31	0.3%
<code>cost_benefit_framing</code>	Rational Appeal	27	0.3%
<code>loss_versus_gain_framing</code>	Framing and Presentation	21	0.2%

Strategy Identifier	Parent Category	Active Turns	Prevalence (%)
reciprocity	Reciprocity and Exchange	14	0.1%
scarcity_appeal	Urgency and Scarcity	8	0.1%
negative_consequence_warning	Threat and Pressure	7	0.1%
authority_endorsement	Social Influence	5	0.0%
deadline_pressure	Urgency and Scarcity	5	0.0%
door_in_the_face	Commitment and Consistency	2	0.0%